The U(1) Higgs Model

I. The Continuum Limit

Abstract. By using rigorous renormalization group methods we construct the continuum limit of the finite-volume lattice U(1) Higgs model in three dimensions. The method relies on a proof of the convergence of the effective action.

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1. Introduction

Recently, renormalization group methods have been used to study lattice regularizations of Euclidean quantum field theories. In particular, Balaban has proved ultra-violet stability for the finite volume lattice Higgs model in three dimensions [Ba 1–4], obtaining bounds independent of the lattice spacing. In this paper we construct the continuum limit of this model in three dimensions. In a succeeding paper [II], the infinite volume limit will also be constructed, and some of the Osterwalder-Schrader axioms verified. This model was constructed previously in two dimensions by Brydges et al. [BFS 1–3].

The U(1) Higgs model is an interacting theory of a vector field $A_\mu(x)$ coupled in a gauge covariant way to a N-component scalar field $\phi(x)$. The classical (Euclidean) action of the model

in d dimensions is

$$S(A,\phi) = \int d^d x \left\{ 1/4 \sum_{\mu,\nu=1}^{d} |F_{\mu\nu}(x)|^2 + 1/2 \sum_{\mu=1}^{d} |D_\mu \phi(x)|^2 + 1/2\, m^2 |\phi(x)|^2 + \lambda |\phi(x)|^4 \right\}. \tag{1.1}$$

The field strength tensor is $F_{\mu\nu}(x) = \partial_\mu A_\nu(x) - \partial_\nu A_\mu(x)$, and the covariant derivative of the scalar field is

$$(D_\mu \phi)_i(x) = \partial_\mu \phi_i(x) - e A_\mu(x)\, (q\phi)_i(x)\,. \tag{1.2}$$

The coupling constants of the theory are e, λ and q is an antisymmetric $N \times N$ matrix. The action (1.1) is invariant under local gauge transformations: define

$$A_\mu^\chi = A_\mu(x) - \partial_\mu \chi(x)\,, \qquad \phi^\chi(x) = \exp[-eq\chi(x)]\phi(x)\,, \tag{1.3}$$

where $\chi(x)$ is an arbitrary real function on R^d. Then

$$S(A,\phi) = S(A^\chi, \phi^\chi)\,. \tag{1.4}$$

We study the quantum field theory corresponding to the action (1.1) by using the lattice approximation introduced by Wilson [W 1]. The model is considered on a finite regular lattice T_ε, with spacing ε, and (1.1) is replaced by a lattice action $S^\varepsilon(T_\varepsilon, A, \phi)$ (gauge-invariant except for the mass and gauge-fixing terms of the vector field) used to define a functional integral

$$Z^\varepsilon(T_\varepsilon) = \int (dA)\,(d\phi)\, \exp[-S^\varepsilon(T_\varepsilon, A, \phi)]\,. \tag{1.5}$$

These cutoff functional integrals can be analyzed using a phase cell expansion, first introduced by Glimm and Jaffe [GJ 1]. On the lattice, the expansion is implemented using a block spin transformation. The lattice is divided into disjoint blocks, each containing L^d sites, for some small integer L. Every field configuration $\{A, \phi\}$ is separated into an average part $\{A_1, \phi_1\}$, which is constant over each block, and a fluctuation part $\{A', \phi'\}$, which varies within blocks. The integral over $\{A', \phi'\}$ is estimated using a standard weak coupling expansion, producing a new effective action $S^{(1),L\varepsilon}(T_{L\varepsilon}, A_1, \phi_1)$ for the average fields. Contributions from configurations where fields are large can be estimated, and a perturbative expansion derived for $S^{(1),L\varepsilon}$ in the small field region ("large" and "small" fields will be defined later). By repeating the block spin transformation k times, we produce an effective action $S^{(k),L^k\varepsilon}$ on the $L^k\varepsilon$-lattice. This program has been carried out by Balaban for the Higgs model in dimensions $d = 2, 3$, giving the ultraviolet stability bound [Ba 1–4].

The renormalization transformation can be understood from a different viewpoint if we re-scale every effective action $S^{(k)}$ to the unit lattice. When $k = 0$, this is the original action; the re-scaling replaces the parameters m^2, λ, e of the theory by $m^2\varepsilon^2$, $\lambda\varepsilon^{4-d}$, $e\varepsilon^{2-d/2}$. So for $d < 4$, the action is a small perturbation of the massless gaussian action given by

$$S_G(A,\phi) = \sum_x \left\{ 1/4 \sum_{\mu,\nu=1}^{d} |\partial_\mu A_\nu(x) - \partial_\nu A_\mu(x)|^2 + 1/2 \sum_{\mu=1}^{d} |\partial_\mu \phi(x)|^2 \right\} \tag{1.6}$$

(∂_μ is the difference derivative on the unit lattice). We can imagine a space of unit lattice actions as in Fig. 1; the renormalization transformation takes every point along a trajectory in this space. The massless gaussian (1.6) at G is driven to a fixed point P in the subspace V.

In our case, the initial action $S^{(0)}$ is at B, outside the gaussian subspace, and is driven along a trajectory away from V (this is ultraviolet asymptotic freedom). Consider a fixed lattice spacing ε_0; when $L^k\varepsilon = \varepsilon_0$, this trajectory is at B', and the effective action (rescaled to the ε_0-lattice) is $S^{(k),\varepsilon_0}$. It is clear that if the initial action $S^{(0)}$ is defined on a smaller

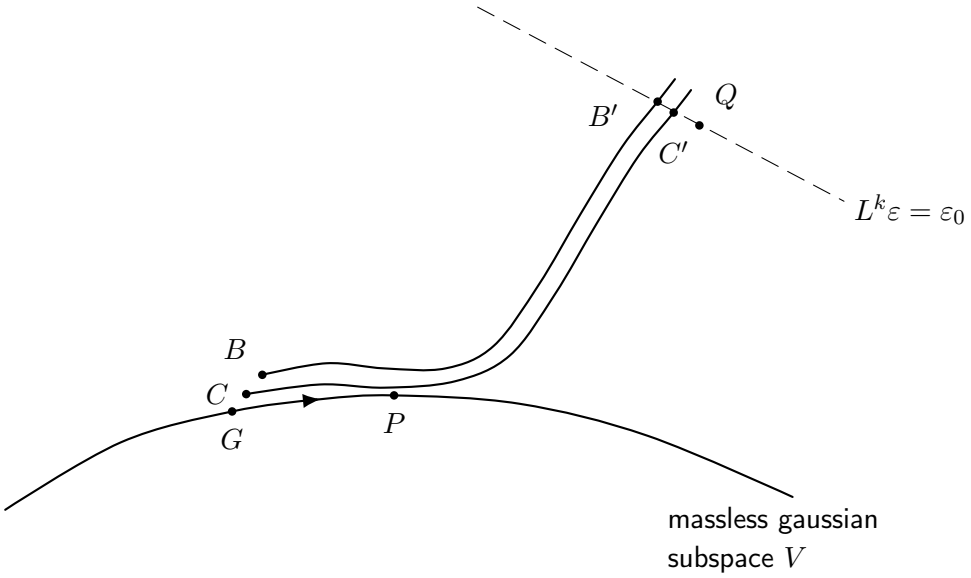


Fig. 1. Renormalization transformation trajectories in the space of unit lattice actions

lattice, it is given by a point C closer to G. So taking the limit $\varepsilon \to 0$ produces a sequence of starting points converging to G. Corresponding to these, there is a sequence of effective actions $\{S^{(k),\varepsilon_0}\}$, $k = 0, \ldots, \infty$, on the ε_0-lattice.

To construct the continuum theory, we need to know that the partition function with a source coupled to the field strength, normalized by its value without the source (Theorem 2.1), converges to a finite limit as the initial points $B, C, \ldots$ converge to G. By using the renormalization transformation, this reduces to proving the convergence of $\{S^{(k),\varepsilon_0}\}$ as $k \to \infty$, up to constants independent of the fields and of the source, or that the points $B', C' \ldots$ converge to some point Q. In this paper we prove that the perturbation expansions for $\{S^{(k),\varepsilon_0}(A, \phi)\}$ converge in this sense provided the fields A, ϕ are small. By taking ε_0 arbitrarily small this allows us to derive convergence of the normalized partition function as $\varepsilon \to 0$. The generating functional with a scalar source is treated in [II].

We use the non-compact form of the action for the vector field, in the Feynman gauge of [Ba 1], and also introduce an explicit vector field mass μ_0, which is kept fixed throughout. However, the methods used in this paper apply to any other model analyzed using the renormalization transformation, for example to the compact formulation of the Higgs model under consideration by Balaban et al. [BBIJ 1], [BIJ 1].

The paper is organized as follows. Section 2 provides a summary of the renormalization transformation notation, and a statement of the main result. Section 3 contains the proof of convergence of the effective action, and hence of the normalized partition function, and in Sect. 4, we derive the technical estimates used in the proof. Section 5 collects the results on the renormalization transformation with the block size $L = 5$ of Sect. 2.2, on the mass counterterm and on the range of the coupling constants which are used here and in [II].

2. Notation and Results

2.1. Definition of the Model. Let Ω be a subset of εZ^d; the set of oriented bonds connecting nearest neighbor sites in Ω is denoted Ω^*. If $b \in \Omega^*$ has endpoints x and y, we write $b = \langle x, y\rangle = \langle b_-, b_+\rangle$. Then a vector field configuration on Ω is a map $A : \Omega^* \to R$; it can also be interpreted as a map $A : \Omega \to R^d$. We identify these meanings by $A(\langle x, x + \varepsilon e_\mu\rangle) = A_\mu(x)$, where e_μ is the unit vector in R^d parallel to the x_μ-axis. We assume that $A_{-b} = -A_b$. A scalar field configuration on Ω is a map $\phi : \Omega \to R^N$; we take $N = 2$, the case to which the

Gaussian domination of Sect. 3.2 and the correlation inequalities used in [II] apply. The scalar and vector fields interact in Ω via a unitary operator on R^N: $U(A) = \exp(\varepsilon e q A)$, where e is the electric charge (coupling constant) and q is an $N \times N$ anti-symmetric matrix with $\|q\| = 1$. The covariant derivative of the scalar field is

$$(D_A^\varepsilon \phi)(b) = \varepsilon^{-1}(U(A_b)\phi(b_+) - \phi(b_-)) \tag{2.1}$$

(we denote by ∂^ε the usual derivative on εZ^d). We introduce an inner product on Ω for n-component functions, any n:

$$\langle f, g\rangle = \sum_{i=1}^{n} \sum_{x\in\Omega} \varepsilon^d f_i(x) g_i(x)\,. \tag{2.2}$$

With this notation, the covariant Laplacian on Ω is

$$1/2\langle\phi, (-\Delta_A^{\varepsilon,\Omega})\phi\rangle = 1/2 \sum_{b\in\Omega^*} \varepsilon^d |(D_A^\varepsilon\phi)(b)|^2\,. \tag{2.3}$$

The regularized action which we consider is

$$\begin{aligned} S^\varepsilon(\Omega, A, \phi) = {} & 1/2\langle A, (-\Delta^{\varepsilon,\Omega} + \mu_0^2)A\rangle + 1/2\langle\phi, (-\Delta_A^{\varepsilon,\Omega} + m^2)\phi\rangle \\ & + \sum_{x\in\Omega} \varepsilon^d (1/2\,\delta m^2(x)|\phi(x)|^2 + \lambda|\phi(x)|^4) + E_0 + E_1\,, \end{aligned} \tag{2.4}$$

and the corresponding partition function is

$$Z^\varepsilon(\Omega) = \int (dA)\,(d\phi)\, \exp[-S^\varepsilon(\Omega, A, \phi)]\,. \tag{2.5}$$

Explicit expressions for the counterterms δm^2 and E_1 will be given later. E_0 is the normalization for the Gaussian measure:

$$\exp[E_0] = \int (dA)\,(d\phi)\, \exp[-1/2\langle A, (-\Delta^{\varepsilon,\Omega} + \mu_0^2)A\rangle - 1/2\langle\phi, (-\Delta^{\varepsilon,\Omega} + m^2)\phi\rangle]\,. \tag{2.6}$$

Here μ_0, m and λ are positive.

Since we are interested in gauge invariant Schwinger functions, we shall introduce sources coupled to gauge invariant operators. Denote by Ω^{**} the set of plaquettes on Ω; a plaquette p is an elementary square of the lattice, written $p = \langle x, x + \varepsilon e_\mu, x + \varepsilon e_\mu + \varepsilon e_\nu, x + \varepsilon e_\nu\rangle$ with $\mu < \nu$. The field strength tensor on Ω is a map $F : \Omega^{**} \to R$ given by $F(p) = \sum_{b\in\partial p} \varepsilon^{-1} A_b$. For a function $g : \Omega^{**} \to R$, we define

$$F(g) = \sum_{p\in\Omega^{**}} \varepsilon^d F(p) g(p)\,. \tag{2.7}$$

Here $g(p) = g_{\mu\nu}(x_p)$, where $g_{\mu\nu}$ are the components of a real 2-form on R^d and x_p is the center of the plaquette p. This is the form in which two lattices are compared in Sect. 4 of [II].

The gauge invariant operator for the scalar field is $|\phi|^2$; of course this must be normal-ordered in the generating functional. So for $h : \Omega \to R$, we have

$$:|\phi|^2:(h) = \sum_{x\in\Omega} \varepsilon^d h(x) \Big\{ \sum_{i=1}^{N} \big[\phi_i(x)^2 - (-\Delta^{\varepsilon,\Omega} + m^2)^{-1}(x,x)\big] - a_\varepsilon \Big\}, \tag{2.8}$$

where a_ε is the term of order e^2, and of order zero in λ, in the formal perturbation expansion of $\langle|\phi(0)|^2\rangle$ for the model (2.4) on εZ^d; the counterterm δm^2 is included to this order, and is given by (3.50)–(3.51) with Ω replaced by εZ^d. This subtraction is needed in $d = 3$.

The one-loop self-energy, after the subtraction of its value at zero momentum, is in perturbation theory $-\frac{1}{4}e^2|p|$ up to bounded terms for $|p|$ large compared with m, μ_0 and small compared with ε^{-1} (the lattice modifies the subtracted self-energy in this range by terms which contribute a bounded amount to the integral), so that on εZ^d the term of order e^2 in the subtracted square grows like $\ln \varepsilon^{-1}$:

$$\Big[\langle|\phi(x)|^2\rangle - \sum_i (-\Delta^\varepsilon + m^2)^{-1}(x,x)\Big]_{e^2\lambda^0} = a_\varepsilon = \frac{e^2}{4\pi^2}\ln\varepsilon^{-1} + O(1) \qquad (N=2)\,. \tag{2.9}$$

Hence without a_ε the term of first order in h, of order e^2 and of order zero in λ, in $\ln[Z^\varepsilon(\Omega,0,h)Z^\varepsilon(\Omega,0,0)^{-1}]$ would have no limit as $\varepsilon \to 0$ unless $\int h = 0$; a_ε is defined so that this term vanishes on εZ^d. The constant a_ε does not depend on h or on Ω, and it multiplies the generating functional by $\exp[-a_\varepsilon \sum_x \varepsilon^d h(x)]$.

The term of order zero in e and λ vanishes by the normal ordering in (2.8). The term of order λ and of order zero in e has degree zero by (3.49), but it vanishes identically, since the tadpole does not depend on the momentum and is cancelled by δm^2 of order λ in (3.51). The remaining terms of first order in h, of orders e^4, $e^2\lambda$, λ^2 and higher, have positive degree (3.49), since in $d = 3$ each further power of e^2 or λ raises the degree by one; their subdivergences are removed by δm^2 of (3.50)–(3.51) and by the gauge invariant combinations of the photon two-point subgraphs (Sect. 3.5). The generating functional with the source h is treated in [II]. The generating functional with source g, h is

$$Z^\varepsilon(\Omega,g,h) = \int (dA)\,(d\phi)\,\exp[-S^\varepsilon(\Omega,A,\phi) + F(g) + :|\phi|^2:(h)]\,. \tag{2.10}$$

We shall assume that the components $g_{\mu\nu}$ of g and h are C^∞ functions on R^d with compact support. With abuse of notation, we also denote their restrictions to Ω by g, h. Throughout this paper, we shall use C to denote a positive constant which is either $O(1)$ or depends only on $\|g\|$, $\|h\|$.

2.2. Block Spin Transformations. Throughout this paper we will use the block spin transformations defined in [Ba 1], and the reader is assumed to be familiar with the idea of this method. We present below a collection of relevant notation and formulae connected with the method.

The block size is L, a small integer, which we take to be $L = 5$, the value used in Sect. 4 of [II], and for any $\Omega \subset \varepsilon Z^d$ we write $\Omega^{(k)} = \Omega \cap L^k \varepsilon Z^d$. To each point $y \in L^k \varepsilon Z^d$ there corresponds a block of L^{kd} sites on εZ^d, which we define as $B^k(y) = \{x \in \varepsilon Z^d : y_i \leqq x_i < y_i + L^k\varepsilon,\ i = 1,\dots,d\}$. The averaging operators used to map configurations on Ω into configurations on $\Omega^{(k)}$ are

$$Q_k A_\mu(y) = L^{-kd} \sum_{x \in B^k(y)} A_\mu(x)\,, \tag{2.11}$$

$$(Q_k(A)\phi)(y) = L^{-kd} \sum_{x \in B^k(y)} U(A(\Gamma^{(k)}_{y,x}))\phi(x)\,, \tag{2.12}$$

where the contour $\Gamma^{(k)}_{y,x}$ connects y to x in the following way; writing x_j as the site for which $x \in B^j(x_j)$, we have $\Gamma^{(k)}_{y,x} = \Gamma_{y,x_{k-1}} \cup \Gamma_{x_{k-1},x_{k-2}} \cup \dots \Gamma_{x_1,x}$ and the contour $\Gamma_{z,w}$ is a union of bonds on εZ^d given by

$$\begin{aligned}\Gamma_{z,w} = {} & \langle z, (z_1,\dots,z_{d-1},w_d)\rangle \cup \langle (z_1,\dots,z_{d-1},w_d), (z_1,\dots,z_{d-2},w_{d-1},w_d)\rangle \\ & \cup \dots \cup \langle (z_1,\dots,w_d), w\rangle\,.\end{aligned} \tag{2.13}$$

For $d = 2$ the blocks $B^1(x_1)$, $B^2(y)$ and a contour $\Gamma^{(2)}_{y,x}$ are shown in Fig. 2. Furthermore, $A(\Gamma) = \sum_{b\in\Gamma} \varepsilon A_b$ and $U(A(\Gamma)) = \exp[eq\,A(\Gamma)]$. The block spin transformation proceeds by

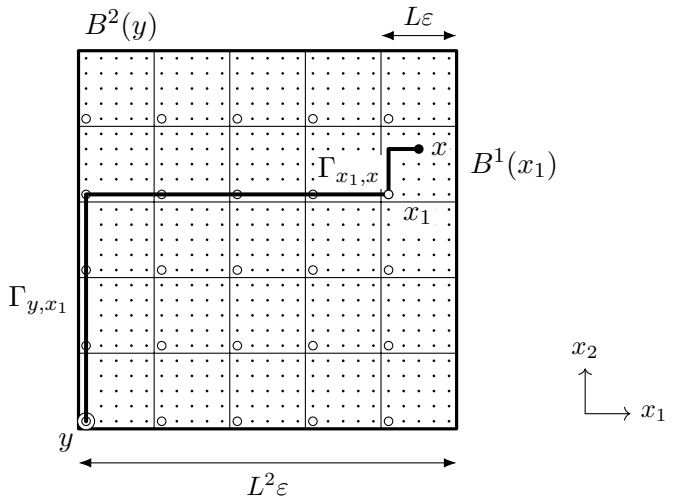


Fig. 2. Blocks $B^1(x_1)$, $B^2(y)$ at $L = 5$ in two dimensions, the block sites of $\Omega^{(1)}$ (small circles) and of $\Omega^{(2)}$ (large circle), and the contour $\Gamma^{(2)}_{y,x} = \Gamma_{y,x_1} \cup \Gamma_{x_1,x}$ of (2.12)–(2.13)

holding fixed these average fields and integrating over fluctuations around them. The covariance of the scalar fluctuation field over blocks of size L^{kd} on Ω is

$$G^{\varepsilon}_k(\Omega, A) = (-\Delta^{\varepsilon,\Omega}_A + m^2 + a_k(L^k\varepsilon)^{-2}Q_k(A)^*Q_k(A))^{-1}\,, \tag{2.14}$$

where $a_k = a(1-L^{-2})(1-L^{-2k})^{-1}$ and a is $O(1)$. The effective Laplacian of the average scalar field on $\Omega^{(k)}$ is

$$\Delta^{(k),L^k\varepsilon}(\Omega, A) = a_k(L^k\varepsilon)^{-2}I - a_k^2(L^k\varepsilon)^{-4}Q_k(A)G^{\varepsilon}_k(\Omega, A)Q_k(A)^*\,. \tag{2.15}$$

We will also need the normalization factor for (2.14):

$$Z^{\varepsilon}_k(\Omega, A) = \left(\frac{a_k(L^k\varepsilon)^{d-2}}{2\pi}\right)^{N/2(L^k\varepsilon)^{-d}|\Omega^{(k)}|} \int (d\phi)\, \exp[-1/2\langle\phi, G^{\varepsilon}_k(\Omega, A)^{-1}\phi\rangle]\,, \tag{2.16}$$

where $(L^k\varepsilon)^{-d}|\Omega^{(k)}|$ is the number of lattice sites in $\Omega^{(k)}$. The covariance of the scalar fluctuation field over blocks of size L^d on $\Omega^{(k)}$ is

$$C^{(k),L^k\varepsilon}(\Omega, A) = (\Delta^{(k),L^k\varepsilon}(\Omega, A) + a(L^{k+1}\varepsilon)^{-2}Q(A)^*Q(A))^{-1}\,, \tag{2.17}$$

where $C^{(0),\varepsilon}(\Omega, A) = G^{\varepsilon}_1(\Omega, A)$. The operator (2.14) can be decomposed into a sum of contributions from scales between Ω and $\Omega^{(k)}$:

$$\begin{aligned} G^{\varepsilon}_k(\Omega, A) = C^{(0),\varepsilon}(\Omega, A) + \sum_{j=1}^{k-1} a_j^2(L^j\varepsilon)^{-4}G^{\varepsilon}_j(\Omega, A)Q_j(A)^* \\ \cdot C^{(j),L^j\varepsilon}(\Omega, A)Q_j(A)G^{\varepsilon}_j(\Omega, A) \equiv \sum_{j=0}^{k-1} G^{\varepsilon}_{(j)}(\Omega, A)\,, \end{aligned} \tag{2.18}$$

with the obvious definition of $G^{\varepsilon}_{(j)}(\Omega, A)$. All the corresponding expressions for the vector field are obtained from (2.14)–(2.18) by setting $m^2 = \mu_0^2$, $N = d$ and $A = 0$.

After k steps of the block spin transformation, the original vector field A on Ω is replaced by a sum of $k+1$ independent fields, coming from fluctuation fields A'_j defined on $\Omega^{(j)}$, and the average field A_k on $\Omega^{(k)}$:

$$A = \sum_{j=0}^{k-1} A'^{(j),\varepsilon} + A^{(k),\varepsilon} = A' + A^{(k),\varepsilon}\,, \tag{2.19}$$

where $A'^{(0),\varepsilon} = A'_0$, $A'^{(j),\varepsilon} = a_j(L^j\varepsilon)^{-2}G_j^{\varepsilon}(\Omega)Q_j^* A'_j$ for $1 \leqq j \leqq k-1$, and $A^{(k),\varepsilon} = a_k(L^k\varepsilon)^{-2}G_k^{\varepsilon}(\Omega)Q_k^* A_k$. When A is given by (2.19), we adopt the following convention:

$$A(\Gamma_{y,x}^{(k)}) = \sum_{j=0}^{k-1} A'^{(j),\varepsilon}(\Gamma_{x_{j+1},x}^{(j+1)}) + A^{(k),\varepsilon}(\Gamma_{y,x}^{(k)}), \tag{2.20}$$

In Sect. 3.3 we will describe in detail the perturbative expansion of the effective action obtained for the fields ϕ_k and A_k on $\Omega^{(k)}$ after applying k steps of the block spin transformation to the original action on Ω. We call this effective action $S^{(k),L^k\varepsilon}(\Omega^{(k)}, A_k, \phi_k, g, h)$, or if $\Omega^{(k)}$ is rescaled to have lattice spacing equal to one, it is written $S^{(k),1}$.

Finally we note how operators transform under rescalings of the lattice. If we rescale εZ^d to ηZ^d, the following relations hold between the operators on those lattices:

$$\begin{aligned}&\{G_k^{\varepsilon}, \Delta^{(k),L^k\varepsilon}, C^{(k),L^k\varepsilon}\}(x,y)\\ &\quad = \{(\varepsilon/\eta)^{2-d}G_k^{\eta}, (\varepsilon/\eta)^{-(2+d)}\Delta^{(k),L^k\eta}, (\varepsilon/\eta)^{2-d}C^{(k),L^k\eta}\}\\ &\qquad \cdot (\eta x/\varepsilon, \eta y/\varepsilon)\,.\end{aligned} \tag{2.21}$$

When $\eta = L^{-k}$ we will omit the lattice superscript.

2.3. Statement of Results. We shall consider the model on a torus T defined by

$$T = \{x \in R^d : -L_\mu \leqq x_\mu < L_\mu,\ \mu = 1, \ldots, d\}, \tag{2.22}$$

i.e. we use periodic boundary conditions on T. We require that $L_\mu M^{-1} \in N$, each $\mu = 1, \ldots, d$, where M is a large integer (this is the same integer that is specified in [Ba 4]); we take M to be a multiple of 50; this is used in Sect. 4 of [II]. We then define lattice spacings $\{\varepsilon_K = L^{-K}\}$, $K = 0, \ldots, \infty$; so for each $K \geqq 0$, T can be fitted exactly by neighboring disjoint blocks of M^d sites on the lattice $\varepsilon_K Z^d$ (henceforth called "large blocks"). We write $T_\varepsilon = T \cap \varepsilon Z^d$.

In [Ba 1–4] uniform upper and lower bounds were derived for $Z^{\varepsilon_K}(T_{\varepsilon_K})$ in $d = 2, 3$. We prove the existence of the continuum limit of $Z^{\varepsilon_K}(T_{\varepsilon_K}, g, 0)\, Z^{\varepsilon_K}(T_{\varepsilon_K}, 0, 0)^{-1}$ as $K \to \infty$ in $d = 3$, and we prove bounds on it uniform in K.

Theorem 2.1. *For the torus T defined by* (2.22) *in dimension $d = 3$, with $L_1 = L_2 = L_3$ in* (i)*, and for g as in Sect. 2.1,*

$$\text{(i)} \qquad \exists \lim_{K\to\infty} Z^{\varepsilon_K}(T_{\varepsilon_K}, g, 0)\, Z^{\varepsilon_K}(T_{\varepsilon_K}, 0, 0)^{-1} = Z(T, g)\,, \qquad Z(T, g) \geqq 1\,, \tag{2.23}$$

$$\begin{aligned}&\text{(ii)} \qquad |\ln Z^{\varepsilon_K}(T_{\varepsilon_K}, 0, 0)| \leqq C|T|\,,\\ &\qquad\qquad 0 \leqq \ln[Z^{\varepsilon_K}(T_{\varepsilon_K}, g, 0)\, Z^{\varepsilon_K}(T_{\varepsilon_K}, 0, 0)^{-1}] \leqq 1/2\, \|g\|^2\,,\end{aligned} \tag{2.24}$$

where $|T| = \prod_{\mu=1}^d 2L_\mu$, $\|g\|^2 = \sum_p \varepsilon_K^d\, g(p)^2$, and C depends only on the coupling constants, L and M.

Theorem 2.1 is proved in Sect. 3.2, statement (i) from Theorem 3.4: the terms of $S^{(k),1}$ which depend neither on the fields nor on g cancel in the ratio, those which depend on g are retained in $S_N^{(k),1}$ and bounded by Theorem 3.4, and Theorem 3.4 is proved in Sects. 3.3–3.6; the first inequality of (2.24) is the ultra-violet stability bound of [Ba 1–4], and the modifications required for $L = 5$ are described in Sect. 5.1.

Only ratios of partition functions are needed in [II]; the generating functional with a scalar source h, for which the constant a_ε of (2.8) is needed, and the comparison of two lattices are considered there (Sects. 2 and 4 of [II]).

3. Convergence in a Finite Volume

3.1. Summary of Previous Results. In this section we will recall some results from the papers [Ba 1–4]. We use the following quantity to define "large" and "small" fields:

$$p(\varepsilon) = b_0(1 + \log \varepsilon^{-1})^p \,, \tag{3.1}$$

where p is a small integer, which we take to be 6, and b_0 is a constant $O(1)$. After rescaling to the unit lattice $T_1^{(k)}$, the effective action is written $S^{(k),1}(T_1^{(k)}, A_k, \phi_k, g, h)$ with A_k, ϕ_k unit lattice fields. Let us denote by $\chi_k(A_k, \phi_k)$ the function which is one when all the following inequalities hold and zero otherwise:

$$\begin{aligned} |A_{k,\mu}(y)| &\leqq p(L^{k-1}\varepsilon)\,(\mu_0 L^{k-1}\varepsilon)^{-1} \,, \\ |\partial A_{k,\mu}(b)| &\leqq p(L^{k-1}\varepsilon) \,, \\ |\phi_k(y)| &\leqq p(L^{k-1}\varepsilon)\,[\lambda (L^{k-1}\varepsilon)^{4-d}]^{-1/4} \,, \\ |D_{A_k}\phi_k(b)| &\leqq p(L^{k-1}\varepsilon) \,, \end{aligned} \tag{3.2}$$

for all y, b and μ, and where

$$D_{A_k}\phi_k(b) = \exp[e(L^k\varepsilon)^{2-d/2} q A_k(b)]\phi_k(b_+) - \phi_k(b_-) \,. \tag{3.3}$$

We can now formulate the basic results of the papers [Ba 1–2].

Theorem 3.1. *Let T be defined by (2.22) in dimension $d = 3$. Then for some integer J depending on the coupling constants, g, h and $|T|$, for all $K \geqq 2J$, and all $J \leqq k \leqq K - J$, and for some $\sigma > 0$,*

$$Z^{\varepsilon_K}(T_{\varepsilon_K}, g, h) \geqq \int (dA_k)\,(d\phi_k)\chi_k(A_k, \phi_k) \exp[-S^{(k),1}(T_1^{(k)}, A_k, \phi_k, g, h) - C(L^k\varepsilon_K)^\sigma |T|] \,, \tag{3.4}$$

$$\begin{aligned} Z^{\varepsilon_K}(T_{\varepsilon_K}, g, h) \leqq & \int (dA_k)\,(d\phi_k)\chi_k(A_k, \phi_k) \exp[-S^{(k),1}(T_1^{(k)}, A_k, \phi_k, g, h) + C(L^k\varepsilon_K)^\sigma |T|] \\ & + \exp[-c\,p(L^k\varepsilon_K)^2 + C|T|] \,. \end{aligned} \tag{3.5}$$

Once again, C depends on g, h and $|T| = \prod_{\mu=1}^d 2L_\mu$, and $c > 0$ is a constant. The bounds (3.4) and (3.5) are weaker than those of [Ba 1–2], and are all that is needed below; they will be used at fixed T only, for $k = K - [\vartheta K]$ with K large (Sect. 3.2).

The bounds on the fields (3.2) are different from those used in [Ba 1–2]. This change is justified in Appendix A, and also the inclusion of sources g, h is presented. The reader is referred to [Ba 1–4] for the proof of Theorem 3.1. The proofs there are written for L sufficiently large. We fix $L = 5$ as in Sect. 2.2; the conditions which [Ba 1–4] impose on L, M, p and b_0, and the verification that they hold for this L, are given in Sect. 5.1. We will also use extensively the decay and regularity results obtained in [Ba 4] for the operators (2.14), (2.15), and (2.17).

When $\Omega = T_\varepsilon$, the torus, all operators are defined with periodic boundary conditions, and for convenience we will omit their volume dependence. When Ω is a proper subset of T_ε and $\Omega^{(k)}$ is a union of large blocks on $T^{(k)}_{L^k\varepsilon}$, we will define $G^\varepsilon_k(\Omega, A)$, $\Delta^{(k),L^k\varepsilon}(\Omega, A)$ and $C^{(k),L^k\varepsilon}(\Omega, A)$ using Neumann boundary conditions on $\partial\Omega$.

Definition 3.2. Let A be a vector field configuration on Ω_η, where $\Omega^{(k)} \subset T_1^{(k)}$ is a union of large blocks and $\eta = L^{-k}$. Then A is *regular* on Ω_η if

$$|\partial^\eta_\mu A_\nu(x)| \leqq C(e(L^k\varepsilon)^{2-d/2})^{\beta-1} \tag{3.6}$$

for all $x \in \Omega_\eta$, $\mu, \nu = 1, \ldots, d$ and some $\beta > 0$ (e is the electric charge).

The following theorem summarises the decay properties of the operators (2.14), (2.15), and (2.17).

Theorem 3.3. *Let A be regular on Ω_η, with $\eta = L^{-k}$, and let $\Omega^{(k)}$ be a rectangular parallelepiped which is a union of large blocks on $T_1^{(k)}$. Then for all $k \leqq K$, some $\delta_0 > 0$, $0 < \alpha < 1$, and for $f : \Omega_\eta \to R^N$,*

$$|\Delta^{(k)}(\Omega, A)(x, y)|\,, \qquad |C^{(k)}(\Omega, A)(x, y)| \leqq C \exp[-\delta_0 |x - y|]\,, \tag{3.7}$$

$$|G_k(\Omega, A)f(x)|\,, \qquad |D^\eta_{A,\mu} G_k(\Omega, A)f(x)| \leqq C \exp[-\delta_0 \operatorname{dist}(x, \operatorname{supp} f)]\, \|f\|\,, \tag{3.8}$$

$$\begin{aligned} &\frac{1}{|x - y|^\alpha} \left|U(A(\Gamma_{y,x}))D^\eta_{A,\mu} G_k(\Omega, A)f(x) - D^\eta_{A,\mu} G_k(\Omega, A)f(y)\right| \\ &\qquad \leqq C \exp[-\delta_0 \operatorname{dist}(\{x, y\}, \operatorname{supp} f)]\, \|f\|\,. \end{aligned} \tag{3.9}$$

Finally, define $\delta C^{(k)}(\Omega, A) = C^{(k)}(\Omega, A) - C^{(k)}(A)$ and $\delta G_k(\Omega, A) = G_k(\Omega, A) - G_k(A)$; then for $x, y \in \Omega_\eta$, $\delta C^{(k)}(\Omega, A)$ and $\delta G_k(\Omega, A)$ satisfy the bounds (3.7) and (3.8)–(3.9) respectively, with the additional factors

$$\exp[-\delta_0 \operatorname{dist}(\{x, y\}, \partial\Omega)] \tag{3.10}$$

and

$$\exp[-\delta_0 \operatorname{dist}(\{x, y\}, \partial\Omega) - \delta_0 \operatorname{dist}(\operatorname{supp} f, \partial\Omega)] \tag{3.11}$$

respectively on the right-hand sides.

Theorem 3.3 is proved in [Ba 4]. It implies the same bounds for the vector field operators (of course with $A = 0$).

3.2. Convergence of the Partition Function. We will use the convergence of the effective action in the small field region to deduce convergence of the partition function. Consider the model defined on the lattices T_{ε_K} and $T_{\varepsilon_{K+n}}$ for some integer n. Then by applying the renormalization transformation k and $k + n$ times respectively we generate two models on the same unit lattice $T_1^{(k)}$, with effective actions $S^{(k),1}$ and $S^{(k+n),1}$. We write

$$S_N^{(k),1}(T_1^{(k)}, A_k, \phi_k, g) = S^{(k),1}(T_1^{(k)}, A_k, \phi_k, g, 0) - S^{(k),1}(T_1^{(k)}, 0, 0, 0, 0) \tag{3.12}$$

for the effective action at $h = 0$ with the constant $S^{(k),1}(T_1^{(k)}, 0, 0, 0, 0)$ removed. This constant cancels in the ratios considered below; the field-independent terms of $S_N^{(k),1}$ which depend on g are retained, and the theorem bounds them.

The following theorem is the core of this paper. We state it for $n = 1$, which is all that is needed below.

Theorem 3.4. *Let $h = 0$ and $n = 1$, let the torus (2.22) be a cube, $L_1 = L_2 = L_3$, let $K \geqq K_0$, where K_0 depends on ϑ, and $k = K - [\vartheta K]$ with $0 < \vartheta < 1/9$. Then*

$$\begin{aligned} &\chi_k(A_k, \phi_k)|S_N^{(k),1}(T_1^{(k)}, A_k, \phi_k, g) - S_N^{(k+1),1}(T_1^{(k)}, A_k, \phi_k, g)| \\ &\qquad \leqq C(L^{-\gamma k}(L^k \varepsilon_K)^{-\beta} + (L^k \varepsilon_K)^\sigma)\, |T|\,, \end{aligned} \tag{3.13}$$

where $0 < \gamma < 1$, $0 < \sigma$, $0 < \beta$ do not depend on ϑ, and C depends on g, ϑ and $|T|$.

Theorem 3.4 is proved in Sects. 3.3–3.6. We can now prove Theorem 2.1 in the form stated in Sect. 2.3; the proof of (i) uses Theorem 3.4 and is therefore given for the cubic torus, $L_1 = L_2 = L_3$; the bounds (ii) hold for every torus (2.22) and do not use Theorem 3.4. We take $h = 0$ throughout, and $N = 2$ as in Sect. 2.1; the argument uses the change of sign of $F(g)$ under $A \to -A$, which has no analogue for $:|\phi|^2:(h)$. We write

$$Z_K(g) = Z^{\varepsilon_K}(T_{\varepsilon_K}, g, 0)\,, \tag{3.14}$$

and $I_K^{(k)}(g)$ for the integral on the right-hand side of (3.4) with the factor $\exp[-C(L^k \varepsilon_K)^\sigma |T|]$ omitted.

We apply the renormalization transformation k times to $Z_K(g)$ and $k+1$ times to $Z_{K+1}(g)$, with $k = K - [\vartheta K]$ as in Theorem 3.4. Since $L^k \varepsilon_K = L^{k+1} \varepsilon_{K+1}$, the fields A_k, ϕ_k have the same bounds in each integral. Writing

$$b = C(L^k \varepsilon_K)^\sigma |T|\,, \qquad \rho = \exp[-c\,p(L^k \varepsilon_K)^2 + C|T|]\,, \tag{3.15}$$

Theorem 3.1 gives

$$e^{-b}\, I_K^{(k)}(g) \leqq Z_K(g) \leqq e^b\, I_K^{(k)}(g) + \rho\,, \tag{3.16}$$

and the same bounds for $Z_{K+1}(g)$ and $I_{K+1}^{(k+1)}(g)$.

On the support of χ_k the effective actions $S^{(k),1}$ and $S^{(k+1),1}$ differ by the constant

$$\Delta = S^{(k),1}(T_1^{(k)}, 0, 0, 0, 0) - S^{(k+1),1}(T_1^{(k)}, 0, 0, 0, 0) \tag{3.17}$$

and by $S_N^{(k),1} - S_N^{(k+1),1}$, which Theorem 3.4 bounds by

$$\theta = C(L^{-\gamma k}(L^k \varepsilon_K)^{-\beta} + (L^k \varepsilon_K)^\sigma)|T|\,. \tag{3.18}$$

Since both integrals are over the same variables with the same factor χ_k and positive integrands,

$$e^{\Delta - \theta} \leqq I_{K+1}^{(k+1)}(g)\, I_K^{(k)}(g)^{-1} \leqq e^{\Delta + \theta}\,, \tag{3.19}$$

where Δ does not depend on g.

From (3.16) we have, for $\rho \leqq \frac{1}{2} Z_K(g)$,

$$|\ln[Z_K(g)\, I_K^{(k)}(g)^{-1}]| \leqq b + 2\rho\, Z_K(g)^{-1}\,. \tag{3.20}$$

Here $Z_K(0) \geqq e^{-C|T|}$ by the ultra-violet stability bound, and $Z_K(g) \geqq Z_K(0)$ by Jensen's inequality, since the measure at $g = 0$ is invariant under $(A, \phi) \to (-A, R\phi)$, $R = \mathrm{diag}(1, -1)$, for which $Rq = -qR$; under this transformation $F(g)$ changes sign, so that the mean of $F(g)$ vanishes. Combining (3.16) and (3.19) for g and for $g = 0$, the constant Δ cancels and we get

$$|\ln[Z_{K+1}(g) Z_{K+1}(0)^{-1}] - \ln[Z_K(g) Z_K(0)^{-1}]| \leqq 4b + 2\theta + 8\rho\, e^{C|T|}\,. \tag{3.21}$$

We take k as in Theorem 3.4, with

$$k = K - [\vartheta K]\,, \qquad 0 < \vartheta < \min\{\gamma(\gamma + \beta)^{-1}, 1/9\}\,, \tag{3.22}$$

so that

$$L^k \varepsilon_K \leqq L\, L^{-\vartheta K}\,, \qquad L^{-\gamma k}(L^k \varepsilon_K)^{-\beta} \leqq L^{-(\gamma(1-\vartheta) - \beta\vartheta)K}\,. \tag{3.23}$$

The right-hand side of (3.21) is then summable in K, where we have used $\exp[-c\,p(\varepsilon)^2] \leqq \varepsilon^\sigma$ for ε small, since $p(\varepsilon)^2$ grows faster than $\ln \varepsilon^{-1}$, and that $|T|$ is fixed. For K sufficiently large, depending on T, g, ϑ and the coupling constants, we have $J \leqq k \leqq K - J$, $K \geqq K_0$, and $\rho \leqq 1/2\, e^{-C|T|} \leqq 1/2\, Z_K(0)$, $1/2\, Z_{K+1}(0)$, so that (3.21) holds for all such K.

Hence $\{Z_K(g) Z_K(0)^{-1}\}$ is a Cauchy sequence and converges to a unique limit as $K \to \infty$, and the limit is at least 1. The first bound in Theorem 2.1(ii), $e^{-C|T|} \leqq Z_K(0) \leqq e^{C|T|}$, is the statement of ultra-violet stability for $K \geqq 2J$, and it holds for the finitely many smaller K once C is enlarged.

For the upper bound we integrate out the scalar field at fixed A. Writing $\phi_1 + i\phi_2 = re^{i\psi}$, we expand the factor of e^{-S^ε} belonging to a bond b in the series

$$\begin{aligned} &\exp[\varepsilon^{d-2} r(b_-) r(b_+) \cos(\psi(b_+) - \psi(b_-) + \varepsilon e A_b)] \\ &\quad = \sum_n I_n(\varepsilon^{d-2} r(b_-) r(b_+))\, e^{in(\psi(b_+) - \psi(b_-) + \varepsilon e A_b)}\,, \end{aligned} \tag{3.24}$$

whose coefficients I_n are non-negative and satisfy $I_{-n} = I_n$. The series converges absolutely, $\sum_n I_n(z) = e^z$, and the radial integrals converge, the quartic term with $\lambda > 0$ dominating the exponent $\varepsilon^{d-2} r(b_-) r(b_+) \leqq \frac{1}{2}\varepsilon^{d-2}(r(b_-)^2 + r(b_+)^2)$. The integral over the angles selects the integer-valued currents J on the bonds with zero divergence, and the integral over the radial variables is positive; so the integral over the scalar field is a sum

$$\sum_J w(J) \cos(\varepsilon e \langle J, A\rangle)\,, \qquad \langle J, A\rangle = \sum_b \varepsilon^d J_b A_b\,, \quad J_b \in \varepsilon^{-d} Z\,, \quad w(J) \geqq 0\,, \tag{3.25}$$

all pairings $\langle \cdot, \cdot \rangle$ here and below being the inner product (2.2). Writing $F(g) = \langle v, A\rangle$, the Gaussian integral over A of each term satisfies

$$\begin{aligned} \int e^{\langle v, A\rangle} \cos(\varepsilon e \langle J, A\rangle)\, d\gamma(A) &= e^{\frac{1}{2}\langle v, C_0 v\rangle - \frac{1}{2}\varepsilon^2 e^2 \langle J, C_0 J\rangle} \cos(\varepsilon e \langle v, C_0 J\rangle) \\ &\leqq e^{\frac{1}{2}\langle v, C_0 v\rangle} \int \cos(\varepsilon e \langle J, A\rangle)\, d\gamma(A)\,, \end{aligned} \tag{3.26}$$

where γ is the Gaussian measure of the vector field and $C_0 = (-\Delta^{\varepsilon, T_\varepsilon} + \mu_0^2)^{-1}$, acting componentwise, is its covariance. Hence

$$Z_K(g) Z_K(0)^{-1} \leqq \exp[1/2 \langle v, C_0 v\rangle]\,. \tag{3.27}$$

Finally

$$\langle v, C_0 v\rangle = \sup_A [2F(g) - \langle A, C_0^{-1} A\rangle] \leqq \sum_p \varepsilon^d g(p)^2 = \|g\|^2\,, \tag{3.28}$$

since on the torus

$$\langle A, C_0^{-1} A\rangle \geqq \sum_p \varepsilon^d F(p)^2 \tag{3.29}$$

the left side being $\sum_p \varepsilon^d F(p)^2 + \sum_x \varepsilon^d (\sum_\mu \partial_\mu^* A_\mu(x))^2 + \mu_0^2 \langle A, A\rangle$ by the lattice identity $d^* d + d d^* = -\Delta$ on the periodic lattice T_ε, which holds on every torus (2.22) since the translations of T_ε are unitary and the difference operators commute, $F(p) = \partial_\mu A_\nu - \partial_\nu A_\mu$ on the plaquette p. This gives the upper bound in (2.24), the right side being bounded uniformly in K for g smooth with compact support.

3.3. The Effective Action. The effective action at the kth step can be written in the following way:

$$\begin{aligned} S^{(k),1}(T^{(k)}, A_k, \phi_k, g, h) = {} & 1/2 \langle A_k, \Delta^{(k)} A_k\rangle - \ln Z_k + 1/2 \langle \phi_k, \Delta^{(k)}(A^{(k)}) \phi_k\rangle - \ln Z_k(A^{(k)}) \\ & + E_0 + P^{(k),1}(T^{(k)}, A_k, \phi_k, g, h)\,, \end{aligned} \tag{3.30}$$

where $A^{(k)}$ and $Z_k(A)$ are the rescaled versions of $A^{(k),\varepsilon}$ and $Z_k^\varepsilon(A)$, and

$$\begin{aligned} E_0 = \ln\{ \int (dA)(d\phi) \exp[& -1/2 \langle A, (-\Delta^{\eta, T_\eta} + \mu_0^2 (L^k \varepsilon)^2) A\rangle \\ & - 1/2 \langle \phi, (-\Delta^{\eta, T_\eta} + m^2 (L^k \varepsilon)^2) \phi\rangle]\}\,. \end{aligned} \tag{3.31}$$

Henceforth we will always denote $\eta = L^{-k}$, $\eta' = L^{-n}\eta = L^{-(k+n)}$ and $|T^{(k)}| = |T| = \prod_{\mu=1}^d 2L_\mu$. The interaction term $P^{(k),1}$ in (3.30) is given by derivatives of a function E_k', given below. The scalar field ϕ is separated into an average and a fluctuation part as follows:

$$\phi = \phi' + a_k G_k(A^{(k)}) Q_k(A^{(k)})^* \phi_k = \phi' + \phi^{(k)}\,. \tag{3.32}$$

Then we define E_k' to be

$$E_k'(e', \lambda', \tau, \theta, A_k, \phi_k, g, h) = -\ln \left\{ \int \prod_{j=0}^{k-1} d\mu_{C^{(j), L^j \eta}}(A_j')\, d\mu_{G_k(A^{(k)})}(\phi')\, e^{V^{(0),\eta}} \right\}, \tag{3.33}$$

where $V^{(0),\eta}$ depends on A', ϕ' and A_k, ϕ_k, as well as on the parameters e', λ', τ, θ. The interaction $P^{(k),1}$ is a sum of derivatives of (3.33) taken with respect to e', λ', τ, θ; it is sufficient to take derivatives up to order $\bar{n} = 13$. These derivatives are represented by connected graphs on T_η; the vertices are those present in $V^{(0),\eta}$, and the propagators are the covariances of A'_j and ϕ' in (3.33). $V^{(0),\eta}$ contains the following vertices (for convenience we write Ω in place of T_η):

$$-\lambda'\lambda(L^k\varepsilon)^{4-d}\sum_{x\in\Omega}\eta^d|\phi'(x)+\phi^{(k)}(x)|^4\,, \tag{3.34}$$

$$-1/2(L^k\varepsilon)^2\sum_{x\in\Omega}\eta^d\,\delta m^2(x;e'e,\lambda'\lambda)\,|\phi'(x)+\phi^{(k)}(x)|^2\,, \tag{3.35}$$

$$\tau(L^k\varepsilon)^{d/2}\sum_{p\in\Omega^{**}}\eta^d g(p)\left(\eta^{-1}\sum_{b\in\partial p}(A'_b+A^{(k)}_b)\right), \tag{3.36}$$

$$\begin{aligned}\theta(L^k\varepsilon)^2\sum_{x\in\Omega}\eta^d h(x)\Big\{\sum_{i=1}^{N}\Big[(\phi'_i(x)+\phi^{(k)}_i(x))^2-(-\Delta^{\eta,\Omega}+m^2(L^k\varepsilon)^2)^{-1}(x,x)\Big]\\ -(L^k\varepsilon)^{d-2}a_\varepsilon\Big\}\end{aligned} \tag{3.37}$$

$$\begin{aligned}-(e(L^k\varepsilon)^{2-d/2})^m\frac{\eta^{m-1}}{m!}\sum_{b\in\Omega^*}\eta^d\big[D^\eta_{A^{(k)}}(\phi'+\phi^{(k)})(b)\\ U(A^{(k)}_b)q^m(\phi'+\phi^{(k)})(b_+)\big]\,(e'A'_b)^m\quad\text{with}\quad 1\leqq m\leqq\bar{n}\,,\end{aligned} \tag{3.38}$$

$$\begin{aligned}(e(L^k\varepsilon)^{2-d/2})^m\frac{\eta^{m-2}}{m!}\sum_{b\in\Omega^*}\eta^d\big[(\phi'+\phi^{(k)})(b_+)q^m(\phi'+\phi^{(k)})(b_+)\big]\,(e'A'_b)^m\\ \text{with } m \text{ even},\quad 2\leqq m\leqq\bar{n}\,.\end{aligned} \tag{3.39}$$

The last term in (3.37) contains no fields and contributes to $P^{(k),1}$ only the constant $\theta\varepsilon^d a_\varepsilon\sum_x h(x)$, with $e'e$ in place of e in a_ε as in (3.35). It is needed when $d = 3$ (Sect. 2.1). The vacuum graphs in which (3.37) is joined by two scalar lines to a one particle irreducible two-point graph of order e^2 built from the vertices (3.38), (3.39) have degree $6 - 2d$ by (3.48), (3.49), which vanishes for $d = 3$.

The logarithm comes from the graph with two vertices (3.38), $m = 1$, whose self-energy, subtracted at zero momentum by (3.51), grows like $e^2|p|$ for $d = 3$, so that its insertion between two scalar lines diverges like $\ln\varepsilon^{-1}$. The graph with the vertex (3.39), $m = 2$, has a local self-energy and is cancelled exactly by the seagull part of the term of order e^2 in (3.35) when the loop carries C_v, and up to the part $F_{k,v}$ of $C_v = G_k + F_{k,v}$ when, as in $P^{(k),1}$, it carries G_k.

The graph with the vertex (3.38), $m = 2$, is bounded uniformly in ε and k. The vertex carries the factor η and a lattice derivative on one scalar leg, and summation by parts along the bonds puts a lattice derivative on the other leg at the cost of a second factor η, up to a local term carrying the difference of the coefficient $G_k(b,b)$ along the bond together with the factor η. When the loop carries C_v this term is absent, $C_v(b,b)$ being constant; the self-energy then vanishes at zero momentum, and the graph contributes nothing to (3.51). When, as in $P^{(k),1}$, the loop carries $G_k = C_v - F_{k,v}$, the difference along the bond is that of $F_{k,v}(b,b)$, which is of order η by Theorem 3.3, the first differences of the kernels of C_v and G_k being integrable over a block for $d = 3$; the local term is then of order η, and the graph is bounded by $C\sum_{x\in T_\varepsilon}\varepsilon^d|h(x)|$.

In the expansion of $\sum_i\langle\phi_i(x)^2\rangle$ with the full propagators C_v, C_s of (3.47) below, four graphs occur at order e^2 besides the insertion of $\delta m^2(x;2,0)$; the graph with a scalar loop at a vertex (3.38) vanishes, since $\operatorname{tr} q = 0$, and three remain. The first contains the vertex (3.39) with $m = 2$ and is cancelled exactly by the corresponding term of (3.35), since that tadpole is local.

The second contains the vertex (3.38) with $m = 2$; it vanishes at zero momentum, contributes nothing to (3.51), and is not cancelled. The third contains two vertices (3.38) with $m = 1$ and is cancelled at zero momentum only, since (3.35) is fixed by the sum (3.51) over y.

The constant a_ε of (2.8) is this term of order e^2 on εZ^d,

$$a_\varepsilon = \Big[\sum_i \langle \phi_i(0)^2 \rangle \Big]_{e^2 \lambda^0} , \tag{3.40}$$

with $\delta m^2(x; 2, 0)$ given by (3.51) with Ω replaced by εZ^d. It does not depend on x, and the sum of the three graphs, of the graph containing (3.35), and of the constant vanishes on εZ^d. On T_ε this sum is a constant bounded by $Ce^{-c\ell} \sum_x \varepsilon^d |h(x)|$ uniformly in ε, ℓ the smallest side of T_ε (Appendix A).

The remaining vertices occur when the renormalization transformation of the scalar field is expanded about $A' = 0$:

$$\phi_k(y) , \tag{3.41}$$

$$-Q_k(A^{(k)})(\phi' + \phi^{(k)})(y) , \tag{3.42}$$

$$\begin{aligned} &- (e(L^k \varepsilon)^{2-d/2})^m (m!)^{-1} \sum_{x \in B^k(y)} \eta^d U(A^{(k)}(\Gamma^{(k)}_{y,x})) \, (e' A'(\Gamma^{(k)}_{y,x}))^m \\ &\cdot q^m(\phi' + \phi^{(k)})(x) \quad \text{with} \quad 1 \leqq m \leqq \bar{n} . \end{aligned} \tag{3.43}$$

These vertices are made by multiplying two of the expressions (3.41)–(3.43), one of which must be (3.43) for some m, at the same $y \in \Omega^{(k)}$. If the expressions are different, the product is multiplied by $-a_k$; if the same, by $-(1/2)a_k$. The resulting vertices are summed over $y \in \Omega^{(k)}$.

Finally, the vacuum energy counterterm $E_1(e'e, \lambda'\lambda)$ is also a part of $V^{(0),\eta}$. We shall see that E_1 is also represented by connected graphs on T_η built from the vertices (3.34), (3.35), and (3.38), (3.39).

Except for the vertex (3.43), the vector fields A'_j enter $V^{(0),\eta}$ in the combination A'. Therefore, in a graph in $P^{(k),1}$, we can sum over the propagators for these fields using (2.18), and get the full propagator G_k for the "field" A'. When vertex (3.43) occurs in a graph, we can use (2.18) and (2.20) to write the propagator as

$$G^\eta_{(j)}(\Gamma^{(j+1)}_{x_{j+1},x}, b) = \sum_{b' \in \Gamma^{(j+1)}_{x_{j+1},x}} \eta \, G^\eta_{(j)}(b', b) . \tag{3.44}$$

When the external field $\phi^{(k)}$ occurs at a vertex v in a graph in $P^{(k),1}$, we can use (3.32) to write it as

$$\phi^{(k)}(v) = \sum_{y \in T^{(k)}} a_k (G_k(A^{(k)}) Q_k(A^{(k)})^*)(v, y) \phi_k(y) . \tag{3.45}$$

If there are s such fields in a graph G, the contribution to $P^{(k),1}$ from G can be written

$$E^{(k)}(G) = \sum_{a_1, \dots, a_s = 1}^{N} \sum_{y_1, \dots, y_s \in T^{(k)}} \phi^{a_1}_k(y_1) \dots \phi^{a_s}_k(y_s) E^{(k)}_{a_1 \dots a_s}(G, A^{(k)}; \{y_i\}) . \tag{3.46}$$

We will omit indices on $E^{(k)}$ from now on. To facilitate our analysis, we will use graphical notation. As usual, we use a wavy line for a vector field propagator, and a straight line for a scalar field propagator. A derivative acting on a propagator is denoted by an arrow at the appropriate vertex. We also introduce the "full" propagators for the fields,

$$C_v = (-\Delta^{\eta,\Omega} + \mu_0^2 (L^k \varepsilon)^2)^{-1} , \qquad C_s = (-\Delta^{\eta,\Omega} + m^2 (L^k \varepsilon)^2)^{-1} , \tag{3.47}$$

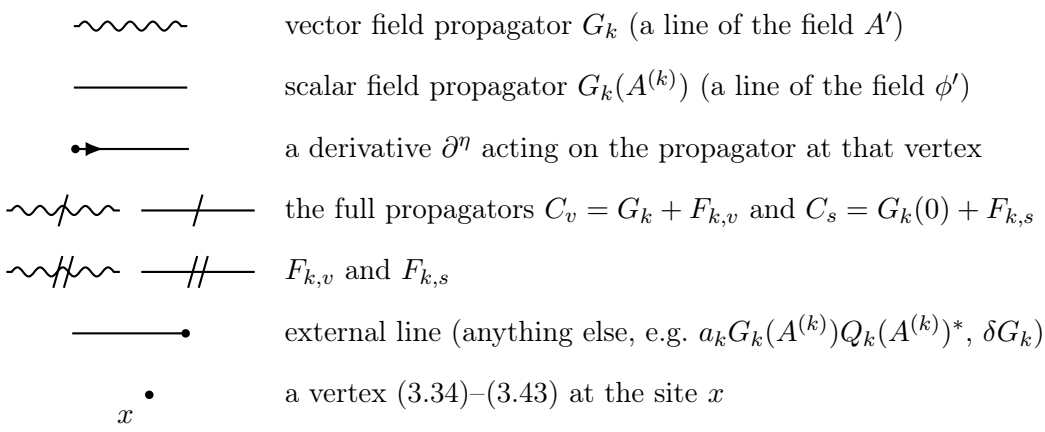


Fig. 3. Graphical notation of Sects. 3.3–3.5

and we define $C_v = G_k + F_{k,v}$ and $C_s = G_k(0) + F_{k,s}$. The propagator C_v (or $F_{k,v}$) will be denoted by a wavy line with one dash (or two dashes) through it, and similarly for C_s and $F_{k,s}$. These conventions are collected in Fig. 3.

In order to isolate divergences in graphs, we need the notion of degree of a graph. A line in a graph is *internal* if it carries one of the propagators G_k, $G_k(A^{(k)})$, C_v, C_s or their derivatives. A line is external if it carries anything else, for example $a_k G_k(A^{(k)}) Q_k(A^{(k)})^*$. An *internal vertex* is the endpoint of at least one internal line. Then the degree of an internal vertex v in a graph G is defined by

$$\begin{aligned} D_G(v) = {} & (1 - d/2) \text{ (number of ends of internal lines at } v \text{ without derivatives)} \\ & + (-d/2) \text{ (number of ends of internal lines at } v \text{ with derivatives)} \\ & + (\text{number of factors } \eta \text{ at } v) \\ & + (\text{value of } m \text{ when } v \text{ contains (3.43)}) \,. \end{aligned} \tag{3.48}$$

There is always one factor η^d at v, coming from the integration over T_η; the vertices (3.34)–(3.43) may produce other factors. The degree of a connected graph G is then

$$D(G) = \sum_{v \in G} D_G(v) - d \,, \tag{3.49}$$

and the degree of an arbitrary graph is the sum of the degrees of its connected components. The divergent graphs are those with non-positive degree.

We write $(-\delta m^2 + \Gamma^\varepsilon)$ for the full one particle irreducible graph with two (undifferentiated) external scalar field lines; here, as for E_1 below, the graphs are built from the vertices (3.34), (3.35), (3.38), (3.39) with $A^{(k)} = 0$, $\phi^{(k)} = 0$ and the propagators of (2.4) on T_ε, which rescaled to T_η are C_v^η, C_s^η; the source vertices (3.36), (3.37) do not occur in them. The divergent graphs in the graphical expansion of $(-\delta m^2 + \Gamma^\varepsilon)$ have order less than or equal to four in the couplings e and λ, and so we define

$$\delta m^2(x) = \sum_{2 \leqq \alpha + 2\beta \leqq 4} e^\alpha \lambda^\beta \, \delta m^2(x; \alpha, \beta) \,. \tag{3.50}$$

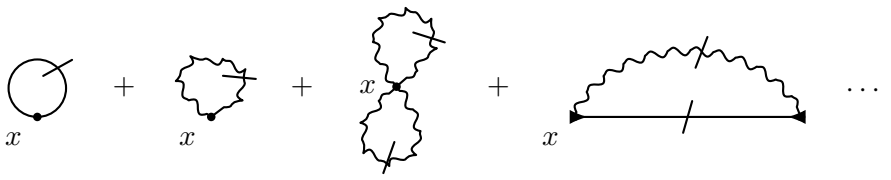
The terms with α odd vanish, since each vertex (3.38), (3.39) carries as many factors e as fields A', the fields A' are contracted in pairs, and (3.51) is solved recursively.

This definition is inserted into Γ^ε, and terms of the same order in e, λ are collected together; this allows us to define $\delta m^2(x; \alpha, \beta)$ by the equation

$$-\delta m^2(x; \alpha, \beta) + \sum_{y \in \Omega} \varepsilon^d \Gamma^\varepsilon(x, y; \alpha, \beta) = 0 \,. \tag{3.51}$$

Here $\Omega = T_\varepsilon$; in Sect. 3.5 the same graphs appear at the scale η, in the rescaled variables.

Equation (3.51) is solved recursively beginning with $\beta = 1$, and it implies a graphical expansion for $\delta m^2(x)$:



(3.52)

The vacuum energy counterterm E_1 is defined by

$$E_1(e', \lambda') = \sum_{1 \leqq \alpha+\beta \leqq \bar{n}} \frac{e'^\alpha \lambda'^\beta}{\alpha!\, \beta!} \frac{\partial^{\alpha+\beta}}{\partial e^\alpha \partial \lambda^\beta} \cdot \left(\ln \int (dA)\, (d\phi)\, \exp[-S^\varepsilon(\Omega, A, \phi) + E_1] \right)_{e=\lambda=0}, \tag{3.53}$$

where $S^\varepsilon(\Omega, A, \phi)$ is given by (2.4). Clearly E_1 is represented by a sum of vacuum energy diagrams on T; the vertices are (3.34), (3.35), (3.38), (3.39) with $A^{(k)} = 0$, $\phi^{(k)} = 0$, and propagators C_v^η, C_s^η.

3.4. Convergence of a Diagram. We now establish Theorem 3.4 for a diagram H with s external scalar field legs, as in (3.46), and t localization centers, which are defined below. Introduce the functions $\{\chi_z\}$, $z \in T^{(k)}$, where $\chi_z(x) = \prod_{\mu=1}^d \chi(x_\mu - z_\mu)$. The positive real function χ is C^∞, supported in $[-2/3, 2/3]$, equal to unity on $[-1/3, 1/3]$, and chosen so that

$$\sum_{z \in T^{(k)}} \chi_z(x) = 1\,, \quad \text{each} \quad x \in T_\eta\,. \tag{3.54}$$

By inserting (3.54) once for each center, at the vertex which carries it, we can rewrite (3.46) as

$$E^{(k)}(H) = \sum_{y_1, \dots, y_s \in T^{(k)}} \sum_{z_1, \dots, z_t \in T^{(k)}} \phi_k(y_1) \dots \phi_k(y_s)\, E^{(k)}(H, A^{(k)}; \{y_i\}, \{z_q\})\,. \tag{3.55}$$

For a graph with vertices (3.35) we first decompose the lines of their counterterm graphs, $C^\eta = \sum_j G^\eta_{(j)}$, as in Sect. 3.5, and apply (3.55) to each term of this decomposition separately. We attach the functions (3.54) to centers rather than to vertices: one center for each field A' at a vertex (3.36), (3.38), (3.39) or (3.43), and one center for each vertex (3.34), (3.37). A vertex (3.35) whose counterterm graph has all its integers $\leqq k-1$ carries the centers of the vertices of that graph, inserted at those vertices, so that the sum over the internal points of the counterterm graph is localised exactly as in the graphs it cancels.

A vertex (3.35) whose counterterm graph carries an integer $\geqq k$ carries none; in that ordering its counterterm graph cancels no subgraph of H, its lines are ordered after the lines with integers $\leqq k-1$, as in (3.96), and its index sums are bounded by (3.98)–(3.100) (Sect. 3.5), the inequalities (3.98) for the counterterm graphs of order four being verified there after the cancellation of their divergent subgraphs at those integers. Every vertex, the internal vertices of the counterterm graphs included, lies within a distance $\frac{2}{3}\sqrt{d}$ of each of its centers.

The number of centers of a graph is the number of its fields A' at the vertices (3.36), (3.38), (3.39), (3.43), plus the number of its vertices (3.34), (3.37), plus, for each vertex (3.35) whose counterterm graph has all integers $\leqq k-1$, the number of centers of that graph. Each field A' at a vertex (3.38), (3.39) or (3.43) carries a factor e and one center, each field $\tilde{A}^{(k)}$ of the expansions below carries a factor e and no center, and each vertex (3.34) carries a factor λ and one center. So the number of centers is $\alpha - E_{\tilde{A}} + \beta$ plus the number of source vertices, α, β

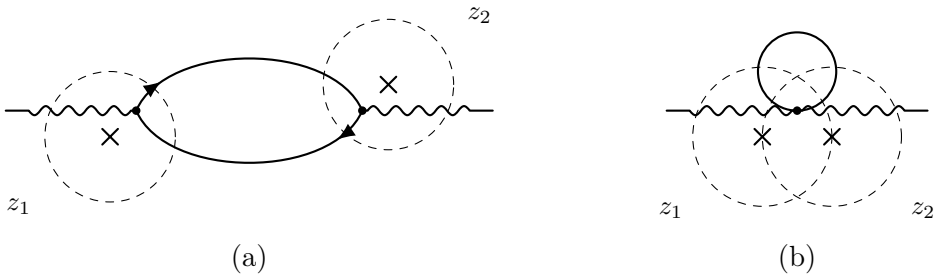


Fig. 4. Two members of the photon two-point class of order e^2, the bubble (a) and the seagull graph (b), with their localization centres z_1, z_2 (crosses) and the supports of χ_{z_1}, χ_{z_2} (dashed); in (b) both centres are carried by the one vertex

being the numbers of factors e and λ of the graph $\tilde{H}$, the factor e of each leg $\tilde{A}^{(k)}$ included, the vertices (3.35) with an integer $\geqq k$ excluded, and $E_{\tilde{A}}$ its number of legs $\tilde{A}^{(k)}$.

A vertex (3.35) with an integer $\geqq k$ contributes to neither count; such a vertex has no partner in the classes of Sect. 3.5, and a term containing it is renormalised at its other divergent subgraphs as before and localised by its remaining centers. The number of centers is therefore the same for every member of a class of Sect. 3.5, the members having the same orders and the same legs, though they may have different numbers of internal vertices, as the photon two-point classes have already at order e^2 (Fig. 4). The vertices of a vacuum graph of E_1 carry no centers; such a graph is not localised by (3.55).

The next theorem expresses the results of this section. We define $\mathrm{dist}(\{v_i\})$ to be the length of the shortest tree graph connecting $\{v_i\}$.

We call a renormalised graph H (Sect. 3.5) *convergent* if the following holds for every graph $\tilde{H}$ obtained from it by the expansions (3.61)–(3.73) below, every ordering $\tilde{l}$ of the internal lines of $\tilde{H}$ and every position of its vertices. Every connected component of every subgraph H_i of (3.84) formed from the lines with integers $j \leqq k-1$ either vanishes or has positive degree after the cancellations of Sect. 3.5, or contains a line whose length is bounded below, or is a tree of two vertices of degree zero.

Such a tree is a scalar line carrying a derivative at each end, between two vertices (3.38) of H or of the counterterm graphs of its vertices (3.35), between such a vertex and a new vertex of (3.71), or between new vertices of (3.71), or a photon line carrying a curl at each end between two vertices (3.36); the summation by parts described after (3.73), or after (3.98) for the lines of H, reduces the tree to degree one, the curl passing in the last case to the source g, whose differences are bounded.

A line whose length is bounded below is bounded by (3.94) below for any N, by (3.81)–(3.83), and the subgraph is treated as one of positive degree. The field-independent vacuum graphs of $P^{(k),1}$ at $g = h = 0$ and the vacuum graphs of E_1 (3.53) are not covered by Theorem 3.5; they are constants, removed in $S_N^{(k),1}$ (Sect. 3.2, Appendix A). The whole graph of order τ^2 is of degree zero and is covered: one curl passes to g by the summation by parts, and its two depths are compared by Proposition 3.6 with $g(x')$ replaced by $g(x)$ as for the external fields there.

For the remaining graphs the lines with integers $j \geqq k$ lie in the counterterm graphs of the vertices (3.35); inside a cluster they are ordered after the lines with integers $j \leqq k-1$, as in (3.96), and their index sums are bounded by (3.98)–(3.100), the powers of $\ln(L^k\varepsilon)^{-1}$ being absorbed in n_1, (3.98) being verified in Sect. 3.5 for the counterterm graphs of order four after the cancellation of their divergent subgraphs at those integers. For the difference (3.57) the legs of such a vertex are expanded by (3.61)–(3.73) like the other lines of H, while the counterterm graph between them is a field-independent constant whose slices are compared by Propositions

3.7 and 3.9 for $k \leqq j \leqq K$ (Sect. 4). A chain of such lines and of the scalar legs joining them whose localised ends lie in different clusters is bounded as a single propagator, (3.95) below.

Theorem 3.5. *Let $d = 3$. With the notation of* (3.55)*, for a convergent renormalised graph H, for $L^k\varepsilon \leqq \eta_2$, where $\eta_2 > 0$ depends on $\bar{n}$ and the coupling constants, for some $n_1, \delta, \gamma, \sigma > 0$ depending only on $\bar{n}$, d, and $\varrho \geqq s/4$, where C depends in addition on the coupling constants and on the sources g, h,*

$$\begin{aligned} &\text{(i)} \qquad |E^{(\theta)}(H, A^{(\theta)}; \{y_i\}, \{z_q\})| \\ &\qquad\qquad \leqq C(L^k\varepsilon)^{\varrho}(p(L^k\varepsilon))^{n_1} \exp[-\delta \operatorname{dist}(\{y_i\}, \{z_q\})]\,, \end{aligned} \tag{3.56}$$

where θ is k or $k+n$;

(ii) *when* $\operatorname{dist}(\{y_i\}, \{z_q\}) \leqq p(L^k\varepsilon)$,

$$\begin{aligned} &|E^{(k+n)}(H, A^{(k+n)}; \{y_i\}, \{z_q\}) - E^{(k)}(H, A^{(k)}; \{y_i\}, \{z_q\})| \\ &\quad \leqq C(L^k\varepsilon)^{\varrho-1/2}(p(L^k\varepsilon))^{n_1}[L^{-\gamma k} + (L^k\varepsilon)^{d+\sigma+1/2}] \\ &\qquad \cdot \exp[-\delta \operatorname{dist}(\{y_i\}, \{z_q\})]\,. \end{aligned} \tag{3.57}$$

Using Theorem 3.5, we can now establish Theorem 3.4 for a convergent renormalised graph H; the remaining graphs are treated in Sect. 3.5. From (3.55) we have

$$\begin{aligned} E^{(k+n)}(H) - E^{(k)}(H) = \sum_{\{y_i\},\{z_q\} \in T^{(k)}} &\phi_k(y_1) \dots \phi_k(y_s)\, \{E^{(k+n)}(H, A^{(k+n)}; \{y_i\}, \{z_q\}) \\ &- E^{(k)}(H, A^{(k)}; \{y_i\}, \{z_q\})\}\,. \end{aligned} \tag{3.58}$$

Define $R = \operatorname{dist}(\{y_i\}, \{z_q\})$; when $R > p(L^k\varepsilon)$, we have

$$\exp[-\delta R] \leqq C(L^k\varepsilon)^{d+\sigma} \exp[-1/2\,\delta R] \tag{3.59}$$

for any σ, since $p > 1$. Therefore using Theorem 3.5 and the bounds (3.2), we have

$$\begin{aligned} &|E^{(k+n)}(H) - E^{(k)}(H)| \\ &\quad \leqq C \sum_{\{y_i\},\{z_q\} \in T^{(k)}} (L^k\varepsilon)^{\varrho-s/4}(p(L^k\varepsilon))^{n_1+s}\{\chi(R > p(L^k\varepsilon))\,(L^k\varepsilon)^{d+\sigma} \exp[-1/2\,\delta R] \\ &\qquad + \chi(R \leqq p(L^k\varepsilon))\,(L^k\varepsilon)^{-1/2}[L^{-\gamma k} + (L^k\varepsilon)^{d+\sigma+1/2}] \exp[-\delta R]\} \\ &\quad \leqq C(L^k\varepsilon)^{-1/2}(p(L^k\varepsilon))^{n_1+s}[L^{-\gamma k} + (L^k\varepsilon)^{d+\sigma+1/2}] \cdot \sum_{\{y_i\},\{z_q\} \in T^{(k)}} \exp[-1/2\,\delta R] \\ &\quad \leqq C(L^k\varepsilon)^{-1/2}(p(L^k\varepsilon))^{n_1+s}[L^{-\gamma k} + (L^k\varepsilon)^{d+\sigma+1/2}]\,(L^k\varepsilon)^{-d}|T| \\ &\quad \leqq C[L^{-\gamma k}(L^k\varepsilon)^{-\beta} + (L^k\varepsilon)^{\sigma'}]\,|T|\,. \end{aligned} \tag{3.60}$$

Here $\beta > d + 1/2$ and $\sigma' < \sigma$.

Notice that the vacuum graphs linear in h of orders e^0, λ and e^2 have degree at most zero and are not covered by Theorem 3.5; they are independent of the fields and are considered in Sect. 3.5 and in Appendix A, where the constant a_ε of (2.8) removes the logarithm of the exchange graph of order e^2. The powers of $\ln(L^k\varepsilon)^{-1}$ which arise from lines of index $j \geqq k$ (Sect. 3.5) are absorbed in the factor $p(L^k\varepsilon)^{n_1}$ by (3.1), the sums over these indices running over at most $K - k + 1 \leqq C(1 + \ln(L^k\varepsilon)^{-1})$ values.

The proof of Theorem 3.5 is similar to the proofs of renormalizability in [Ba 3]. We shall first prove (3.57). There is a smallest cube $\square$ on $T^{(k)}$ which is a union of large blocks and which contains every vertex y_i and z_q. We denote by $\square'$, $\square''$ the smallest cubes on $T^{(k)}$ containing the

large blocks closer than $2p(L^k\varepsilon)$, $p(L^k\varepsilon)$ to $\square$. Let B be the value of A_k at some point x_0 in $\square'$, and define $\tilde{A}^{(k)}_\mu(x) = A^{(k)}_\mu(x) - B_\mu$ for x in $\square'$. Using Eq. (A.5) from Appendix A we have

$$B_\mu = (1 + \mu_0^2 (L^k\varepsilon)^2 a_k^{-1})\, a_k G_k Q_k^* B_\mu \,. \tag{3.61}$$

Therefore writing $\tilde{A}_{k,\mu}(w) = A_{k,\mu}(w) - (1 + \mu_0^2(L^k\varepsilon)^2 a_k^{-1}) B_\mu$, we have

$$\tilde{A}^{(k)}_\mu(x) = \sum_{w \in T^{(k)}} (a_k G_k Q_k^*)(x,w)\, \tilde{A}_{k,\mu}(w) \,. \tag{3.62}$$

From the bounds (3.2) and using Theorem 3.3, we deduce that

$$\begin{aligned} |\tilde{A}_{k,\mu}(w)| &\leqq C p(L^k\varepsilon)\, [|w - x_0| + L^k\varepsilon] \,, \\ |\tilde{A}^{(k)}_\mu(x)| \leqq C p(L^k\varepsilon)\, [|x - x_0| + 1] \,, &\qquad |\partial^\eta_\nu \tilde{A}^{(k)}_\mu(x)| \leqq C p(L^k\varepsilon) \,. \end{aligned} \tag{3.63}$$

For $L^k\varepsilon$ sufficiently small the field $(1-\theta)\tilde{A}^{(k)}$ is regular in the sense of Definition 3.2, by (3.63), $|\partial^\eta \theta| \leqq C[p(L^k\varepsilon)]^{-1}$ and (3.1), so that Theorem 3.3 applies to $G_k(\square', (1-\theta)\tilde{A}^{(k)})$. The second line follows from (3.62) by writing $\tilde{A}_k(w) = \tilde{A}_k(x_1) + (\tilde{A}_k(w) - \tilde{A}_k(x_1))$, x_1 a site of $T^{(k)}$ nearest to x, since $a_k G_k Q_k^* 1$ is constant by (A.5).

In the expression $E^{(k)}(H, A^{(k)}; \{y_i\}, \{z_q\})$, we replace each propagator $G_k(A^{(k)})$ by the sum $G_k(\square', A^{(k)}) - \delta G_k(\square', A^{(k)})$. Multiplying out the result gives a sum of graphs of the same form as H, but with appropriately changed propagators. A line carrying $\delta G_k(\square', A^{(k)})$ (or its derivative) is treated as external. We now make a gauge transformation of the external field $A^{(k)}$ in each propagator and vertex function in the graph, except in the propagators $\delta G_k(\square', A^{(k)})$. $A^{(k)}_\mu(x)$ is replaced by $\tilde{A}^{(k)}_\mu(x)$ for each $x \in \square'$; by gauge covariance this is equivalent to a unitary transformation at each vertex connected to an external field or an external line. Specifically, for each graph produced by the expansion, there is a subset of vertices $\{v_i\}$ such that the only dependence of the graph on B is the following factor at each vertex v_j:

$$U(B(\Gamma^{(k)}_{x_0,v_j})) = \exp[e(L^k\varepsilon)^{2-d/2} q B(\Gamma^{(k)}_{x_0,v_j})] \,. \tag{3.64}$$

We next want to write each graph as a (non-local) polynomial in the field $\tilde{A}^{(k)}$. We do this by expanding around $\theta\tilde{A}^{(k)} = 0$; θ is a C^∞ function with $\theta(x) = 1$ for $\mathrm{dist}(\square'', x) \leqq 1/3\, p(L^k\varepsilon)$, and $\theta(x) = 0$ for $\mathrm{dist}(\square'', x) > 2/3\, p(L^k\varepsilon)$ (Fig. 5). For future reference, we give the expansion for a general fluctuation field A' about a background field B. The covariant derivative becomes

$$D^\eta_{A'+B}\phi(b) = D^\eta_B \phi(b) + F_{1,k}(A')\, U(B)\, \phi(b_+) \,, \tag{3.65}$$

where

$$F_{1,k}(A') = \sum_{j=1}^{\bar{n}} \frac{\eta^{j-1}}{j!}\, (e(L^k\varepsilon)^{2-d/2} q A')^j + \eta^{-1} R_{\bar{n}+1}(\eta\, e(L^k\varepsilon)^{2-d/2} q A') \,. \tag{3.66}$$

The remainder term can be written as $R_n(x) = (x^n/n!)\, V_n(x)$, and $V_n(z)$ is an analytic function of z given by $V_n(z) = n \int_0^1 (1-t)^{n-1} e^{tz}\, dt$. Therefore taking $A' = \theta\tilde{A}^{(k)}$ and using (3.63),

$$|\eta^{-1} R_{\bar{n}+1}(\eta\, e(L^k\varepsilon)^{2-d/2} q A')| \leqq \frac{C}{(\bar{n}+1)!}\, (e(L^k\varepsilon)^{2-d/2} p(L^k\varepsilon)^2)^{\bar{n}+1} \,. \tag{3.67}$$

Similarly the averaging function becomes

$$(Q_k(A' + B)\phi)(y) = (Q_k(B)\phi)(y) + (F_{2,k}(A', B)\phi)(y) \,, \tag{3.68}$$

where

$$(F_{2,k}(A', B)\phi)(y) = \sum_{x \in B^k(y)} \eta^{d+1} F_{1,k}(\eta^{-1} A'(\Gamma^{(k)}_{y,x}))\, U(B(\Gamma^{(k)}_{y,x}))\, \phi(x) \,. \tag{3.69}$$

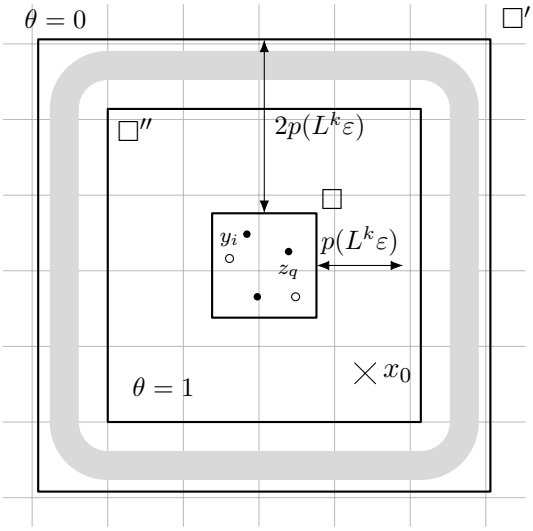


Fig. 5. The cubes $\square \subset \square'' \subset \square'$ on $T^{(k)}$ in the proof of Theorem 3.5, the point x_0 at which $B = A_k(x_0)$ is taken, and the cut-off θ, which interpolates between 1 and 0 across the shaded band

Together these imply an expansion for $G_k(\square', A' + B)$:

$$G_k(\square', A' + B) = \sum_{m=0}^{\bar{n}} G_k(\square', B)\,[V_k(A', B)G_k(\square', B)]^m + G_k(\square', B) \\ \cdot\,[V_k(A', B)G_k(\square', B)]^{\bar{n}}\, V_k(A', B)\, G_k(\square', A' + B)\,, \tag{3.70}$$

where the vertex function is

$$-V_k(A', B) = D_B^{\eta *} F_{1,k}(A')\, U(B) + U^*(B) F_{1,k}(A')^* D_B^{\eta} + |F_{1,k}(A')|^2 \\ + a_k F_{2,k}(A', B)^* Q_k(B) + a_k Q_k(B)^* F_{2,k}(A', B) + a_k F_{2,k}(A', B)^* F_{2,k}(A', B)\,. \tag{3.71}$$

These equations are used to expand the propagators, vertex functions and covariant derivatives, taking $A' = \theta\tilde{A}^{(k)}$ and $B = (1-\theta)\tilde{A}^{(k)}$; at the vertices of H, which lie where $\theta = 1$, this is $B = 0$. The result is a sum of new graphs, all with vertices inside $\square'$ apart from the vertices (3.35) without centers, which we treat first, with vector fields $\tilde{A}^{(k)}$ at the new vertices, and with scalar propagators $G_k(\square', (1-\theta)\tilde{A}^{(k)})$; the factors $U((1-\theta)\tilde{A}^{(k)})$ at the new vertices are unitary and are equal to one within $p(L^k\varepsilon)/3$ of $\square''$.

The scalar legs of the vertices (3.35) without centers form chains of at most $\bar{n}+1$ legs through such vertices, whose ends are localised vertices of H or points y_i, all within $\square$; for a maximal chain with a vertex x outside $\square'$ one of its legs is longer than a fixed multiple of $p(L^k\varepsilon)$, so that, keeping a part $\delta_0/2$ of each propagator for the decay in the points y_i, z_q, Theorem 3.3 bounds the part of the sum over x outside $\square'$ by $C\exp[-\frac{1}{2}\delta_0 c p(L^k\varepsilon)] \leqq C(L^k\varepsilon)^N$ for any N, by (3.1), which is absorbed in the term $(L^k\varepsilon)^{d+\sigma+1/2}$ of (3.57). For $x \in \square'$ the vertex (3.35) is gauge invariant and its legs are expanded like the other lines of H, its counterterm graph being independent of the fields. Note that these graphs have the same structure as those already present in $P^{(k),1}$. Finally, we write every scalar field propagator as a sum

$$G_k(\square', (1-\theta)\tilde{A}^{(k)}) = G_k(0) + \delta G_k(\square'', 0) + \delta G_k(\square', \square'', (1-\theta)\tilde{A}^{(k)})\,, \tag{3.72}$$

where $\delta G_k(\square', \square'', A) = G_k(\square', A) - G_k(\square'', A)$. Using (3.62) we can hold the fields $\tilde{A}_k$ fixed at some points $\{w_l\}$ in $T^{(k)}$. So if we denote by $\{\tilde{H}\}$ the graphs obtained from H by all the

previous expansions, we can write

$$E^{(k)}(H, A^{(k)}; \{y_i\}, \{z_q\}) = \sum_{\{\tilde{H}\}} \Big\{ \sum_{w_1,\dots,w_r \in T^{(k)}} \tilde{A}_k(w_1) \dots \tilde{A}_k(w_r) \cdot E^{(k)}(\tilde{H}; \{y_i\}, \{z_q\}, \{w_l\}) \Big\} . \tag{3.73}$$

The subgraphs of $\tilde{H}$ whose vertices are new are trees of degree at least zero, and the degree is zero only for a tree of two vertices whose line carries a derivative at each end. A subgraph containing a new vertex v and both lines at v has degree larger by at least one than the subgraph of H from which it arises, since $D_G(v) \geqq 1$ when both ends at v are internal; a subgraph containing one line l at v only has degree at least that of the subgraph obtained by removing l and v, since the end at v contributes at least $\frac{d}{2}$ and the new end at the other vertex u of l at least $-\frac{d}{2}$; when l alone remains this is $\frac{d}{2} + D_G(u) - d \geqq 0$, with equality exactly when l carries a derivative at v and at u, whether u is a vertex of H or a new vertex. Both cases of degree zero are treated by the summation by parts below, performed at the new end.

Such a tree is a connected component of a subgraph H_i only in the orderings in which its line precedes every line adjacent to it at either end, which then carry integers at least as large. In these orderings we sum by parts at one end of the line, covariantly with respect to the field $\tilde{A}^{(k)}$ at that vertex. The factor (3.64) is a pure gauge for the constant field B of (3.61), and the covariant difference applied to it times a leg is the factor times the covariant difference of the leg, which Theorem 3.3 bounds; so it is not differentiated.

The difference then acts on the fields $\theta\tilde{A}^{(k)}$, $(1-\theta)\tilde{A}^{(k)}$, on the localisation function χ_z and on the external fields at that vertex, the source g at a vertex (3.36) included, which are bounded together with their first differences by (3.2), (3.63) and $|\partial^\eta \theta| \leqq C[p(L^k\varepsilon)]^{-1}$, by a power of $p(L^k\varepsilon)$, or on the photon line or the other scalar line at that vertex, whose integers are not smaller. The tree then has degree one, and every subgraph H_i containing the adjacent line contains the tree and keeps its degree, so that no subgraph of degree at most zero is created.

The divergent subgraphs of H therefore occur in $\tilde{H}$ unchanged in the terms of order zero in $\theta\tilde{A}^{(k)}$, and in the terms with fields $\tilde{A}^{(k)}$, from (3.65), (3.68) at the vertices of H and from (3.70) with $m \geqq 1$, as subgraphs of degree -2 and -1 with one or two legs $\tilde{A}^{(k)}$, the images of the one-vertex loops of H of degree -2 and -1 and of its two-point subgraphs of degree -1.

These are the classes of Sect. 3.5 with legs $\tilde{A}^{(k)}$ in place of legs A'; their degrees are those of (3.48)–(3.49), the factor η^{j-1} of a new vertex with j fields being counted and the legs not. The loops with one or three legs vanish on the lines $G_k(0)$ and $\delta G_k(\Box'', 0)$, since $\operatorname{tr} q = \operatorname{tr} q^3 = 0$, and the lines $\delta G_k(\Box', \Box'', (1-\theta)\tilde{A}^{(k)})$ carry the factor $\exp[-\delta_0 p(L^k\varepsilon)]$. Every vertex of such a subgraph lies where $\theta = 1$, a new vertex outside that region being joined to H by a line bounded by (3.94) below; so $B = 0$ at its vertices, and when its lines carry $G_k(0)$ its divergences cancel exactly as in Sect. 3.5, the terms of (3.72) with a line δG_k being treated there as remainders.

A new vertex at a distance larger than $p(L^k\varepsilon)/3$ from $\Box''$ is joined to a vertex of H by at most $\bar{n}+1$ lines of its chain, one of which has length at least $(p(L^k\varepsilon)/3 - \frac{2}{3}\sqrt{d})/(\bar{n}+1)$; this line, of integer j', is bounded by (3.94) for any N by (3.81), and the other subgraphs of new vertices are treated as above.

There is a similar expansion of $E^{(k+n)}(H, A^{(k+n)}; \{y_i\}, \{z_q\})$ with the same graphs $\tilde{H}$, so the proof reduces to the following proposition.

Proposition 3.6. *With the notation of* (3.73)*, for a graph $\tilde{H}$ obtained by* (3.61)–(3.73) *from a convergent renormalised graph H,*

$$|E^{(k+n)}(\tilde{H}; \{y_i\}, \{z_q\}, \{w_l\}) - E^{(k)}(\tilde{H}; \{y_i\}, \{z_q\}, \{w_l\})| \leqq C(L^k\varepsilon)^{\varrho - 1/2} \cdot (p(L^k\varepsilon))^{n_2} [L^{-\gamma k} + (L^k\varepsilon)^{d+\sigma+1/2}] \exp[-\delta \operatorname{dist}(\{y_i\}, \{z_q\}, \{w_l\})] . \tag{3.74}$$

To see that (3.57) follows from (3.74) consider

$$\begin{aligned}&|E^{(k+n)}(H, A^{(k+n)}; \{y_i\}, \{z_q\}) - E^{(k)}(H, A^{(k)}; \{y_i\}, \{z_q\})| \\ &\quad \leqq \sum_{\{\tilde{H}\}} \Big\{ \sum_{\{w_l\} \in T^{(k)}} |\tilde{A}_k(w_1)| \dots |\tilde{A}_k(w_r)| \cdot |E^{(k+n)}(\tilde{H}; \{y_i\}, \{z_q\}, \{w_l\}) \\ &\qquad - E^{(k)}(\tilde{H}; \{y_i\}, \{z_q\}, \{w_l\})| \Big\} .\end{aligned} \tag{3.75}$$

Using the bounds (3.63) and recalling that $x_0 \in \square'$, so $|x_0 - y_i|$, $|x_0 - z_q| \leqq Cp(L^k \varepsilon)$ each i, q, we can use the exponential tree decay in (3.74) to sum over $\{w_l\}$ and get (3.57) with $n_1 = n_2 + 2r$ (the number of graphs $\tilde{H}$ depends only on $\bar{n}$ and d).

We now prove Proposition 3.6. Every internal line in $\tilde{H}$ carries a propagator G_k, $G_k(0)$ or one of their derivatives, which may be decomposed by writing $G_k^\eta = \sum_{j=0}^{k-1} G_{(j)}^\eta$, see (2.18). The resulting product of sums is multiplied out and rewritten as a sum over orderings $\tilde{l} = \{l(1), \dots, l(m)\}$ of the m internal lines. This is followed by a sum over integers $\{j_{l(p)}\}$, $p = 1, \dots, m$, compatible with this ordering, meaning

$$j_{l(1)} \leqq j_{l(2)} \leqq \dots \leqq j_{l(m)} . \tag{3.76}$$

A term with two or more integers equal is arbitrarily assigned to one of the orderings with which it is compatible. Denoting by $\tilde{H}(j)$ the graph with specified integers $j = \{j_{l(p)}\}$ on each internal line, and omitting the external vertices, we have

$$E^{(k)}(\tilde{H}) = \sum_{\tilde{l}} \sum_{\substack{j \text{ compatible} \\ \text{with } \tilde{l}}} E^{(k)}(\tilde{H}(j)) . \tag{3.77}$$

There is a similar representation for $E^{(k+n)}(\tilde{H})$; for convenience we write the decomposition of $G_{k+n}^{\eta'}$ in the form $G_{k+n}^{\eta'} = \sum_{j=-n}^{k-1} G_{(j)}^{\eta'}$, where

$$G_{(-n)}^{\eta'} = C^{(0),\eta'} , \qquad G_{(j)}^{\eta'} = a_{j+n}^2 (L^j \eta)^{-4} G_{j+n}^{\eta'} Q_{j+n}^* C^{(j+n), L^j \eta} Q_{j+n} G_{j+n}^{\eta'} . \tag{3.78}$$

We divide the internal lines into two sets S and S^c; in S all integers are zero or positive, while in S^c all integers are negative. Hence we can write

$$E^{(k+n)}(\tilde{H}) = \sum_{S^c} E_{S^c}^{(k+n)}(\tilde{H}) , \tag{3.79}$$

where $E_{S^c}^{(k+n)}(\tilde{H})$ contains a sum over orderings and integers compatible with S. Before proceeding, we state the required bounds on propagators. For a propagator $G(x, y)$, we define a "Hölder derivative" by

$$(\partial_\alpha(x, y) G)(z) = |x - y|^{-\alpha} \{G(x, z) - G(y, z)\} . \tag{3.80}$$

Proposition 3.7. *For $x, y, z \in T_\eta$, $0 < \alpha < 1$, and all $0 \leqq j \leqq k - 1$,*

$$\begin{aligned}&|G_{(j)}^\eta(x, y)| , \qquad |\partial_\mu^\eta G_{(j)}^\eta(x, y)| \leqq C\{(L^j \eta)^{2-d}, (L^j \eta)^{1-d}\} \\ &\qquad \cdot \exp[-\delta_0 (L^j \eta)^{-1} |x - y|] ,\end{aligned} \tag{3.81}$$

$$|G_{(j)}^\eta(\Gamma_{x_{j+1}, x}^{(j+1)}, b)| \leqq C (L^j \eta)^{3-d} \exp[-\delta_0 (L^j \eta)^{-1} \mathrm{dist}(B^j(x), b)] , \tag{3.82}$$

$$\begin{aligned}&|\partial_\alpha(x, y) G_{(j)}^\eta(z)| , \qquad |\partial_\alpha(x, y) \partial_\mu^\eta G_{(j)}^\eta(z)| \\ &\quad \leqq C\{(L^j \eta)^{2-d-\alpha}, (L^j \eta)^{1-d-\alpha}\} \exp[-\delta_0 (L^j \eta)^{-1} \mathrm{dist}(\{x, y\}, z)] .\end{aligned} \tag{3.83}$$

Furthermore, if we replace $G^{\eta}_{(j)}$ by $G^{\eta'}_{(j)}$ throughout, and replace $\Gamma^{(j+1)}_{x_{j+1},x}$ by $\Gamma^{(j+n+1)}_{x_{j+n+1},x}$, then the bounds are valid for $-n \leqq j \leqq k-1$.

Proposition 3.7 follows immediately from Theorem 3.3 and the scaling properties of the operators. We will first establish (3.74) for a graph with no remainder terms from the expansion (3.66), no final terms from (3.70) and no propagators δG_k. These restrictions will be dropped later. For a fixed ordering $\tilde{l}$, we define a sequence of subgraphs $H_1, \ldots, H_m$ as follows:

$$\begin{aligned} H_1 &= \{\text{the line } l(1) \text{ and its two vertices}\}\,, \\ H_{i+1} &= H_i \cup \{\text{the line } l(i+1) \text{ and its two vertices}\}\,. \end{aligned} \tag{3.84}$$

Some of the subgraphs H_i may be divergent. In Sect. 3.5, we will show how graphs may be added together to form renormalised graphs. We will assume below that this has been done already, and that we are analysing a convergent renormalised graph, so that every connected component of every H_i which does not vanish after the cancellations of Sect. 3.5 has positive degree, or is reduced to positive degree by the summation by parts described after (3.73), or contains a line bounded by (3.94).

Consider first a term on the right-hand side of (3.79) with S^c non-empty. We get an upper bound for this expression by replacing every propagator by the bounds given in Proposition 3.7 and Theorem 3.3, and bounding vertex functions appropriately (the sources, g, h are bounded by C). By extracting a small part of each propagator, we get the exponential decay on the right-hand side of (3.74). Let x, y be the endpoints of the graph $H_1 = l(1)$; there is a factor $\exp[-\delta_0 L^{k-j_{l(1)}}|x-y|]$ present. All the other propagators attached to y carry similar exponential factors, but with $j_{l(1)}$ replaced by some $j_{l(p)} \geqq j_{l(1)}$. These propagators are "transferred" to x by using the inequality

$$\exp[-\delta_0 L^{k-j_{l(p)}}|y-z| - \delta_1 L^{k-j_{l(1)}}|x-y|] \leqq \exp[-\delta_1 L^{k-j_{l(p)}}|x-z|]\,, \tag{3.85}$$

where δ_1 is a fraction of δ_0 chosen so that after all these transfers there is some decay left on $l(1)$. We then sum over y, giving

$$\sum_y \eta^d \chi(y) \exp[-\delta_1 L^{k-j_{l(1)}}|x-y|] \leqq C[L^{k-j_{l(1)}}]^{-d}\,. \tag{3.86}$$

Combining this factor with the power of $L^{j_{l(1)}-k}$ already present from the propagator on $l(1)$ (and any extra factors η at x and y), we get altogether the exponent $D(H_1)$. If $l(2) \in S^c$, we can sum over $j_{l(1)} \leqq j_{l(2)}$, since $D(H_1) > 0$:

$$\sum_{j=-n}^{j_{l(2)}} (L^{j-k})^{D(H_1)} \leqq C(L^{j_{l(2)}-k})^{D(H_1)}\,. \tag{3.87}$$

Graphically, we have shrunk $l(1)$ to a point in $\tilde{H}$; the remaining vertices are summed with the constraint $-n \leqq j_{l(2)} \leqq j_{l(3)} \leqq \cdots \leqq j_{l(m)} \leqq k-1$. We continue doing this until S^c is exhausted, so that $l(i+1) \in S$. This gives

$$\sum_{j=-n}^{0} (L^{j-k})^{D(H_i)} \leqq C(L^{-k})^{D(H_i)} \leqq CL^{-\gamma k}(L^{j_{l(i+1)}-k})^{D(H_i)-\gamma}\,. \tag{3.88}$$

For γ small enough, $D(H_i) - \gamma > 0$ and we continue the process, eventually shrinking $\tilde{H}$ to one point x. The final sum over $j_{l(m)}$ is then bounded by C, and the sum over x by $|\Box'| \leqq C(p(L^k\varepsilon))^d$. The vertices in $\tilde{H}$ give the factor $(L^k\varepsilon)^{\varrho}$, and (3.88) gives $L^{-\gamma k}$. Finally, there is a sum over orderings of the lines, again depending only on $\bar{n}$.

So we reduce to the case with S^c empty, i.e. we must bound $|E^{(k)}(\tilde{H}) - E_\phi^{(k+n)}(\tilde{H})|$. This is done by replacing one by one every factor in $E_\phi^{(k+n)}(\tilde{H})$ by the corresponding factor in $E^{(k)}(\tilde{H})$, and bounding the error at each step. First, we replace the propagators on the external lines, using the following proposition proved in Sect. 4. When $x' \in T_{\eta'}$, we denote by x that point in T_η for which $x' \in B^n(x)$.

Proposition 3.8. *For $x', y' \in T_{\eta'}$, $0 < \alpha < 1$, and γ sufficiently small,*

$$\begin{gathered}
|a_{k+n} G_{k+n}^{\eta'} Q_{k+n}^*(x', z) - a_k G_k^\eta Q_k^*(x, z)|\,, \\
|a_{k+n} \partial_\mu^{\eta'} G_{k+n}^{\eta'} Q_{k+n}^*(x', z) - a_k \partial_\mu^\eta G_k^\eta Q_k^*(x, z)|\,, \\
|(\partial_\alpha(x', y') a_{k+n} G_{k+n}^{\eta'} Q_{k+n}^*)(z) - (\partial_\alpha(x, y) a_k G_k^\eta Q_k^*)(z)|\,, \\
|(\partial_\alpha(x', y') a_{k+n} \partial_\mu^{\eta'} G_{k+n}^{\eta'} Q_{k+n}^*)(z) - (\partial_\alpha(x, y) a_k \partial_\mu^\eta G_k^\eta Q_k^*)(z)| \\
\leqq C L^{-\gamma k} \exp[-\delta_0 \{|x - z|, \mathrm{dist}(\{x, y\}, z)\}]\,.
\end{gathered} \tag{3.89}$$

If we replace such a propagator in $E_\phi^{(k+n)}(\tilde{H})$, the error is the same graph with a difference of propagators on one line. Using the method presented, this error is bounded, and Proposition 3.8 gives the desired factor $L^{-\gamma k}$. All such propagators are replaced in this way. The sources $g_{\mu\nu}(x')$, $h(x')$ and the partition of unity function $\chi(x')$ can be replaced by $g_{\mu\nu}(x)$, $h(x)$ and $\chi(x)$ respectively, since their derivatives are uniformly bounded. The operator (3.64) can be replaced using the following bound:

$$|U(B(\Gamma_{x_0,x'}^{(k+n)})) - U(B(\Gamma_{x_0,x}^{(k)}))| \leqq C e (L^k \varepsilon)^{2-d/2} L^{-k} \frac{p(L^k \varepsilon)}{\mu_0 L^k \varepsilon}\,. \tag{3.90}$$

Now we replace propagators on the internal lines, using the following proposition (also proved in Sect. 4):

Proposition 3.9. *For $0 \leqq j \leqq k-1$, $0 < \alpha < 1$, and γ sufficiently small,*

$$\begin{aligned}
&|G_{(j)}^{\eta'}(x', y') - G_{(j)}^\eta(x, y)|\,, \qquad |\partial_\mu^{\eta'} G_{(j)}^{\eta'}(x', y') - \partial_\mu^\eta G_{(j)}^\eta(x, y)| \\
&\quad \leqq C L^{-\gamma k} \{(L^j \eta)^{2-d-\gamma}, (L^j \eta)^{1-d-\gamma}\} \exp[-\delta_0 (L^j \eta)^{-1} |x - y|]\,,
\end{aligned} \tag{3.91}$$

$$\begin{aligned}
&|G_{(j)}^{\eta'}(\Gamma_{x_{j+n+1},x}^{(j+n+1)}, b') - G_{(j)}^\eta(\Gamma_{x_{j+1},x}^{(j+1)}, b)| \\
&\quad \leqq C L^{-\gamma k} (L^j \eta)^{3-d-\gamma} \exp[-\delta_0 (L^j \eta)^{-1} \mathrm{dist}(B^j(x), b)]\,,
\end{aligned} \tag{3.92}$$

$$\begin{aligned}
&|(\partial_\alpha(x', y') G_{(j)}^{\eta'})(z') - (\partial_\alpha(x, y) G_{(j)}^\eta)(z)|\,, \\
&|(\partial_\alpha(x', y') \partial_\mu^{\eta'} G_{(j)}^{\eta'})(z') - (\partial_\alpha(x, y) \partial_\mu^\eta G_{(j)}^\eta)(z)| \\
&\quad \leqq C L^{-\gamma k} \{(L^j \eta)^{2-d-\alpha-\gamma}, (L^j \eta)^{1-d-\alpha-\gamma}\} \exp[-\delta_0 (L^j \eta)^{-1} \mathrm{dist}(\{x, y\}, z)]\,.
\end{aligned} \tag{3.93}$$

The cancellations of Sect. 3.5 are performed on each lattice separately, so that the graphs to be compared are the renormalised graphs of the two models. Each line of a member, the lines inside the counterterm graphs of its vertices (3.35) included, corresponds by (3.51) to a line with the same endpoints in every member containing the subgraph it belongs to; each class being written on each lattice as the sum of terms of positive degree, (3.104)–(3.106), multilinear in the kernels of its lines, we replace the kernel of one line by the difference of the two kernels in every term at once; the terms remain of positive degree, and the error is the class with one difference of propagators.

The zero-momentum parts of the classes which vanish by gauge covariance vanish on each lattice, and the classes are otherwise written as sums of terms of positive degree, (3.104)–(3.106), whose convergence factors are replaced last by (3.109); the lines correspond by (3.51)–(3.52).

The differences (3.91)–(3.93) are then introduced line by line, and each lowers the degree of every subgraph H_i containing the line by γ, which is smaller than the degrees after the cancellations for γ small.

First we fix the ordering $\tilde{l}$ of the internal lines. If we replace a propagator $G^{\eta'}_{(j)}(x',y')$ by $G^{\eta}_{(j)}(x,y)$, the error is the same graph with a difference of propagators on one line. Redoing the analysis, we see that the degrees of some subgraphs have been reduced by γ; for γ small enough, the exponents $D(H_i)-\gamma$ are still positive, and the bound proceeds as before. This replacement is made for every internal line, and every ordering $\tilde{l}$. Having done this, we can replace the sums of internal vertices x' over $T_{\eta'}$ by sums over $x \in T_\eta$, and this gives exactly $E^{(k)}(\tilde{H})$, proving Proposition 3.6.

The remainder terms we neglected give large positive powers of $L^k\varepsilon$, so it is sufficient to bound $E^{(k)}(\tilde{H})$ and $E^{(k+n)}(\tilde{H})$ separately. We note that Proposition 3.7 also holds for $G^{\eta}_{(j)}(\square')$ and $G^{\eta}_{(j)}(\square', \tilde{A}^{(k)})$, since Theorem 3.3 gives bounds on these operators also. The remainders from the expansions (3.66) and (3.70) give factors $((L^k\varepsilon)^{2-\alpha-d/2})^{\bar{n}+1}$, for any $\alpha > 0$. When a graph contains a vertex with extra powers of η or η', we can extract $\eta^\gamma = L^{-\gamma k}$. The propagator $\delta G_k(\square', A^{(k)})(x,y)$ is bounded by $C\exp[-\delta_0|x-y| - \delta_0 p(L^k\varepsilon)]$, since $x, y \in \square$. The propagators $\delta G_k(\square', \square'', (1-\theta)\tilde{A}^{(k)})$ and $\delta G_k(\square'', 0)$ give a factor $\exp[-\delta_0 \operatorname{dist}(\{x,y\}, \partial\square'')]$. If $x, y \in \square$, this gives $\exp[-\delta_0 p(L^k\varepsilon)]$. If $x, y \notin \square$, then δG_k must come from the expansion (3.70), so there is a string of propagators, one of which is δG_k, which begins and ends in $\square$. By extracting a small exponential factor from each of these propagators, we get $\exp[-\delta \operatorname{dist}(\square, \partial\square')]$.

So we have established Proposition 3.6. The factor $(L^k\varepsilon)^\varrho$ comes from the vertices (3.34)–(3.43). Examining these vertices, we see that in each of them we may give a factor $(L^k\varepsilon)^{1-d/4}$ to every field $\phi^{(k)}(x)$. Therefore $\varrho \geqq s/4$. The bound (3.90) gives $(L^k\varepsilon)^{-1/2}$.

In order to derive (3.56) we group the points $\{y_i\}$, $\{z_q\}$ into clusters, two points lying in the same cluster when they are joined by a chain of points with consecutive distances at most $(13/3)\sqrt{d}$. Since the clusters are determined by these points alone, they are the same for every member of the renormalised class and for every position of its vertices.

A line of index $j \leqq k-1$ joining vertices which carry centers in different clusters, or an external line $a_k G_k Q_k^*$ joining a vertex to a point y_i of another cluster, is longer than $3\sqrt{d}$ when both its ends are vertices carrying centers at them, and longer than a fixed positive constant in every case. The former we bound by

$$C_N (L^{j-k})^N \exp[-\tfrac{1}{2}\delta_0|x-y|]\,, \tag{3.94}$$

for any N, and sum its index independently of the ordering; the latter by $C\exp[-\frac{1}{2}\delta_0|x-y_i|]$, and a line $G_k(A^{(k)})$ by $C\exp[-\delta_0|x-y|]$ from Theorem 3.3. We treat the cut line as an external line of both subgraphs, with the factors (3.64) of each at its ends.

The lines of index $j \geqq k$ have at least one end at a vertex without centers (Sect. 3.5) and are therefore never cut. A vertex (3.35) without centers has exactly two scalar legs, and each maximal chain of such legs through such vertices is regarded as one line between its localised ends, which is cut when these lie in different clusters; its length then exceeds a fixed positive constant c_0, and we bound the cut chain, counterterm graphs included, as a single propagator at unit scale by

$$C(1+K-k)^{n_3} \exp[-\tfrac{1}{2}\delta_0|x-y|]\,, \qquad |x-y| \geqq c_0\,, \tag{3.95}$$

n_3 depending on $\bar{n}$ only and the power of $1+K-k$ being absorbed in n_1 by (3.1); such a cut chain is treated as an external line of both pieces, while a chain whose ends lie in one cluster is ordered with the lines of its piece by (3.96). The bound follows from the bounds $|G_k(A^{(k)})(x,y)| \leqq C(|x-y|+\eta)^{2-d}e^{-\delta_0|x-y|}$ and $|D_{A^{(k)}}G_k(A^{(k)})(x,y)| \leqq C(|x-y|+\eta)^{1-d}e^{-\delta_0|x-y|}$ of [Ba 4], which Theorem 3.3 summarises, on each of the at most $\bar{n}+1$ legs and from (3.98)–(3.100) on each counterterm graph. Each vertex (3.35) of the chain carries the sum over the counterterm graphs of its order defined by (3.51). In it the subgraphs with integers $\leqq k-1$ of non-positive

degree are cancelled by the lower-order terms of δm^2, and a photon two-point subgraph of the graph of order e^4 whose lines carry integers $\geqq k$ is cancelled against its partner at those integers as in Sect. 3.5, by gauge covariance, before (3.98) is applied; the remainder then satisfies (3.98) in every ordering, as shown in Sect. 3.5 for the graphs of order four.

When every integer of the counterterm graph is $\geqq k$ we read (3.98)–(3.100) with $m_1 = 0$; the first stage is then $\sum_{j=k}^{j_{l(2)}} (L^{j-k})^{D(H_1)}$, bounded by $C(L^{j_{l(2)}-k})^{D(H_1)}$ when $D(H_1) > 0$, by C when H_1 is a one-vertex loop of degree -1 or -2, and by $1 + K - k$ when $D(H_1) = 0$.

The last case occurs for a scalar line carrying a derivative at each end between two vertices (3.38), which the summation by parts of Sect. 3.5 reduces to degree one, the derivative passing to the photon line, the only other line at that vertex, whose integer is not smaller, and for a photon loop at a vertex (3.38) with $m = 2$ whose difference acts on a leg of the chain, of degree zero by the factor η in (3.38); its index sum is bounded by C, since this factor is L^{-j} times the factor $L^j\eta$ counted for it in (3.48), j the integer of the loop (the second exception noted after (3.98)). In each case the sums over the intermediate vertices converge in $d = 3$, the chain having at least two legs and its outer legs being separated by $|x - y| \geqq c_0$. The counterterm vertices whose lines have indices $\leqq k - 1$ carry the centers of the vertices they replace, so that the cancellations of Sect. 3.5 are performed within each cluster.

The graph then factorizes into a product of localised subgraphs, one for each cluster, each with its own cube $\square'$ and background B; the expansions (3.61)–(3.73) are performed in the cube $\square'$ of the piece, which does not change the degrees or the classes.

Each piece is a convergent renormalised graph in the sense of Theorem 3.5. An ordering of its internal lines, continued by the lines of the other pieces, is an ordering of $\tilde{H}$ whose first subgraphs H_i are those of the piece, and the cancellations of Sect. 3.5 take place inside the piece, since a vertex, an internal vertex of a counterterm graph included, carries centers of one cluster only, the product $\chi_z(v)\chi_{z'}(v)$ vanishing when $|z - z'| > \frac{4}{3}\sqrt{d}$. When the two members of a pair are cut no cancellation is needed: each piece is a vertex with cut lines, the cut line with its endpoints fixed is a common factor of all members, and the sum over the index of a cut line is bounded by (3.94).

A renormalised graph is the sum over the members of its class, and the members differ only at the divergent subgraphs, where one of the alternatives of Sect. 3.5 is chosen; when the centers of an alternative lie in one cluster, the sum over the alternatives is performed inside that cluster, and the sum over the class is the product over the clusters of these sums.

When they lie in different clusters, the alternative has an internal line longer than $3\sqrt{d}$, which is cut like any other line and bounded by (3.94), its partners with the same centers vanishing or being cut as well. In each piece the methods outlined yield the desired bound, and the factors $\exp[-\frac{1}{2}\delta_0|x - y|]$ of the cut lines give the decay in (3.56).

3.5. Cancellation of Divergences. Divergent subdiagrams vanish either because of some symmetry requirement (gauge covariance or the approximate Euclidean symmetry of the lattice), or because they are cancelled by similar divergent diagrams in the counterterms δm^2 and E_1. We consider together all graphs of the same order and type in $P^{(k),1}$; for each ordering $\tilde{l}$, these graphs can be combined to form renormalised graphs, which are convergent. This property, which is a consequence of the renormalizability of the model, is proved in [Ba 3]. The steps of that proof, and the points at which $L = 5$ enters, are listed in Sect. 5.1.

We shall present below a more precise version of this statement. Recall that graphs in δm^2 and E_1 are defined with the propagators C_v^η and C_s^η; these may be decomposed as $C^\eta = \sum_{j=0}^{K} G_{(j)}^\eta$, where $G_{(j)}^\eta$ is given by (2.18) for $0 \leqq j \leqq K - 1$, and $G_{(K)}^\eta = C^\eta - G_K^\eta$. In Sect. 4 we will show that $G_{(j)}^\eta$ and $G_{(j)}^{\eta'}$ satisfy Propositions 3.7 and 3.9 for $k \leqq j \leqq K$. Therefore in any graph containing propagators C^η, we may introduce this decomposition and again order the lines. For any ordering $\tilde{l}$, we again get a sum of terms; each term has the property that for

some $m_1 \leqq m$, we sum over integers with the constraint

$$0 \leqq j_{l(1)} \leqq \cdots \leqq j_{l(m_1)} \leqq k-1 < j_{l(m_1+1)} \leqq \cdots \leqq j_{l(m)} \leqq K\,. \tag{3.96}$$

Then the renormalised graphs $\{H_{\rm ren}\}$ produced after cancelling divergences for this ordering $\tilde{l}$ have the following properties;

$$\text{(i)} \qquad D(H_i) > 0 \quad \text{for} \quad 1 \leqq i \leqq m_1\,, \tag{3.97}$$

(ii) for $m_1 < m$, and some $\{m_i\}$ with $m_1 < m_2 < \ldots < m_n = m$, $D(H_{m_i+l}) > D(H_{m_i})$ each $l = 1, \ldots, m_{i+1} - m_i - 1$, each $i = 1, \ldots, n-1$,

$$D(H_{m_{i+1}}) \leqq D(H_{m_i}) \quad \text{each} \quad i = 1, \ldots, n-1\,. \tag{3.98}$$

The integers $\{m_i\}$ correspond to divergent subgraphs encountered between integers k and K. In (i) there is one exception: a subgraph consisting of a single line with a derivative at each end, added as a connected component by itself, has degree zero by (3.48)–(3.49). This occurs for a scalar line joining two vertices (3.38), two vertices of (3.71) carrying D^η, or one of each, and for the photon line joining two vertices (3.36) in the graph of order τ^2.

A second exception is a component of degree zero by (3.48) at a vertex (3.38) or (3.39) with $m \geqq 2$: a photon loop at a vertex (3.38) with $m = 2$, or the second of two photon loops at a vertex (3.39) with $m = 4$. Here the explicit factor η at the vertex is $L^{-j}(L^j\eta)$, j the integer of the line last added, so that the sum (3.99) for the component carries the factor L^{-j} and is bounded by C; we count η in the degree as in (3.48) and keep L^{-j} as a convergence factor. At zero background such a loop cancels exactly against its partner in δm^2, the photon propagator not depending on the background, and only its slices with integers $\geqq k$ remain. For the difference (3.57) these components are compared between the two models as described after (3.108).

In these cases we sum by parts covariantly at one endpoint, moving the covariant difference $D^*_{A^{(k)}}$ (the curl for (3.36)) onto the other factors at that vertex, namely the external propagator, the fields $\tilde{A}^{(k)}$ and θ of Sect. 3.4, the functions g, h, χ, or another internal line; the "unitary" factors (3.64), a pure gauge for the constant field B of (3.61), pass through the covariant difference and are not differentiated.

The line then has degree one, and the degree of every connected subgraph containing it is unchanged or increased by one: the end of the line now counts $1 - \frac{d}{2}$ in place of $-\frac{d}{2}$, a subgraph H_i not containing the line does not count that end, and if the difference has fallen on another internal line l' at the vertex, then l' follows the line in the ordering, the line having been added as a component by itself, so that every H_i containing l' also contains the line; the same summation by parts is performed in the partner of the line in δm^2. A derivative on an external propagator is bounded by Theorem 3.3, a derivative on g, h or χ by a constant, a derivative on θ by $C[p(L^k\varepsilon)]^{-1}$, and a derivative on $\tilde{A}^{(k)}$ by (3.63).

In the pair (3.103) below the summation by parts at the vertex y leaves the term with the difference on the photon line and the term with the difference on the external factor; in the first the term linear in $y-x$, and in the second the term at zero momentum, $\partial_\rho\phi(x)\sum_y \partial_x G(x,y)G_v(x,y)$, G_v the slice of the photon line, have degree zero, and both are removed by the reflections through the centers of the blocks of scale $j_{l(2)}+1$ described after (3.108), up to the terms of degree zero with an explicit factor L^{-i} described there, the remainders having degree α by (3.9).

We can now extend the bound (3.56) to a graph $H_{\rm ren}$. Using the methods of Sect. 3.4, we shrink lines until H_{m_1} is one vertex, leaving the sum

$$\sum_{j=0}^{k-1} (L^{j-k})^{D(H_{m_1})} \leqq C\,. \tag{3.99}$$

Continuing the procedure, we shrink lines until H_{m_2} is one vertex, giving the sum

$$\sum_{j=k}^{j_{l(m_2+1)}} (L^{j-k})^{D(H_{m_2})-D(H_{m_1})} \leqq \begin{cases} C & \text{if} \quad D(H_{m_2}) < D(H_{m_1}) \\ C \ln(L^k\varepsilon)^{-1} & \text{if} \quad D(H_{m_2}) = D(H_{m_1}) \,. \end{cases} \tag{3.100}$$

We get a similar bound for each m_i, $i = 2, \ldots, n$. Therefore the graph H_{ren} is bounded as before, with the possible addition of some power of $\ln(L^k\varepsilon)^{-1}$. Notice that when $j \geqq k$, we cannot "transfer" external propagators. However, this never arises for graphs in E_1, and graphs in δm^2 are attached to external propagators at one vertex only.

Before proceeding, we must specify how the partition of unity (3.54) is to be introduced in the counterterms. First we decompose C^η on each line to get a set of graphs satisfying (3.96). When all the integers in a graph in δm^2 are less than k, we introduce (3.54) at each vertex of the graph. If any integer on a line is bigger than $k-1$, we do not introduce (3.54) at any vertex in the graph. For E_1, we again use (3.54) everywhere if $m_1 = m$; if $m_1 < m$, we introduce (3.54) only at one vertex connected to $l(m)$. Of course the exponential tree decay in (3.56) and (3.57) involves only the external vertices and those internal vertices with partitions of unity.

For the pairs of graphs which cancel because of gauge covariance we attach the functions (3.54) in Sect. 3.4 to one center for each field A', in the same way in both members, so that the cancellation is organised at each value of the localisations, the remainder being a graph with a difference acting on a localisation function or on a field. The members of such a pair have the same legs and the same orders in e and λ, so that the rule of Sect. 3.4 gives them the same number of centers.

At order e^2 the graphs which cancel at zero momentum are the photon self-energy with two vertices (3.38), the seagull (3.39) and the graphs with the vertices (3.42), (3.43) of the same order; on the lattice ηZ^d a constant field A' is the gradient of a linear function, so that the operator of the scalar field at the background $A'+c$, the term $a_k Q_k(A'+c)^* Q_k(A'+c)$ included, is unitarily equivalent to the one at A' for every A' and every constant c; hence the second-order kernel $K_{\mu\nu}(x,y)$ of these graphs satisfies $\sum_y K_{\mu\nu}(x,y) = 0$ at each x: the current $J_\mu(x; A'+c)$ equals $J_\mu(x; A')$ by the unitary equivalence, and the identity is its derivative in c.

On T_η the propagators are the periodic sums of those on ηZ^d, each block of T_η being a block of ηZ^d; the operators on ηZ^d are the limits of those on the tori as the sides grow, the masses being positive, and the bounds of Theorem 3.3, whose constants do not depend on the torus, hold for them as well; so the sum at zero momentum on T_η reduces to the terms in which a scalar line winds around T, each bounded by $C \exp[-\delta_0 \min_\mu L_\mu (L^k\varepsilon)^{-1}]$, smaller than every power of $L^k\varepsilon$, and these are carried as remainders. One difference lowers the divergence from linear to logarithmic only; the term linear in $y - x$ is removed by the reflection argument given below for the scalar self-energy, which applies since each vertex (3.38) with $m = 1$ is invariant under the reflection while $y - x$ changes sign; the remainder has degree α, up to the terms of degree zero with an explicit factor L^{-i} described there.

We will prove (3.97) and (3.98) for one linearly divergent self-energy diagram for the scalar field. This will show the idea of the general proof; the reader is referred to [Ba 3] for an exhaustive list of how divergences cancel. The graph in δm^2 corresponding to the diagram is the following:



(3.101)

When δm^2 occurs in a graph, the last three diagrams on the right-hand side of (3.101) contribute to terms with $m_1 < m$. After shrinking to a vertex all lines with integers less than

k, the remaining subgraphs (composed of lines with two dashes) have non-positive degree. So (3.98) holds in these cases.

The same reasoning applies, for $d = 3$, to the other graphs in δm^2 of order two and to the sunset graph of order λ^2, if at least one line of the graph carries an integer $j \geqq k$. Such a graph is attached to the rest of the renormalised graph at one vertex. As its lines are added in any order compatible with (3.96) the degree changes by $+2$ and -3, or by 0 and -1, or, when the scalar line is $l(1)$ with integer $j \leqq k-1$ and has been summed by parts as after (3.98), by $+1$ and -2 if the difference falls on the photon line and by $+1$ and -1 if it falls on an external factor, for the photon exchange graph, by -1 for the tadpoles of order λ and e^2, and by $+2$, -1, -1 for the sunset, whose subgraphs have degrees 2, 1, 0 in every ordering.

For a graph of order $e^{2p}\lambda^q$ with the vertices (3.34), (3.38), (3.39) and undifferentiated external legs the degree of the two-point graph at zero momentum is, for $d = 3$ by (3.48)–(3.49),

$$D = p + q - 2 \leqq 0\,, \tag{3.102}$$

where $\sum m = 2p$, the factors η^{m-1} of (3.38) and η^{m-2} of (3.39) being counted in (3.48); the inequality is (3.50). A graph of order two with a difference on an external leg has degree zero at zero momentum and no partner in δm^2, which by (3.51) contains the two-point function with undifferentiated legs only; its term at zero momentum, $\partial^\eta_\mu \tilde\phi(x)\tilde\phi(x)\sum_y \partial^\eta_{x,\mu} G(x,y)G_v(x,y)$, is removed by the reflection through the centers of the blocks described after (3.108), exactly as the second term of the summation by parts after (3.98), and the remainder has degree α. In each of the cases listed above every partial sum of the degree changes is at least the degree of the graph, so that (3.98) holds.

Indeed (3.98) for an ordering is equivalent to $D(H_m) \leqq D(H_t)$ for every $t \geqq m_1$. Given the latter, take m_{i+1} to be the first $t > m_i$ with $D(H_t) \leqq D(H_{m_i})$; since $t = m$ qualifies, the sequence ends at m. Conversely, (3.98) gives $D(H_m) \leqq D(H_{m_i})$ for each i, and $D(H_t) > D(H_{m_i}) \geqq D(H_m)$ for $m_i < t < m_{i+1}$. The degree of a counterterm graph attached at one vertex adds to that of the rest, and so do the degrees of several such graphs.

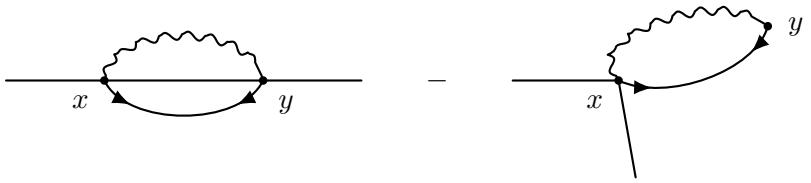
The graph of order λ^2 consisting of a tadpole with a tadpole on its loop combines, after the expansion (3.70)–(3.72) of Sect. 3.4 and in the terms with the propagator $G_k(0)$, with the graph carrying $-\delta m^2(y;0,1)$ at the inner vertex y into a tadpole whose inner loop is, up to sign, $F_{k,s}(y,y) = \sum_{j\geqq k} G^\eta_{(j)}(y,y)$, since by (3.51) the two cancel exactly for the propagator C^η_s; the terms with two or more factors $\tilde A^{(k)}$ on the inner loop have positive degree, and the term with one factor vanishes, q being antisymmetric (both the trace of q and the sum over the two legs of the quartic vertex vanish). For the combined graph (3.97) and (3.98) hold, the inner loop being the last line and lowering the degree from one to zero; in $\delta m^2(x;0,2)$ itself only the sunset remains.

For the other graphs in δm^2 of order four, a subgraph of negative degree may precede the other lines of the graph, and the inequalities (3.98) are to be read after this subgraph has been expanded as in (3.104) at the indices k and above; (3.98) then holds for these graphs as well.

For the graphs of E_1, on the other hand, (3.98) does not hold in general: a vacuum graph of positive degree with no divergent subgraph (for $d = 3$, for instance five quartic vertices $a,\dots,e$ with lines $ab, ab, ac, ad, bc, bd, ce, ce, de, de$, of degree 2, whose two lines ab form a subgraph of degree 1) admits orderings with $D(H_m) > D(H_{m_1})$, and the sum over the largest infrared integer, of the form (3.100), is then of order L^{K-k}, that is of order $(L^k\varepsilon)^{-1}$, since $\varepsilon = L^{-K}$. We do not need (3.98) for the graphs of E_1: E_1 enters $V^{(0),\eta}$ additively, so that only its expectation, E_1 itself, survives in (3.33); its graphs are field-independent, and they enter only the constant $S^{(k),1}(T^{(k)},0,0,0,0)$, which is removed in the normalized action $S^{(k),1}_N$ of Sect. 3.2 and cancels in the ratios (3.19)–(3.21).

The first graph on the right-hand side of (3.101) combines with a divergent diagram in $P^{(k),1}$

to give the following difference:

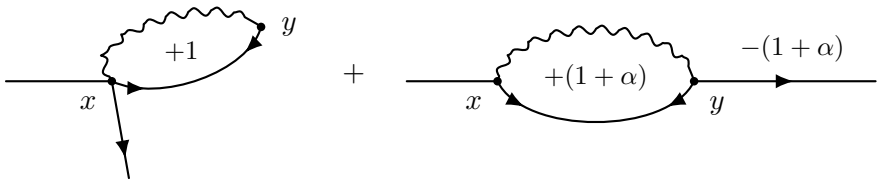


(3.103)

We can expand the external scalar field $\phi(y)$ as follows:

$$\phi(y) = \phi(x) + \sum_{\mu=1}^{d} (y_\mu - x_\mu)\,(\partial_\mu^\eta \phi)(x) + \sum_{b \in \Gamma_{x,y}} \eta\,|b_x - b|^\alpha\,(\partial_\alpha(b, b_x)\,\partial^\eta \phi)\,, \tag{3.104}$$

where $0 < \alpha < 1$, and b_x is the bond at x parallel to b. Substituting this into (3.103) gives



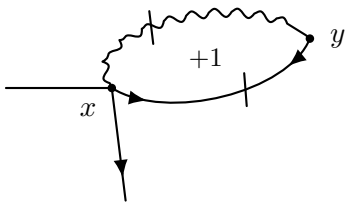
(3.105)

We have indicated the presence of additional convergence factors for the graphs, and additional derivatives on propagators. When the field $\phi(y)$ is $\chi(y)G^\eta_{(j)}(y,z)$ for some j and z, we can use (3.83) to bound the last term in (3.105). The localization function gives terms involving (3.81), so the bound is

$$\begin{aligned} &C|x-y|^{1+\alpha}\{(L^j\eta)^{1-d-\alpha} + (L^j\eta)^{1-d} + (L^j\eta)^{2-d}\} \\ &\qquad \cdot \exp[-\delta_0 (L^j\eta)^{-1} \mathrm{dist}(\Gamma_{x,y}, z)]\,. \end{aligned} \tag{3.106}$$

The factor $|x-y|^{1+\alpha}$ means that the last graph in (3.105) has degree $+\alpha$, and so it is convergent. Of course this extra degree of convergence has been obtained at the price of a derivative of order $(1+\alpha)$ on an external line. However, this means that we have postponed consideration of this divergence until it occurs in a bigger subgraph. Because the model is superrenormalizable, sufficiently large subgraphs always have positive degree; therefore the divergence will cause no problems at a later stage.

If we consider the following graph



(3.107)

we see that it contains the factor

$$\begin{aligned} &\sum_{\mu=1}^{d} \partial^\eta_{\mu,x} \partial^\eta_{\mu,y} C^\eta_s(x,y)\,(y_{\mu'} - x_{\mu'}) \\ &\qquad = \delta^\eta(x-y)\,(y_{\mu'} - x_{\mu'}) - m^2 (L^k \varepsilon)^2 C^\eta_s(x,y)\,(y_{\mu'} - x_{\mu'})\,, \end{aligned} \tag{3.108}$$

where δ^η is the lattice δ-function. So whenever a graph with the propagator C^η_s contains the first term in (3.105), we may introduce a similar graph with (3.107) at the vertex x, and the

error is the last term in (3.108). This would apply to C_s^η only: for that propagator the first graph in (3.105) would reduce to a term with δ^η, which vanishes since $y-x$ vanishes at $y=x$, and to an error of degree two with the factor $(L^k\varepsilon)^2$. But the first graph in (3.105) carries the propagators G_k of $P^{(k),1}$, for which (3.108) acquires the term $-a_k P_k G_k$, and for the slices $G^\eta_{(j)}$ of a fixed ordering, to which (3.97)–(3.98) refer, it has no analogue. We remove the first graph in (3.105) instead as follows.

The first graph in (3.105), the term linear in $y-x$ in (3.104), has the coefficient $\sum_y \Pi(x,y)(y-x)_\rho$, where Π is the kernel of the first member of the pair at zero background and x, y are the bonds of the two vertices; the two bonds are parallel, the photon propagator being diagonal, so that $y-x$ is also the difference of their midpoints. Here ϕ denotes the external field in the gauge of (3.61)–(3.64), $\tilde\phi(x) = U(-B(\Gamma^{(k)}_{x_0,x}))\phi^{(k)}(x)$, the factors (3.64) being part of it; since B is constant, $\partial^\eta\tilde\phi = U(-B(\Gamma))D_{A^{(k)}}\phi^{(k)} - e(L^k\varepsilon)^{2-d/2}\tilde A^{(k)} q\tilde\phi$ up to terms of higher order in η, so that the differences of $\tilde\phi$ and of $\partial^\eta\tilde\phi$ over a block of scale $j+1$ are bounded by (3.2) and (3.9) up to terms carrying the factor $e(L^k\varepsilon)^{2-d/2}p(L^k\varepsilon)^2$, by (3.63).

Let j be the larger of the two integers of the loop and c the center of the block of scale $j+1 \leqq k$ containing x. Since L is odd ($L=5$, Sect. 2.2), c is a lattice point, and the reflection $x \to 2c-x$, which maps the bond $(x, x+\eta e_\mu)$ onto the bond $(2c-x-\eta e_\mu, 2c-x)$ with the sign of A_μ reversed, maps the blocks of every scale $i \leqq j+1$ onto blocks; the operators G_i, Q_i and $C^{(i)}$ of (2.14)–(2.18) which make up the scalar slices $G^\eta_{(i)}$, $i \leqq j$, involve the blocks of the scales i and $i+1$ only, so that these slices are invariant.

Each vertex (3.38) with $m=1$ is invariant: its reflected form carries $\phi(b_-)$ in place of $U(A_b)\phi(b_+)$, D^η_μ and A'_μ changing sign together, and the difference of the two forms, $(D^\eta_A\phi(b))\cdot q\,[U(A_b)\phi(b_+) - \phi(b_-)]\,A'_b$, vanishes identically, since $U(A_b)\phi(b_+) - \phi(b_-)$ is $\eta D^\eta_A\phi(b)$ and q is antisymmetric. Hence the kernel is even while $(y-x)_\rho$ is odd, and $\sum_{x\subset B}\sum_y \Pi(x,y)(y-x)_\rho = 0$ for every block B of scale $j+1$, the sum running over the bonds x with both endpoints in B. The bonds with one endpoint in B we reflect instead through the center c_f of the face of B which they cross. Since $2c_{f,\mu} = 2y_\mu + 2L^{j+1}\eta - \eta$ and $2c_{f,\nu} = 2y_\nu + (L^{j+1}-1)\eta$ for $\nu \neq \mu$, y the corner of B and μ the direction normal to the face, twice every coordinate of c_f is congruent to $(L^i-1)\eta$ modulo $L^i\eta$ for $i \leqq j+1$, so that this reflection maps the lattice onto itself, the blocks of the scales $i \leqq j+1$ onto blocks, and the bonds crossing the face onto themselves with reversed orientation. With the external factors frozen at a site nearest c_f the kernel is even and $(y-x)_\rho$ odd, so that these terms vanish as well, up to the terms of the photon averages (2.11) described below, and the remainder is again a difference of the external factors over a distance at most $L^{j+1}\eta$.

The photon slices are not invariant: the average (2.11) over the bonds with b_- in a block goes over to the average over the bonds with b_+ in the reflected block, and the two differ by η times the block average of $\partial_\mu A_\mu$. The terms so produced are of degree zero by (3.48)–(3.49) and carry the explicit factor $\eta = L^{-i}(L^i\eta)$, i the scale of the slice, so that their sums over i are bounded by C in (3.56). For the difference (3.57) these terms, and the components of the second exception noted after (3.98), whose sums over the integers are dominated by the finest slices, are compared between the two models at the scale of the lattice spacings, where the corresponding slices are the same operators up to the mass terms and the periods; their difference is bounded by CL^{-k}.

The same reflection removes the term at zero momentum of the second case of the summation by parts after (3.98), $\partial^\eta_\rho\tilde\phi(x)\sum_y \partial_x G(x,y)G_v(x,y)$, the derivative ∂_x changing sign in place of $(y-x)_\rho$ while $G_v(x,y)$ does not.

We replace the external factors $\tilde\phi(x)$, $\partial^\eta_\rho\tilde\phi(x)$ and the function (3.54) attached at x by their values at c, having first written the function (3.54) attached at the internal vertex y as its value at x plus a difference, which is bounded by $C|y-x|$ and gives a term of degree one. The terms with the values at c vanish, and the remainder carries a difference of these factors over a distance at most $L^{j+1}\eta$, bounded by $C(L^{j+1}\eta)^\alpha$ by (3.2) and (3.9), so that its degree

exceeds that of the linear term by α. The differences between the propagators of (3.70)–(3.72) and the slices of $G_k(0)$, and the terms with a factor $\tilde{A}^{(k)}$, have higher degree and are treated as remainders.

This is the lattice form of the Euclidean symmetry invoked at the beginning of this section.

The inequalities (3.97) and (3.98) can be established for all renormalized graphs in $P^{(k),1}$ which contain an external field or a source vertex, with the two exceptions noted after (3.98) and, for the graphs of δm^2 of order four, in the sense explained before (3.103). Excepted are also the graphs which vanish identically because a scalar loop at zero background carrying an odd total number of photon legs vanishes, each vertex (3.38), (3.39) with m photon legs carrying q^m and the trace of an odd power of q being zero, q being antisymmetric; among them are the current one-point function and the graph of order e^3 with three vertices (3.38) with $m = 1$. Excepted are, finally, the graphs linear in h and independent of the fields A, ϕ of degree at most zero, namely the tadpole of order zero, whose partner is the normal ordering constant of (3.37), the graph of order λ and the graphs of order e^2, each with its partner in δm^2 (see (3.51)–(3.52)).

In these graphs, and in the graphs $\tilde{H}$ derived from them in Sect. 3.4, every line with integer $j \geqq k$ lies in a δm^2 vertex or in the normal ordering constant of (3.37). We would now like to see how this allows us to deduce Theorem 3.5 for such a graph. The bound (3.56) relies solely on the cancellation of divergences. In this connection, it should be noted that only through the expansion (3.70) can vertices approach $\partial\Box'$. But this expansion never produces divergent subgraphs, so the vertices in a divergent subgraph never see a "sharp" boundary. In order to establish (3.57) we must extract convergence factors as before. Since some subgraphs have degree α, we can extract only $L^{-\gamma k}$ with $\gamma < \alpha$. The only new factors to be replaced in $E_\phi^{(k+n)}$ are the convergence factors $|x-y|^\alpha$, etc. These must be replaced *after* all the other factors. To do so, we use the following bounds:

$$\begin{aligned} ||x'-y'|^\alpha - |x-y|^\alpha| &\leqq CL^{-\gamma k}\{|x'-y'|^{\alpha-\gamma} + |x-y|^{\alpha-\gamma}\}\,, \\ |(x'_\mu - y'_\mu) - (x_\mu - y_\mu)| &\leqq CL^{-\gamma k}\{|x'-y'|^{1-\gamma} + |x-y|^{1-\gamma}\}\,, \\ \Big|\sum_{b'\in b}\eta'|b'_- - x'|^\alpha - \eta|b_- - x|^\alpha\Big| &\leqq CL^{-\gamma k}\sum_{b'\in b}\{\eta'|b'_- - x'|^{\alpha-\gamma} + \eta|b_- - x|^{\alpha-\gamma}\}\,. \end{aligned} \tag{3.109}$$

Then by redoing the analysis of Sect. 3.4, we can establish Theorem 3.4 for any non-vanishing renormalized graph in $P^{(k),1}$ which, after the expansion (3.61)–(3.73), contains an external scalar field, an explicit factor of $\tilde{A}^{(k)}$ or a source vertex. At $h = 0$, the case to which Theorem 3.4 is restricted in Sect. 3.2, the graphs linear in h are absent. The remaining graphs either contain a line δG_k of (3.72), in which case either both ends of δG_k lie in $\Box$, so that Theorem 3.3 gives the factor $\exp[-\delta_0 p(L^k\varepsilon)] \leqq C(L^k\varepsilon)^{d+\sigma+1/2}$ by (3.59), or δG_k lies on a string of propagators of (3.70) beginning and ending in $\Box$, from which the factor $\exp[-\delta_0\,\mathrm{dist}(\Box,\partial\Box')]$ is extracted as in Sect. 3.4; in both cases (3.56) holds for these graphs, and (3.57) holds for $E^{(k)}$ and $E^{(k+n)}$ separately; or they contain no field at all and do not enter $S_N^{(k),1}$.

3.6. Convergence of $S^{(k)}$. To complete the proof of Theorem 3.4, we now consider the remaining terms in (3.30). The normalization factor E_0 can be written

$$\begin{aligned} \exp(E_0) &= Z_k^\eta Z_k^\eta(0)\int(dA_k)\exp[-1/2\langle A_k, \Delta^{(k)}A_k\rangle]\int(d\phi_k)\exp[-1/2\langle\phi_k, \Delta^{(k)}(0)\phi_k\rangle] \\ &= Z_k^\eta Z_k^\eta(0) N_k N_k(0)\,. \end{aligned} \tag{3.110}$$

Hence (3.30) can be written

$$\begin{aligned} &S^{(k),1}(T^{(k)}, A_k, \phi_k, g, h) \\ &\quad = 1/2\langle A_k, \Delta^{(k)}A_k\rangle + \ln N_k + 1/2\langle\phi_k, \Delta^{(k)}(A^{(k)})\phi_k\rangle + \ln N_k(0) \\ &\qquad - \ln[Z_k(A^{(k)})Z_k(0)^{-1}] + P^{(k),1}(T^{(k)}, A_k, \phi_k, g, h)\,. \end{aligned} \tag{3.111}$$

Examining the representation (2.15) for the quadratic term $\langle \phi_k, \Delta^{(k)}(A^{(k)})\phi_k\rangle$, we see that it has the form of a graph in $P^{(k),1}$ with two external scalar fields, and so the results of Sect. 3.4 apply.

The operator $\Delta^{(k)}$ is diagonal in the Fourier representation on $T^{(k)}$, and in Sect. 4 we prove the following proposition concerning its Fourier transform $\Delta^{(k)}(p)$.

Proposition 3.10. $|\Delta^{(k)}(p)| \leqq C$ *uniformly in* k,

$$|\Delta^{(k)}(p) - \Delta^{(k+n)}(p)| \leqq CL^{-2k}|\Delta^{(k)}(p)|\,. \tag{3.112}$$

Theorem 3.4 is stated for $n = 1$, the right-hand side of (3.22) being summable in K; Proposition 3.10, (3.113) and (3.114) below hold for every n, (3.114) for k so large that $CL^{-2k} \leqq \frac{1}{2}$.

From Proposition 3.10 we get immediately

$$|\langle A_k, (\Delta^{(k)} - \Delta^{(k+n)})A_k\rangle| \leqq CL^{-2k}(p(L^k\varepsilon))^2(\mu_0 L^k\varepsilon)^{-2}(L^k\varepsilon)^{-d}|T|\,, \tag{3.113}$$

where we used the bounds (3.2). Furthermore,

$$\begin{aligned} |\ln N_k - \ln N_{k+n}| &= 1/2d\Big|\sum_p \ln[\Delta^{(k+n)}(p)\Delta^{(k)}(p)^{-1}]\Big| \\ &\leqq C\sum_p |(\Delta^{(k+n)}(p) - \Delta^{(k)}(p))\Delta^{(k)}(p)^{-1}| \\ &\leqq C\sum_p L^{-2k} \leqq CL^{-2k}(L^k\varepsilon)^{-d}|T|\,. \end{aligned} \tag{3.114}$$

The convergence of $\ln N_k(0)$ follows similarly. Finally we consider the term $\ln[Z_k(A^{(k)})$ $Z_k(0)^{-1}]$. We introduce the fields $(j/R)A^{(k)}$, $j = 1, \ldots, R$, where $R = 1 + [(\mu_0 L^k\varepsilon)^{-1}]$. Then we have

$$\begin{aligned} \ln[Z_k(A^{(k)})Z_k(0)^{-1}] &= 1/2N \ln\det[G_k(A^{(k)})G_k(0)^{-1}] \\ &= 1/2N\sum_{j=1}^{R} \ln\det[G_k(j/RA^{(k)})G_k((j-1)/R\,A^{(k)})^{-1}]\,. \end{aligned} \tag{3.115}$$

We will expand $G_k(j/RA^{(k)})$, using (3.70) with $B = ((j-1)/R)A^{(k)}$ and $A' = (1/R)A^{(k)}$. The field A' is bounded by $Cp(L^k\varepsilon)$, so we have

$$\begin{aligned} &\ln\det[G_k(j/RA^{(k)})G_k((j-1)/R\,A^{(k)})^{-1}] \\ &\quad = \ln\det[I + G_k^{1/2}(j/RA^{(k)})V_k(A', B)G_k^{1/2}(j/RA^{(k)})]\,. \end{aligned} \tag{3.116}$$

For a symmetric operator D with $\|D\| \leqq \frac{1}{2}$ and any $n \geqq 1$, we have

$$\Big|\ln\det(I + D) - \sum_{p=1}^{n}(-1)^{p-1}p^{-1}\operatorname{tr} D^p\Big| \leqq \frac{2\operatorname{tr}|D|^{n+1}}{n+1}\,. \tag{3.117}$$

Here $\|D\| \leqq Ce(L^k\varepsilon)^{2-d/2}p(L^k\varepsilon)$ by (3.71), the bound on A' and the bounds $\|D^{\eta}_{A'+B}G_k^{1/2}\| \leqq 1$ and $\|Q_k(A'+B)G_k^{1/2}\| \leqq C$, G_k being taken at the field $A' + B = (j/R)A^{(k)}$ as in (3.116), which follow from $D^*_{A'+B}D_{A'+B} \leqq G_k^{-1}$ and $a_kQ_k^*Q_k \leqq G_k^{-1}$ in the units of (3.111), and $\|G_k(A'+B)^{1/2}\| \leqq C$ by (3.8) and Schur's test, together with (3.65) and (3.69), by which D^{η}_B and $Q_k(B)$ differ from $D^{\eta}_{A'+B}$ and $Q_k(A'+B)$ by operators of norm at most $Ce(L^k\varepsilon)^{2-d/2}\sup|A'|$, so that $\|D\| \leqq \frac{1}{2}$ for $L^k\varepsilon$ small. We use (3.117) to expand (3.116) up to order $2\bar{n}$. A general term in the expansion is

$$(-1)^{p-1}p^{-1}\operatorname{tr}[V_k(A', B)G_k(j/RA^{(k)})]^p\,. \tag{3.118}$$

Graphically, this is a scalar loop with external vector fields. The expansion (3.118) is not suited to a comparison of the two models order by order in p, and we compare the two determinants scale by scale instead. Let $C^{(i)}(A) = C^{(i),L^i\eta}(T_\eta, A)$, $i = 0, \dots, k-1$, be the covariances (2.17) of the scalar fluctuation fields at the successive steps of the renormalization transformation on T_η in the background A. Performing the Gaussian integration over the fluctuation fields step by step, $\det G_k(A)$ is, up to a factor independent of A, the product over i of $\det C^{(i)}(A)$, by the composition law of the renormalization transformation (Sect. 2 of [Ba 1]). Completing the square in the Gaussian integral over the field of the ith step, whose quadratic form is $C^{(i)}(A)^{-1}$ of (2.17), gives $Z_{i+1}(A) = c_i Z_i(A) \det C^{(i)}(A)^{N/2}$ with c_i independent of A, so that

$$\ln[Z_k(A) Z_k(0)^{-1}] = 1/2\, N \sum_{i=0}^{k-1} \ln\det[C^{(i)}(A) C^{(i)}(0)^{-1}]\,, \tag{3.119}$$

and likewise for the model on $T_{\varepsilon_{K+1}}$ after $k+1$ steps, where the ith step of the model on T_{ε_K} and the $(i+1)^{\text{st}}$ step of the model on $T_{\varepsilon_{K+1}}$ act at the same scale and on the same lattice.

Proposition 3.11. *Let $0 < \gamma' < \min\{\gamma, \alpha\}$, where γ is the exponent of (3.89) and α the Hölder exponent of (3.9), and let $\sigma' > \sigma + d$. On the support of χ_k, for $L^k\varepsilon$ small and $i = 0, \dots, k-1$,*

$$\begin{aligned} &|\ln\det[C^{(i)}(A^{(k)}) C^{(i)}(0)^{-1}] - \ln\det[C'^{(i+1)}(A^{(k+1)}) C'^{(i+1)}(0)^{-1}]| \\ &\leqq Cp(L^k\varepsilon)^c (L^k\varepsilon)^{-d} \left[L^{-\gamma' k} (L^i\eta)^{1-\gamma'} + (L^k\varepsilon)^{\sigma'} (L^i\eta)\right] |T|\,, \end{aligned} \tag{3.120}$$

where $C'^{(i+1)}$ denotes the covariance at the $(i+1)^{\text{st}}$ step for the model on $T_{\varepsilon_{K+1}}$; c depends only on σ' and d, and C on the coupling constants, N and σ', the dependence on the torus being through $(\min_\mu L_\mu)^{-1}$ only. The term $i = 0$ of that model, which has no counterpart, satisfies

$$|\ln\det[C'^{(0)}(A^{(k+1)}) C'^{(0)}(0)^{-1}]| \leqq Cp(L^k\varepsilon)^c (L^k\varepsilon)^{-d} L^{-k-1} |T|\,. \tag{3.121}$$

Proof. The ith term of the sum is, up to the constant factor of (3.115), $-\sum_y \{\ln[C^{(i)}(A)^{-1}] - \ln[C^{(i)}(0)^{-1}]\}(y,y)$, the trace over the components of ϕ being understood and the sum running over the sites y of the lattice of the ith step. At each site y we subtract from the background its value $A^{(k)}(y)$ by the gauge transformation with the gauge function

$$\chi_y(x) = \psi(x-y)\, A^{(k)}(y) \cdot (x-y)\,, \tag{3.122}$$

where ψ is equal to one on the cube of side $2p(L^k\varepsilon)$ around the origin and to zero outside the cube of side $\rho = \frac{1}{2} \min_\mu L_\mu (L^k\varepsilon)^{-1}$, with $|\partial^\eta \psi| \leqq C\rho^{-1}$ and $|\partial^\eta \partial^\eta \psi| \leqq C\rho^{-2}$.

Then $\partial^\eta \chi_y = A^{(k)}(y)$ on every bond of the inner cube, the cube of side $p(L^k\varepsilon)$ around y, and, by (3.2),

$$|\partial^\eta \chi_y| \leqq Cp(L^k\varepsilon)(\mu_0 L^k\varepsilon)^{-1}\,, \qquad |\partial^\eta \partial^\eta \chi_y| \leqq Cp(L^k\varepsilon)\mu_0^{-1} \tag{3.123}$$

everywhere, the constant in the second bound being proportional to $(\min_\mu L_\mu)^{-1}$, so that $\tilde{A} = A^{(k)} - \partial^\eta \chi_y$ and the fields $\tilde{A}_{\text{in}} + s\tilde{A}_{\text{out}}$, $0 \leqq s \leqq 1$, introduced below are regular in the sense of Definition 3.2 for $L^k\varepsilon$ sufficiently small, and Theorem 3.3 applies to them. Since $\chi_y(y) = 0$, the diagonal element at y is unchanged. The transformed background $\tilde{A}$ vanishes at y; it is bounded by $Cp(L^k\varepsilon)|x-y|$ on the inner cube, by the bound on $\partial^\eta \tilde{A}^{(k)}$ in (3.63), which is a bound on $\partial^\eta A^{(k)}$ since the two fields differ by a constant, and by $Cp(L^k\varepsilon)(\mu_0 L^k\varepsilon)^{-1}$ outside it, by (3.2) and the first bound in (3.123).

We write $\tilde{A} = \tilde{A}_{\text{in}} + \tilde{A}_{\text{out}}$, $\tilde{A}_{\text{in}} = \theta\tilde{A}$, where θ is a C^∞ function as in (3.72), equal to one on the cube of side $p(L^k\varepsilon)/2$ around y and to zero outside the inner cube, with $|\partial^\eta \theta| \leqq Cp(L^k\varepsilon)^{-1}$. Then

$$|\tilde{A}_{\text{in}}| \leqq Cp(L^k\varepsilon)^2\,, \qquad |\partial^\eta \tilde{A}_{\text{in}}| \leqq Cp(L^k\varepsilon)\,, \tag{3.124}$$

so that $\tilde{A}_{\rm in}$ is regular in the sense of Definition 3.2 for $L^k\varepsilon$ small, while $\tilde{A}$ is gauge equivalent to $A^{(k)}$; and $\tilde{A}_{\rm out}$ vanishes within distance $p(L^k\varepsilon)/4$ of y. We put

$$X = C^{(i)}(0)^{-1}\,, \qquad Y = C^{(i)}(\tilde{A}_{\rm in})^{-1}\,, \qquad X' = C^{(i)}(\tilde{A})^{-1}\,, \tag{3.125}$$

so that

$$\{\ln X' - \ln X\}(y,y) = \{\ln X' - \ln Y\}(y,y) + \{\ln Y - \ln X\}(y,y)\,. \tag{3.126}$$

The resolvents $(Y+t)^{-1}$, $(X'+t)^{-1}$ have kernels bounded by $C(\gamma_0+t)^{-1}\exp[-\delta|x-x'|]$ uniformly in $t \geqq 0$: the conditions (4.38), (4.39) hold for Y and X' by (3.7) and Schur's test, with γ_0 the inverse of the bound on $\|C^{(i)}(A)\|$ which (3.7) gives, and the conjugation by $\exp[\delta(L^i\eta)^{-1}d(x,y)]$, d the distance on the torus, changes Y and X' by operators of norm at most $\gamma_0/2$ for δ sufficiently small, so that the conjugated resolvents have norm at most $2(\gamma_0+t)^{-1}$, as in Sect. 4.

For the first term we use the identity

$$\ln X' - \ln Y = -\int_0^\infty (Y+t)^{-1}(Y-X')(X'+t)^{-1}\,dt\,. \tag{3.127}$$

$Y - X'$ vanishes when $\tilde{A}_{\rm out} = 0$. Interpolating along $\tilde{A}_{\rm in} + s\tilde{A}_{\rm out}$, $0 \leqq s \leqq 1$, whose curvature is bounded by Cp, the derivative in s is a chain with one vertex of size $Cep\mu_0^{-1}(L^k\varepsilon)^{1-d/2}$ supported on $\operatorname{supp}\tilde{A}_{\rm out}$, between lines decaying at the rate δ by the bounds of Theorem 3.3; so the kernel of $Y - X'$ is bounded by

$$C(L^k\varepsilon)^{-1}\exp[-\delta\,\operatorname{dist}(x,\operatorname{supp}\tilde{A}_{\rm out}) - \delta\,\operatorname{dist}(x',\operatorname{supp}\tilde{A}_{\rm out})]\,, \tag{3.128}$$

the power of $(L^k\varepsilon)^{-1}$ being absorbed below by the exponential.

The decay is at the rate δ in the units of the step, that is $\delta(L^i\eta)^{-1}$ on $T_1^{(k)}$, and $\operatorname{dist}(y,\operatorname{supp}\tilde{A}_{\rm out}) \geqq p(L^k\varepsilon)/4$; so the diagonal element at y is bounded by

$$C\exp[-\delta p(L^k\varepsilon)(16L^i\eta)^{-1}] \leqq C(L^k\varepsilon)^{\sigma'+d}(L^i\eta)^{d+1}\,, \tag{3.129}$$

since $p(x) \geqq C\log x^{-1}$ for x small, with C as large as we please, by (3.1); and its sum over the $(L^i\eta)^{-d}(L^k\varepsilon)^{-d}|T|$ sites of the step is bounded by $C(L^k\varepsilon)^{\sigma'}(L^i\eta)|T|$.

For the second term we expand by the formula

$$\ln(X-V) - \ln X = -\int_0^\infty (X+t)^{-1}\sum_{m\geqq 1}[V(X+t)^{-1}]^m\,dt\,, \qquad V = X - Y\,, \tag{3.130}$$

which converges since

$$\|V\| \leqq Ce(L^k\varepsilon)^{2-d/2}(L^i\eta)\,p(L^k\varepsilon)^2 < \gamma_0/2 \tag{3.131}$$

for $L^k\varepsilon$ sufficiently small, $|\tilde{A}_{\rm in}|$ being at most $Cp(L^k\varepsilon)^2$. Expanding V in powers of $\tilde{A}_{\rm in}$ gives graphs whose lines are the fine propagators $G_i(0)$ on T_η, which enter $\Delta^{(i)}$ through (2.15), and the resolvents $(X+t)^{-1}$ on the lattice of the step, and whose vertices are the coefficients of the expansion of the covariant Laplacian and of the averaging operators in $\tilde{A}_{\rm in}$ about $B = 0$, as in (3.65)–(3.71).

The expansion of V in powers of $\tilde{A}_{\rm in}$ converges, the series for $C^{(i)}(s\tilde{A}_{\rm in})^{-1}$ in powers of s converging for $|s|$ up to a constant multiple of the inverse of the bound on the vertices of (3.65)–(3.71), which is

$$Ce(L^k\varepsilon)^{2-d/2}(L^i\eta)\,p(L^k\varepsilon)^2\,; \tag{3.132}$$

its terms of order n are bounded by C^n times the $n^{\rm th}$ power of this quantity. By the invariance under $(A,\phi) \to (-A, R_0\phi)$, R_0 an orthogonal matrix with $R_0 q R_0^{-1} = -q$ (for $N = 2$, $R_0 =$

diag$(1,-1)$), the element is an even function of the background, so the expansion begins at second order.

We bound each term separately. In a chain the fine line $G_i(0)$ enters either through the profile $a_i G_i(0) Q_i^*$, whose kernel is bounded by (3.8), or between two vertices, where the derivatives of the vertices are bounded together with the line as operators, $\|DG_i(0)D^*\| \leqq 1$, $\|DG_i(0)\| \leqq C$, and the same bounds hold after conjugation by $\exp[\delta(L^i\eta)^{-1}d(x,y)]$ for δ small, since the conjugation changes D by an operator of norm $C\delta$ and $G_i(0)^{-1}$ by an operator B with $\|G_i(0)^{1/2}BG_i(0)^{1/2}\| \leqq C\delta$, $\|DG_i(0)^{1/2}\|$ being at most one, and the bounds hold with the constants doubled, so that the chain decays from y at a rate $\delta' < \delta$ and the factor $|z-y|$ in $|\tilde{A}_{\rm in}(z)| \leqq Cp(L^k\varepsilon)|z-y|$ is absorbed. We keep the terms up to an even order $2n_0$ depending on σ'; the term of order $m \geqq 2$ is bounded by

$$C^m \big(e(L^i\eta L^k\varepsilon)^{2-d/2}(L^i\eta)^{d/2}p(L^k\varepsilon)\big)^m \tag{3.133}$$

per site of the step.

The terms of order greater than $2n_0$ in $\tilde{A}_{\rm in}$, together with the remainder of the series in V, are bounded at each site by

$$C\big(e(L^k\varepsilon)^{2-d/2}(L^i\eta)\,p(L^k\varepsilon)^2\gamma_0^{-1}\big)^{2n_0+1}\,, \tag{3.134}$$

whose sum over the sites is at most $C(L^k\varepsilon)^{\sigma'}(L^i\eta)|T|$ for n_0 large depending on σ' and d. Summing the terms of orders 2 to $2n_0$ over the sites gives $Cp(L^k\varepsilon)^c(L^k\varepsilon)^{-d}(L^i\eta)|T|$ for each model, which gives the last statement.

For the difference of the two models we compare the two expansions term by term. The backgrounds, after the subtraction of their values at y, differ on the inner cube by at most

$$CL^{-\gamma k}p(L^k\varepsilon)^2|x-y| + Cp(L^k\varepsilon)^2\eta^\alpha|x-y|\,, \tag{3.135}$$

by the second line of (3.89) applied to (3.62) and by (3.9): the first differences of $A^{(k+1)}$ at the two spacings η, η' differ at a common site by η^α times the Hölder quotient of $\partial^{\eta'}A^{(k+1)}$, which is bounded on the inner cube by $Cp(L^k\varepsilon)^2$. The second term is $p(L^k\varepsilon)L^{-\alpha k}$ times the size of the backgrounds and the first is $p(L^k\varepsilon)L^{-\gamma k}$ times it; both are at most $p(L^k\varepsilon)L^{-\gamma' i}$ for every $i \leqq k$, since $\gamma' < \gamma$ and $\gamma' < \alpha$, and the power of $p(L^k\varepsilon)$ is absorbed in $p(L^k\varepsilon)^c$.

The lines through which the background enters the two expansions are of two kinds. The profiles $a_iG_i(0)Q_i^*$ and $a_{i+1}G'_{i+1}(0)Q'^*_{i+1}$ differ, by Proposition 3.8 applied after the rescaling (2.21) to the model with i steps on T_η and to the model with $i+1$ steps on $T_{\eta'}$, both read on the lattice of the step, by $CL^{-\gamma i}$ relative to their size, in the supremum and in the supremum of the first difference at the common sites.

A line $G_i(0)$ between two vertices carries at most two differences, one at each end, from the hopping vertices. If it carries at most one, it acts on a bounded function u on T_η, and we compare $G_i(0)u$ and $\partial^\eta G_i(0)u$ at the common sites with the corresponding quantities of the finer model, the difference of the two functions u being propagated along the chain. For equal u, extended to $T_{\eta'}$ as a constant on the cells of side η, we decompose both lines by (2.18) into their slices at the scales $L^j\eta$, $j < i$; the slice of the finer line at the scale η' has no counterpart, is bounded by Proposition 3.7 alone and is of relative size $L^{-2(i+1)}$, and $L^{-(i+1)}$ for its first difference.

By Proposition 3.9 in the form (3.91)–(3.93), the kernels of corresponding slices and their first differences differ at the common sites by $CL^{-\gamma j}(L^j\eta)^{2-d}$ and $CL^{-\gamma j}(L^j\eta)^{1-d}$, with the exponential decay at the scale $L^j\eta$, so that the sums over z of the differences of the slice kernels, applied to u, are bounded by

$$CL^{-\gamma j}(L^j\eta)^2 \sup|u|\,, \qquad CL^{-\gamma j}L^j\eta \sup|u|\,, \tag{3.136}$$

together with the oscillation of the finer kernel and of its first difference within the cells of side η, which are bounded by $C\eta(L^j\eta)^{1-d}$ and $C\eta(L^j\eta)^{-d}$, the mixed second difference of a slice being bounded by $C(L^j\eta)^{-d}$ with the decay of (3.81), which follows from (3.8) applied to each of the two profiles in (2.18) and the scaling (2.21), as (3.81) does; the comparison of the first differences in Proposition 3.9 is at the two spacings η, η' at the common sites; they contribute $CL^{-j}(L^j\eta)^2 \sup|u|$ and $CL^{-j}(L^j\eta)\sup|u|$. Since

$$\sum_{j<i} L^{-\gamma j} L^{-2(i-j)} \leqq CL^{-\gamma i}\,, \qquad \sum_{j<i} L^{-\gamma j} L^{-(i-j)} \leqq CL^{-\gamma i}\,, \tag{3.137}$$

and the oscillation terms are summed in the same way, the two lines differ by $CL^{-\gamma i}$ relative to their size.

The oscillation of the functions within the cells of side η must also be controlled. The function following an averaging vertex is constant on the blocks of the step up to the phase along the staircase (2.13), which varies within a cell by at most $C\eta$ times the background and by $C\eta$ times the first difference of the background summed along the staircase. A hopping vertex is of the form $\partial^* M + M^*\partial$, where M is multiplication by a coefficient of the expansion; by the bounds on $\tilde{A}_{\mathrm{in}}$ and on the Hölder quotient of $\partial^\eta A^{(k)}$ given below, M is $C^{1+\alpha}$ at the scale of the step, with norm $Cp(L^k\varepsilon)^c$ times its size.

We let the difference act on the function next to the vertex, so that the function following the vertex is $\partial^*(Mf)$ or $M^*\partial f$, where f is the function produced by the preceding line, and by (3.8), (3.9) a line applied to a bounded function is $C^{1+\alpha}$ at the scale of the step. Hence, by induction along the chain from the profile at its end, every f is $C^{1+\alpha}$ and every function following a hopping vertex is bounded and C^α at the scale of the step, with norm $Cp(L^k\varepsilon)^c$ times its size; its oscillation within a cell of side η is therefore at most $Cp(L^k\varepsilon)^c L^{-\alpha i}$ times its size, the power of $p(L^k\varepsilon)$ going into the prefactor and $L^{-\alpha i} \leqq L^{-\gamma' i}$ into the rate, since $\gamma' < \alpha$.

If the line carries two differences, both its neighbours are hopping vertices, and the function following each of them is the product of a coefficient of the expansion, a smooth function of the background, with a line of the chain, which is $C^{1+\alpha}$ at the scale of the step by (3.8), (3.9); the difference at one end acts on this function, so that the line carries one difference and is compared as before, the two difference quotients of the function differing by η^α times its Hölder quotient.

The operators $C^{(i)}(0)$ and $C'^{(i+1)}(0)$, which act on the same lattice, differ by Lemma 4.5 by CL^{-i} times a kernel decaying exponentially at the scale of the step; so do their inverses, which we write X_0 and X_0', by $X_0 - X_0' = X_0(C'^{(i+1)}(0) - C^{(i)}(0))X_0'$ and the kernel bounds (3.7), and so do the resolvents $(X_0+t)^{-1}$, $(X_0'+t)^{-1}$ of the two models, by the resolvent identity and the bound $C(\gamma_0+t)^{-1}$, which give $CL^{-i}(\gamma_0+t)^{-2}$ times such a kernel, integrable in t. So each line differs between the models by $CL^{-\gamma' i}$ relative to its size, and

$$(L^i\eta)L^{-\gamma' i} = L^{-\gamma' k}(L^i\eta)^{1-\gamma'}\,. \tag{3.138}$$

The outer terms and the remainders of the two expansions were each bounded above by $C(L^k\varepsilon)^{\sigma'}(L^i\eta)|T|$; this is the second term in the bracket.

It remains to compare the vertices of the two expansions and the block average terms, which live on the two fine lattices T_η and $T_{\eta'}$, $\eta' = \eta/L$, and are of the form (3.65)–(3.71), with the lines $G_i(0)$, $G'_{i+1}(0)$ between them and with the coefficients a_i, a_{i+1} of (2.14), which differ by CL^{-2i}. We compare each vertex together with the line which follows it, and propagate the discrepancy of the two models along the chain.

The hopping vertices contain difference quotients at the two scales. At the common sites the two quotients differ, in the units of $T_1^{(k)}$, by η^α times the Hölder quotient of the finer line, that is by $L^{-\alpha i}$ times the quotient at the scale of the step, which is bounded by (3.8), (3.9); the first differences of the two backgrounds differ there by η^α times the Hölder quotient of $\partial^{\eta'} A^{(k+1)}$,

which is bounded on the inner cube by $Cp(L^k\varepsilon)^2$ by (3.9) and (3.2) after a constant is subtracted from A_k as in (3.61)–(3.63), and by $\partial^\eta(A^{(k)} - A^{(k+1)})$ at those sites, which Proposition 3.8 applied to (3.62) bounds by $CL^{-\gamma k}p(L^k\varepsilon)(1+|x-y|)$, the factor $1+|x-y|$ being absorbed by the decay from y as before.

The coefficient of the finer vertex is the finer background on the bond issuing from the common site, which differs from its value on the coarser bond by $C\eta$ times the first difference of the finer background, as for the vertex of order two below; and the finer function is evaluated at the end of the finer bond, which differs from the end of the coarser bond by the oscillation of the function within the cell, controlled above.

The vertices of order $\geqq 2$ in the hopping term are of two kinds. Those with one difference, (3.38), carry the explicit factors η^{m-1}, η'^{m-1} relative to the vertex of first order and are bounded by their size in each model, at the price of a factor L^{-i}, the fine spacing being L^{-i} in the units of the step. The vertex (3.39) of order two carries no such factor; it is compared directly, like the local terms below, through the difference of the backgrounds at the common sites and the oscillation of the finer background within the finest cells, which is bounded by $C\eta$ times its first difference. The vertices of order $m \geqq 4$ without a difference carry the factors η^{m-2}, η'^{m-2} and are bounded by their size in each model, which is smaller by CL^{-i} than that of the vertex of order two.

The staircase (2.13) of the finer lattice from a site of the step to a site x of $T_{\eta'}$ runs within distance η of the staircase of the coarser lattice to the corner of the finest cell containing x, the cells of side η of T_η, its corners being displaced by at most η, and the phase of the finer model along a bond of the coarser staircase is the sum of the finer background over the bonds subdividing it.

The two phases therefore differ by the difference of the two backgrounds at the common sites, by $C\eta$ times the first difference of the finer background summed along the staircase, which also bounds the flux through the strip between the displaced legs, and by $C\eta$ times the finer background along the end pieces inside the finest cell, which is bounded on the inner cube by $C\eta\, p(L^k\varepsilon)|x-y|$. The staircase has length at most $CL^i\eta$, so that each of these is of the order of a vertex of first order with the explicit factor L^{-i}, the fine spacing in the units of the step. The same applies to the phases of Q_i^* and Q_i at the two ends of each chain.

The block averages Q_i of the two models are compared after summing over each finest cell: the average of a function over a block differs from the average of its values at the corners of the finest cells by the oscillation of the function in the cells, which is bounded by $C\eta$ times its first difference, and the averages at the corners are compared at the common sites. The two ends of each chain, Q_i^* and Q_i with their phases, are treated in the same way.

The local terms $aL^{-2}Q^*Q$ of (2.17), with Q the average over one step on the lattice of the step, have the same coefficient in both models and staircases (2.13), which pass through sites common to both lattices, so that their difference is bounded by the difference of the backgrounds at these sites, with the factor $L^{-\gamma k}$ of (3.89), together with $C\eta$ times the first difference of the finer background, summed along the staircase.

Each vertex is a sum of at most C terms with coefficients bounded by C^m, so the discrepancy propagates along the chain with the factor $L^{-\gamma' i}$ relative to the size of each term, and the result has the form of the first term in the bracket. At $i = 0$, where the two steps have different structure, the bound follows from the two single-model bounds, since $L^{-\gamma' k}\eta^{1-\gamma'} = L^{-k}$. This proves Proposition 3.11.

Summing the bound of Proposition 3.11 over i, and absorbing the powers of $p(L^k\varepsilon)$ as in (3.60), we obtain

$$\begin{aligned} &|\ln[Z_k(A^{(k)})Z_k(0)^{-1}] - \ln[Z_{k+1}(A^{(k+1)})Z_{k+1}(0)^{-1}]| \\ &\qquad \leqq C(L^{-\gamma' k}(L^k\varepsilon)^{-\beta-1} + (L^k\varepsilon)^\sigma)|T|\,, \end{aligned} \tag{3.139}$$

since $\sum_i (L^i\eta)^{1-\gamma'} \leqq C$, $\sum_i L^i\eta \leqq C$ and $\beta + 1 > d$, β being larger than $d + \frac{1}{2}$ in (3.60); the

term $i = 0$ of the second model contributes $CL^{-k}p(L^k\varepsilon)^c(L^k\varepsilon)^{-d}|T|$, which is of the same form. This is (3.13) with γ' in place of γ and $\beta + 1$ in place of β, admissible values of the exponents of Theorem 3.4.

Together with Sects. 3.4 and 3.5, this completes the proof of Theorem 3.4.

4. Technical Estimates

By using multiple reflection representations, the propagators G_k^η and $G_k^\eta(\Omega)$ can be written in terms of the operator defined by (2.14) with free boundary conditions (and $A = 0$, of course), as long as Ω is a rectangular parallelepiped which is a union of blocks of L^{kd} sites. Such representations are given explicitly in [Ba 4], so it is sufficient to prove Propositions 3.8 and 3.9 for the operator with free boundary conditions, which we write as G_k for simplicity. The basis for our proof is an explicit Fourier representation for $a_k G_k Q_k^*$. We introduce a Fourier transform on ηZ^d by

$$\tilde{f}(p) = \sum_x \eta^d e^{-ipx} f(x)\,, \qquad f(x) = (2\pi)^{-d} \int_{|p| \leqq \pi/\eta} e^{ixp} \tilde{f}(p)\,. \tag{4.1}$$

By applying this to the equation defining G_k, it follows in a straightforward way (see [Ba 4]) that

$$(a_k G_k Q_k^*)(x,y) = (2\pi)^{-d} \int_{|p'| \leqq \pi} dp' \Delta^{(k)}(p') \sum_l e^{i(p'+l)(x-y)} \frac{u_k^\eta(p'+l)}{\Delta^\eta(p'+l)}\,, \tag{4.2}$$

where $x \in \eta Z^d$, $y \in Z^d$, $p' \in [-\pi, \pi)$, $l \in 2\pi Z^d$ and $-\pi(L^k - 1) \leqq l_\mu \leqq \pi(L^k - 1)$ for L odd. We shall take L odd throughout this section, as is the case for $L = 5$. Also

$$u_k^\eta(p) = \prod_{\mu=1}^d [(e^{-ip_\mu} - 1)\eta(e^{-i\eta p_\mu} - 1)^{-1}]\,, \tag{4.3}$$

$$\Delta^\eta(p) = 4\eta^{-2} \sum_{\mu=1}^d \sin^2(1/2\eta p_\mu) + m^2(L^k\varepsilon)^2\,, \tag{4.4}$$

$$\Delta^{(k)}(p') = \Big(a_k^{-1} + \sum_l |u_k^\eta(p'+l)|^2 \Delta^\eta(p'+l)^{-1}\Big)^{-1}\,. \tag{4.5}$$

We will first prove a bound on the difference between $\Delta^{(k)}(p')$ and the following operator:

$$\bar{\Delta}^{(k)}(p') = \Big(a_k^{-1} + \sum_l |u_k^\eta(p'+l)|^2 D^{-1}(p'+l)\Big)^{-1}\,, \tag{4.6}$$

where the operator D satisfies

$$\text{(i)} \qquad |D^{-1}(p) - (|p|^2 + m^2(L^k\varepsilon)^2)^{-1}| \leqq CL^{-2k}\,, \tag{4.7}$$

$$\text{(ii)} \qquad 0 \leqq C|p|^2 \leqq D(p)\,. \tag{4.8}$$

Lemma 4.1.

$$|\Delta^{(k)}(p') - \bar{\Delta}^{(k)}(p')| \leqq CL^{-2k}\Delta^{(k)}(p')\,. \tag{4.9}$$

Proof. We first check that $\Delta^\eta(p)$ satisfies (4.7) and (4.8). By using $x^2 \geqq \sin^2 x \geqq x^2 - x^4/3$, we have

$$|\Delta^\eta(p) - (|p|^2 + m^2(L^k\varepsilon)^2)| \leqq C\eta^2 \sum_{\mu=1}^d |p_\mu|^4 \leqq CL^{-2k}|p|^4\,, \tag{4.10}$$

from which (4.7) follows. The bound (4.8) is obvious. Since Δ^η and D are positive, we have $\Delta^{(k)}(p')$, $\bar\Delta^{(k)}(p') \leqq a_k$. Hence

$$\begin{aligned} |\Delta^{(k)}(p') - \bar\Delta^{(k)}(p')| &\leqq C\Delta^{(k)}(p')|\Delta^{(k)}(p')^{-1} - \bar\Delta^{(k)}(p')^{-1}| \\ &\leqq C\Delta^{(k)}(p') \sum_l |u_k^\eta(p'+l)|^2 L^{-2k}\,, \end{aligned} \tag{4.11}$$

where we used (4.7). Since $\sum_l |u_k^\eta(p'+l)|^2 = 1$, the bound (4.9) follows.

To prove convergence of $\Delta^{(k)}(p')$, we notice that the composition law for renormalization transformations allows us to write $\Delta^{(k+n)}(p')$ in the form (4.6), with

$$D^{-1}(p) = a_n^{-1} L^{-2k} + \sum_{l'} |u_n^{\eta'}(p+l')|^2 \Delta^{\eta'}(p+l')^{-1}\,, \tag{4.12}$$

where $l' \in 2\pi L^k Z^d$ and $|l'_\mu| \leqq \pi(L^{k+n} - L^k)$ for L odd. Also

$$u_n^{\eta'}(p) = \prod_{\mu=1}^d [\eta^{-1}(e^{-i\eta p_\mu} - 1)\eta'(e^{-i\eta' p_\mu} - 1)^{-1}]\,. \tag{4.13}$$

Here we have used

$$u_{k+n}^{\eta'}(p) = u_k^\eta(p)\, u_n^{\eta'}(p)\,, \qquad \sum_l |u_k^\eta(p'+l)|^2 = 1\,, \qquad a_k^{-1} + a_n^{-1} L^{-2k} = a_{k+n}^{-1}\,, \tag{4.14}$$

and the periodicity of u_k^η under $2\pi L^k Z^d$. Here $\Delta^{\eta'}(p)$ denotes (4.4) with η' in place of η and with the same mass term $m^2(L^k\varepsilon)^2$.

Lemma 4.2. *The operator* (4.12) *satisfies* (4.7) *and* (4.8).

Proof.

$$\begin{aligned} D^{-1}(p) - (|p|^2 + m^2(L^k\varepsilon)^2)^{-1} &= a_n^{-1}L^{-2k} + |u_n^{\eta'}(p)|^2\Delta^{\eta'}(p)^{-1} \\ &\quad - (|p|^2 + m^2(L^k\varepsilon)^2)^{-1} + \sum_{l'\neq 0} |u_n^{\eta'}(p+l')|^2 \Delta^{\eta'}(p+l')^{-1}\,, \end{aligned} \tag{4.15}$$

when $l' \neq 0$, $\Delta^{\eta'}(p+l') \geqq C|p+l'|^2 \geqq CL^{2k}$. Therefore

$$\begin{aligned} |1 - |u_n^{\eta'}(p)|^2| &\leqq CL^{-2nd} \sum_{x,w\in B^n(y)} \sin^2 1/2p(x-w) \\ &\leqq CL^{-2nd} \sum_{\mu,\nu=1}^d p_\mu p_\nu \Big\{ \sum_{x,w\in B^n(y)} (x-w)_\mu (x-w)_\nu \Big\} \\ &\leqq CL^{-2nd} |p|^2 L^{nd} L^{n(d-1)} L^{-2(k+n)} L^{3n} \\ &\leqq CL^{-2k}|p|^2\,. \end{aligned} \tag{4.16}$$

Therefore using (4.7) for $\Delta^{\eta'}(p)$ and (4.16),

$$\begin{aligned} |D^{-1}(p) - (|p|^2 + m^2(L^k\varepsilon)^2)^{-1}| &\leqq CL^{-2k}\Big\{1 + \sum_{l'\neq 0} |u_n^{\eta'}(p+l')|^2\Big\} \\ &\leqq CL^{-2k}\,. \end{aligned} \tag{4.17}$$

To prove (4.8) we see that

$$\begin{aligned} D^{-1}(p) &\leqq a_n^{-1}L^{-2k} + C|p|^{-2} \sum_{l'} |u_n^{\eta'}(p+l')|^2 \\ &\leqq |p|^{-2}[C + a_n^{-1}|p|^2 L^{-2k}] \leqq C|p|^{-2}\,. \end{aligned} \tag{4.18}$$

Combining Lemmas 4.1 and 4.2 gives

Lemma 4.3.

$$|\Delta^{(k)}(p') - \Delta^{(k+n)}(p')| \leqq CL^{-2k}\Delta^{(k)}(p')\,. \tag{4.19}$$

Using the representation (4.2), we shall now establish the last bound in (3.89). The rest of Proposition 3.8 is simpler. We have the identity

$$\begin{aligned}
&(\partial_\alpha(x',y')\partial_\mu^{\eta'} a_{k+n} G_{k+n}^{\eta'} Q_{k+n}^*)(z) \\
&\quad = (2\pi)^{-d}\int dp' \sum_l \sum_m e^{i(p'+l+m)(x'-z)}|x'-y'|^{-\alpha}\{1-\exp[i(p'+l+m)(y'-x')]\} \\
&\qquad \cdot (\eta')^{-1}\{\exp[i\eta'(p'+l+m)_\mu]-1\}\{\Delta^{(k+n)}(p')u_{k+n}^{\eta'}(p'+l+m)\Delta^{\eta'}(p'+l+m)^{-1}\}\,,
\end{aligned} \tag{4.20}$$

where $l \in 2\pi Z^d$ is the same as in (4.2). Also $m \in 2\pi L^k Z^d$ and $|m_\mu| \leqq \pi L^k(L^n-1)$ for L odd. The corresponding expression for G_k^η is obtained from (4.20) by replacing x', y', η' by x, y, η and $(k+n)$ by (k), and taking $m=0$. Here $x' \in B^n(x)$ and $y' \in B^n(y)$, as in Proposition 3.8, so that $|x-x'|,\ |y-y'| \leqq \sqrt{d}\,L^{-k}$. We have the following bounds:

$$|u_{k+n}^{\eta'}(p'+l+m)| \leqq C\prod_{\mu=1}^{d}|p'_\mu|\,|(p'+l+m)_\mu|^{-1}\,, \tag{4.21}$$

$$|\Delta^{(k+n)}(p')\Delta^{\eta'}(p'+l+m)^{-1}| \leqq C(|p'|^2+m^2(L^k\varepsilon)^2)\,|p'+l+m|^{-2}\,. \tag{4.22}$$

To prove (4.22) we keep only the term $l=0$ in (4.5) with $k+n$ in place of k, and use $|u_{k+n}^{\eta'}(p')| \geqq C$ for $|p'_\mu| \leqq \pi$, $\Delta^{\eta'}(p') \leqq |p'|^2+m^2(L^k\varepsilon)^2$ and $\Delta^{\eta'}(p'+l+m) \geqq C|p'+l+m|^2$, which follow from (4.3) and (4.4). We will only need the weaker bound

$$\Delta^{(k+n)}(p')\Delta^{\eta'}(p'+l+m)^{-1} \leqq C|p'+l+m|^{-2}\,, \tag{4.23}$$

where C depends on d and m, since $L^k\varepsilon \leqq 1$. Also

$$(\eta')^{-1}|\exp[i\eta'(p'+l+m)_\mu]-1| \leqq C|(p'+l+m)_\mu| \tag{4.24}$$

and

$$|x'-y'|^{-\alpha}|1-\exp[i(p'+l+m)(y'-x')]| \leqq C|p'+l+m|^\alpha\,, \tag{4.25}$$

so the sum over l, m with $(l,m) \neq (0,0)$ is bounded by

$$\sum_{(l,m)\neq(0,0)} |p'+l+m|^{\alpha-1}\prod_\mu |p'_\mu|\,|(p'+l+m)_\mu|^{-1} \leqq C \quad \text{for} \quad \alpha<1\,, \tag{4.26}$$

uniformly in p', while the term with $l=m=0$ is bounded by $C|p'|^{\alpha-1}$, which is integrable over $|p'| \leqq \pi$.

We first bound the terms in (4.20) with $m \neq 0$ as follows:

$$\begin{aligned}
&|(4.20);\ m\neq 0| \\
&\quad \leqq C(2\pi)^{-d}\int dp' \sum_{m\neq 0}\sum_l |p'+l+m|^{\alpha-1}\prod_\mu |p'_\mu|\,|(p'+l+m)_\mu|^{-1} \\
&\quad \leqq CL^{-\gamma k}(2\pi)^{-d}\int dp' \sum_{m\neq 0}\sum_l |p'+l+m|^{\alpha-1+\gamma}\prod_\mu |p'_\mu|\,|(p'+l+m)_\mu|^{-1} \\
&\quad \leqq CL^{-\gamma k} \quad \text{for} \quad \alpha+\gamma<1\,.
\end{aligned} \tag{4.27}$$

To analyze the $m = 0$ term in (4.20), we successively replace each factor by the corresponding one in the expression for $(\partial_\alpha(x,y)\partial_\mu^\eta a_k G_k^\eta Q_k^*)(z)$ and bound the error. We must always be careful to keep enough negative powers of momentum so that the sum over l is bounded. First we have

$$\begin{aligned} &|\exp[i(p'+l)x] - \exp[i(p'+l)x']| \\ &\quad \leqq C|p'+l|^\gamma\,|x-x'|^\gamma \leqq CL^{-\gamma k}|p'+l|^\gamma\,. \end{aligned} \tag{4.28}$$

So keeping $\gamma + \alpha < 1$, the error produced by the above replacement is bounded by $CL^{-\gamma k}$. Next, we have

$$\begin{aligned} &|(\eta')^{-1}[\exp[i\eta'(p'+l)_\mu] - 1] - \eta^{-1}[\exp[i\eta(p'+l)_\mu] - 1]| \\ &\quad \leqq 2(\eta')^{-1}\sin^2 1/2\eta'(p'+l)_\mu + 2\eta^{-1}\sin^2 1/2\eta(p'+l)_\mu \\ &\qquad + |(\eta')^{-1}\sin\eta'(p'+l)_\mu - \eta^{-1}\sin\eta(p'+l)_\mu| \\ &\quad \leqq C|p'+l|^2L^{-k} \leqq C|p'+l|^{1+\gamma}L^{-\gamma k}\,. \end{aligned} \tag{4.29}$$

In fact the proof gives the bound $C|(p'+l)_\mu|^2L^{-k}$ for each μ, which is the form we use in Lemma 4.4.

For the Hölder derivatives, suppose $|x-y| \leqq 4\sqrt{d}\,L^{-k}$. Then

$$|x-y|^{-\alpha}|1-\exp[i(p'+l)(y-x)]| \leqq C|p'+l|^{\alpha+\gamma}|x-y|^\gamma \leqq CL^{-\gamma k}|p'+l|^{\alpha+\gamma}\,. \tag{4.30}$$

Furthermore, when $|x-y| > 4\sqrt{d}\,L^{-k}$, we have

$$||x-y| - |x'-y'|| \leqq 2\sqrt{d}\,L^{-k} \leqq 1/2\,|x-y|\,. \tag{4.31}$$

Therefore it follows easily that

$$\begin{aligned} &||x-y|^{-\alpha}\{1-\exp[i(p'+l)(y-x)]\} - |x'-y'|^{-\alpha}\{1-\exp[i(p'+l)(y'-x')]\}| \\ &\qquad \leqq CL^{-\gamma k}|p'+l|^{\alpha+\gamma}\,, \end{aligned} \tag{4.32}$$

where we have assumed $\alpha+\gamma < 1$ and $\gamma \leqq \alpha$. Next we need a lemma.

Lemma 4.4.

$$|u_k^\eta(p'+l) - u_{k+n}^{\eta'}(p'+l)| \leqq CL^{-\gamma k}|p'+l|^\gamma\,|u_k^\eta(p'+l)|\,. \tag{4.33}$$

Proof.

$$\begin{aligned} &u_{k+n}^{\eta'}(p'+l) - u_k^\eta(p'+l) \\ &\quad = u_k^\eta(p'+l)\prod_{\mu=1}^d[\eta'(e^{-i\eta'(p'+l)_\mu}-1)^{-1}] \\ &\qquad \cdot\Big[\prod_{\mu=1}^d\eta^{-1}(e^{-i\eta(p'+l)_\mu}-1) - \prod_{\mu=1}^d(\eta')^{-1}(e^{-i\eta'(p'+l)_\mu}-1)\Big]\,. \end{aligned} \tag{4.34}$$

By repeated use of the identities $xy - zw = 1/2(x-z)(y+w) + 1/2(x+z)(y-w)$ and $xy + zw = 1/2(x-z)(y-w) + 1/2(x+z)(y+w)$, we can write the difference inside the last bracket of (4.34) as a sum of 2^{d-1} terms. Each term is a product of d factors, at least one of which is the left-hand side of (4.29). So inserting the appropriate bounds, we get

$$\begin{aligned} |(4.34)| &\leqq C|u_k^\eta(p'+l)|\prod_{\mu=1}^d|(p'+l)_\mu|^{-1}L^{-\gamma k}\sum_{\nu=1}^d|(p'+l)_\nu|^{1+\gamma}\prod_{\mu'\neq\nu}|(p'+l)_{\mu'}| \\ &\leqq CL^{-\gamma k}|p'+l|^\gamma\,|u_k^\eta(p'+l)|\,, \end{aligned} \tag{4.35}$$

as required.

Therefore we can replace $u_{k+n}^{\eta'}(p'+l)$ by $u_k^{\eta}(p'+l)$ and bound the error. Lemma 4.3 allows us to replace $\Delta^{(k+n)}(p')$ by $\Delta^{(k)}(p')$. Finally, from (4.7)

$$|\Delta^{\eta'}(p'+l)^{-1} - \Delta^{\eta}(p'+l)^{-1}| \leqq CL^{-2k} \leqq CL^{-\gamma k}|p'+l|^{-2+\gamma}\,. \tag{4.36}$$

Therefore every term in (4.20) for $m = 0$ can be replaced, and so combining our bounds with Theorem 3.3 we deduce (3.89).

To prove Proposition 3.9 we use the representation (2.18) and Proposition 3.8. We also need convergence properties of the operator $C^{(k)}$ defined in (2.17). Introduce the operator

$$C_\Omega^{(k)}(s) = (s\Delta^{(k)} + (1-s)\Delta^{(k+n)} + aL^{-2}Q^*Q)^{-1}\,, \tag{4.37}$$

where $0 \leqq s \leqq 1$, $\Omega \subset Z^d$ is a rectangular parallelepiped composed of blocks of L^d sites, and where $\Delta^{(k)}$, $\Delta^{(k+n)}$ are defined with periodic boundary conditions on Ω. Using the general results of Chap. 5 of [Ba 4], $C_\Omega^{(k)}(s)$ has uniform exponential decay if the following conditions hold:

$$\text{(i)} \qquad C_\Omega^{(k)}(s)^{-1} \geqq \gamma_0 I\,, \tag{4.38}$$

$$\text{(ii)} \qquad |C_\Omega^{(k)}(s)^{-1}(x,y)| \leqq C\exp[-\delta_0|x-y|]\,. \tag{4.39}$$

The condition (4.39) is immediate, since Q^*Q is a short-range operator and $\Delta^{(k)}$ has uniform exponential decay (see Theorem 3.3). To prove (4.38), we use a Fourier representation on Ω. Since Ω is a torus, the allowed momenta satisfy $p'_\mu \in 2\pi|\Omega_\mu|^{-1}Z$, $|p'_\mu| \leqq \pi$, where Ω_μ is the dimension of Ω in the μth coordinate direction. So $\Delta^{(k)}$ has the representation

$$\langle\phi, \Delta^{(k)}\phi\rangle = |\Omega|^{-1}\sum_{p'} |\tilde{\phi}(p')|^2\Delta^{(k)}(p')\,, \tag{4.40}$$

where $|\Omega| = \prod_\mu |\Omega_\mu|$. It is easy to see that $\Delta^{(k)}(p') \geqq C|p'|^2$, and likewise for $\Delta^{(k+n)}(p')$. Furthermore, if Δ denotes the Laplacian on the unit lattice, $-\Delta(p') = \sum_\mu 4\sin^2(1/2p'_\mu) \leqq |p'|^2$, so that (4.40) gives

$$\langle\phi, \Delta^{(k)}\phi\rangle \geqq C\langle\phi, -\Delta\phi\rangle = C\sum_{\langle x,y\rangle\subset\Omega} |\phi(x)-\phi(y)|^2\,, \tag{4.41}$$

where the sum runs over the nearest neighbour pairs of Ω. The operator Q^*Q is the orthogonal projection, with respect to the inner product (2.2), onto the functions which are constant on each block of L^d sites. It does not commute with translations of Ω, so it has no representation of the form (4.40). Instead we use the Poincaré inequality on a block B: if $\sum_{x\in B}\psi(x) = 0$, then

$$\sum_{x\in B}|\psi(x)|^2 \leqq C\sum_{\langle x,y\rangle\subset B}|\psi(x)-\psi(y)|^2\,, \tag{4.42}$$

since the Laplacian on B with free boundary conditions is bounded below by a positive constant on the functions of mean zero.

We apply this to $\psi = (I - Q^*Q)\phi$, which has mean zero on each block and the same differences as ϕ within each block, and sum over the blocks, dropping the pairs which join different blocks. Therefore we have

$$\begin{aligned}&\langle\phi, (s\Delta^{(k)} + (1-s)\Delta^{(k+n)} + aL^{-2}Q^*Q)\phi\rangle\\ &\quad\geqq C\|(I-Q^*Q)\phi\|^2 + aL^{-2}\|Q^*Q\phi\|^2 \geqq \gamma_0\langle\phi,\phi\rangle\,,\end{aligned} \tag{4.43}$$

for some $\gamma_0 > 0$, as required. We now prove the required convergence properties of $C^{(k)}$.

Lemma 4.5.

$$|C^{(k)}(x,y) - C^{(k+n)}(x,y)| \leqq CL^{-k}e^{-\delta_0|x-y|}\,. \tag{4.44}$$

Proof. We prove (4.44) in a finite volume Ω with periodic boundary conditions; the result for free boundary conditions then holds by continuity. First,

$$\begin{aligned} C^{(k)}(x,y) - C^{(k+n)}(x,y) &= \int_0^1 ds\, d/ds\, C_\Omega^{(k)}(s)(x,y) = 1/2 \int_0^1 ds \\ &\cdot \Big\{ \langle \phi(x)\phi(y); \sum_{z,w\in\Omega} \phi(z)[\Delta^{(k+n)}(z,w) - \Delta^{(k)}(z,w)]\phi(w)\rangle_s \Big\}\,, \end{aligned} \tag{4.45}$$

where the expectation $\langle\cdot\rangle_s$ is taken with respect to the covariance $C_\Omega^{(k)}(s)$. Explicit computation gives

$$\sum_{z,w\in\Omega} \{C_\Omega^{(k)}(s)(x,z)[\Delta^{(k+n)}(z,w) - \Delta^{(k)}(z,w)]C_\Omega^{(k)}(s)(w,y)\}\,. \tag{4.46}$$

Combining Lemma 4.3 and the uniform exponential decay of $\Delta^{(k)}$ and $C_\Omega^{(k)}(s)$ gives

$$\begin{aligned} |(4.46)| &\leqq CL^{-k} \sum_{z,w\in\Omega} \exp[-\delta_0|x-z| - \delta_0|z-w| - \delta_0|w-y|] \\ &\leqq CL^{-k}\exp[-\delta_0|x-y|]\,, \end{aligned} \tag{4.47}$$

where the bound CL^{-2k} on the kernel of $\Delta^{(k+n)} - \Delta^{(k)}$ follows from Lemma 4.3, and we have taken its geometric mean with the exponential decay of Theorem 3.3, which reduces the rate δ_0 of (4.47), reduced once more in the last line by the sum over z, w.

We are finally ready to prove Proposition 3.9. We will prove the first bound in (3.91); the other bounds follow in the same way. For $j > 0$, we have the expansion

$$\begin{aligned} G_{(j)}^{\eta'}(x',y') = a_{j+n}^2 (L^j\eta)^{-4} &\sum_{z,w\in L^j\eta Z^d} (L^j\eta)^{2d} G_{j+n}^{\eta'} Q_{j+n}^*(x',z) \\ &\cdot C^{(j+n),L^j\eta}(z,w) Q_{j+n} G_{j+n}^{\eta'}(w,y')\,. \end{aligned} \tag{4.48}$$

We now replace $a_{j+n}Q_{j+n}G_{j+n}^{\eta'}(w,y')$ by $a_jQ_jG_j^\eta(w,y)$ in (4.48); using Proposition 3.8 and the scaling of operators (2.21), the error from this replacement is bounded by

$$\begin{aligned} &C(L^j\eta)^{-4} \sum_{z,w} (L^j\eta)^{2d} (L^j\eta)^{2-d} \exp[-\delta_0 (L^j\eta)^{-1}|x'-z|](L^j\eta)^{2-d} \\ &\quad \cdot \exp[-\delta_0(L^j\eta)^{-1}|z-w|] \cdot L^{-\gamma j}(L^j\eta)^{2-d} \exp[-\delta_0(L^j\eta)^{-1}|w-y'|] \\ &\leqq C(L^j\eta)^{2-3d} L^{-\gamma j} \sum_{z,w} (L^j\eta)^{2d} \exp[-\delta_0(L^j\eta)^{-1}\{|x'-z| + |z-w| + |w-y'|\}] \\ &\leqq C(L^j\eta)^{2-d} L^{-\gamma j} \exp[-\delta_0(L^j\eta)^{-1}|x'-y'|] \\ &\leqq CL^{-\gamma k}(L^j\eta)^{2-d-\gamma} \exp[-\delta_0(L^j\eta)^{-1}|x-y|]\,. \end{aligned} \tag{4.49}$$

Clearly we can replace $a_{j+n}G_{j+n}^{\eta'}Q_{j+n}^*(x',z)$ by $a_jG_j^\eta Q_j^*(x,z)$, and $C^{(j+n),L^j\eta}(z,w)$ by $C^{(j),L^j\eta}(z,w)$, and bound the error in the same way. Hence we deduce the required bound. When $j = 0$, we bound each term separately using Proposition 3.7, and write the factor η^{2-d} as $L^{-\gamma k}L^{-k(2-d-\gamma)}$.

Finally, we must establish Propositions 3.7 and 3.9 for $G_{(K)}^\eta = C^\eta - G_K^\eta$. It can be written

$$C^\eta - G_K^\eta = C^\eta(a_K(L^K\eta)^{-2}Q_K^*Q_K)G_K^\eta = (L^K\eta)^{-2}C^\eta Q_K^* a_K Q_K G_K^\eta\,. \tag{4.50}$$

Furthermore we obtain the Fourier representation for $C^\varepsilon Q_K^*$ from (4.2) by setting $k = K$, $\eta = L^{-K}$, dividing by a_K and letting $a_K \to 0$, so that $\Delta^{(K)}(p')a_K^{-1} \to 1$:

$$C^\varepsilon Q_K^*(x,y) = (2\pi)^{-d} \int_{|p'| \leqq \pi} dp' \sum_l e^{i(p'+l)(x-y)} u_K^\varepsilon(p'+l) \Delta^\varepsilon(p'+l)^{-1}\,, \tag{4.51}$$

where $u_K^\varepsilon(p) = \prod_{\mu=1}^d \{(\exp[-ip_\mu] - 1)\varepsilon(\exp[-i\varepsilon p_\mu] - 1)^{-1}\}$ and $l \in 2\pi Z^d$ with $|l_\mu| \leqq \pi/\varepsilon$. We can extend $\Delta^\varepsilon(p)^{-1}$ to an analytic function for $|\operatorname{Im} p_\mu| \leqq m/2$, each $\mu = 1, \ldots, d$. The function $u_K^\varepsilon(p)$ also has such an extension, and therefore we may extract a decay factor $\exp[-\delta_0|x-y|]$ from (4.51). The sum over l is bounded as before. In the same way, we get exponential decay for $(\partial_\alpha(x,y)\partial_\mu^\varepsilon C^\varepsilon Q_K^*)(z)$. We can easily extend Proposition 3.8 to include (4.51), since the integrand is even simpler than in (4.2). Finally, Proposition 3.9 holds for $G^\eta_{(K)}$ from the convergence of $C^\eta Q_K^*$ and $a_K Q_K G_K^\eta$, and the scalings of the operators.

5. Supplementary Results

5.1. The Renormalization Transformation with $L = 5$. In the proofs of the bounds (3.4)–(3.5) in [Ba 1–3] the block size L is a parameter, and the proofs are written for L sufficiently large; in this paper $L = 5$ (Sect. 2.2). We describe here, in the notation of this paper, the objects of those proofs through which the value of L enters, and the changes which the value $L = 5$ requires. The changes affect the proofs and not the statements: the bounds (3.4)–(3.5), the ultra-violet stability bound of Theorem 2.1(ii) and the decay and regularity results of [Ba 4] summarised in Theorem 3.3, whose constants do not depend on the scale, are unchanged.

At the step k the fields live on the unit lattice $T_1^{(k)}$, and the large blocks of Sect. 2.3 have side M_b in its units.[1] The small field region is built from the function

$$r(\eta) = R(1 + \ln \eta^{-1})^\varrho\,, \qquad \varrho = 2\,, \tag{5.1}$$

decreasing in η, which measures distances at the scale $L^k\varepsilon$: a sequence of regions $\Lambda_j^{(k)}$, $j = -1, 0, 1, \ldots$, each a union of large blocks, is defined so that the complement of $\Lambda_{j+1}^{(k)}$ is the union of the large blocks within distance $r(L^k\varepsilon)$ of the complement of $\Lambda_j^{(k)}$. So $\Lambda_{j+1}^{(k)}$ lies one corridor inside $\Lambda_j^{(k)}$, and a point of $\Lambda_{j+1}^{(k)}$ lies at distance exceeding $r(L^k\varepsilon) - 1$ from the complement of $\Lambda_j^{(k)}$, the distance from a lattice point to a set of large blocks differing by less than one unit from the distance between sets of large blocks. Distances between sets of large blocks are taken in the supremum norm, and the depth of a point in a region is its distance from the complement of the region.[2] The regions of the step $k-1$, read on the lattice of the step k, are written $\Lambda_j^{(k-1)\prime}$; the passage loses one block of the lattice $T_1^{(k-1)}$, of side $L \leqq M_b$ in its units, so that a corridor of the step $k-1$ has width exceeding

$$w_k = (r(L^{k-1}\varepsilon) - M_b)/L \tag{5.2}$$

in the units of $T_1^{(k)}$.[3] The propagator of the step k is $G_k(\Omega, A^{(k)})$ on the domain $\Omega = B^k(\Lambda_2^{(k-1)\prime})$, the background field $A^{(k)}$ being regular in the sense of Definition 3.2 with $|\partial^\eta A^{(k)}| \leqq Cp(L^k\varepsilon)$, and the vertices of the perturbation expansion lie in $B^k(\Lambda_7^{(k-1)\prime})$, five corridors inside Ω.[4] On a domain Ω which is any union of large blocks the bounds (3.7)–(3.9), and the bounds with

[1] M_b is the integer M of Sect. 2.3; the large blocks are the "big blocks" of [Ba 1–3].

[2] (2.7)–(2.8) of [Ba 2], where ϱ is written r; the regions are numbered so that $\Lambda_{j+1} \subset \Lambda_j$, as in Lemma 2.4 of [Ba 2].

[3] The primed regions are defined in Sect. 2 of [Ba 2].

[4] Sect. 1 of [Ba 3]; the bound on $\partial^\eta A^{(k)}$ is (2.60) of [Ba 2], Lemma 2.3 there, and the localization of the vertices comes from the factors (2.52) of [Ba 2].

the additional factors (3.10)–(3.11) on the difference of the propagators of two nested domains, hold at points at depth at least R_1 in Ω, where R_1 is a constant depending on d, a and M_b, with no condition on the shape of Ω; we call them the propagator bounds at depth R_1.[5]

The value of L enters these proofs and the proof of Theorem 3.1 in two ways. The first is through constants, rates of decay, convergent sums over the scales, and inequalities of the form

$$CL^{-\alpha} \leqq c \tag{5.3}$$

between a power of L^{-1} and constants which do not depend on L. Of this kind are also the two conditions on the exponents p, p_1 and the constants b_0, b_1 of the large field bounds on the scalar and the vector field, which bound b_0 below once p, p_1, b_1 and L are chosen, the condition that the large block size be sufficiently large, and the conditions $\varrho > 1$ and $R > R_1$ on (5.1).[6] All of them are satisfiable for every $L \geqq 2$, and none is a condition of the form "L sufficiently large". Appendix B lists them, with the values at $L = 5$ of the quantities through which L enters, the ratios (B.1), the decay rates and the scale sums.

The second is through four statements of containment between regions of different scales, which do not hold at $L = 5$, and through a regular partition of the lattice into cubes of side between $r(L^k\varepsilon)$ and $2r(L^k\varepsilon)$, which does not exist on some of the tori (2.22). An object of size $\alpha r(L^k\varepsilon)$ built at the step k is required to fit inside β corridors of the step $k-1$, of total width exceeding βw_k; since $r(L^{k-1}\varepsilon)/r(L^k\varepsilon) \to 1$ as $L^k\varepsilon \to 0$, this holds at every small scale only if $\alpha L \leqq \beta$ (Appendix B). We replace the four containments by three changes and the partition by a new one, as follows (Fig. 6).

First, the cube step in the proof of the basic estimate of [Ba 3]. That estimate concerns a renormalized class G_{ren} of graphs of the expansion of Sect. 3.3 at the step k, the graphs of one order obtained from one another by replacing divergent subgraphs by their counterterm vertices, each vertex v localized in a unit cube $\square(v)$ of the lattice and each external leg carrying an external field. The sum $E(G_{\mathrm{ren}})$ of the values of the graphs of the class satisfies

$$|E(G_{\mathrm{ren}})| \leqq C\,[e(L^k\varepsilon)^{2-d/2}]^{\alpha}[\lambda(L^k\varepsilon)^{4-d}]^{\beta}\exp[-\delta_0\, d(\{\square(v)\})]\,\|h\phi_{\mathrm{ext}}\|_{1,\alpha_0}\|h'A_{\mathrm{ext}}\|_{1,\alpha_0}\,, \tag{5.4}$$

where α, β are the numbers of factors e, λ of the class, $d(\{\square(v)\})$ is the length of the shortest tree joining the cubes of the vertices, ϕ_{ext}, A_{ext} are the external scalar and vector fields, functions of the positions of all the external legs, h, h' are functions equal to one on the cube of the vertex of a leg and to zero outside a neighbourhood of it, and

$$\|f\|_{1,\alpha_0} = \sup_x |f(x)| + \sup_b |D_{B,\mu}f(b)| + \sup_{x,x',\mu} |x-x'|^{-\alpha_0}|U(B(\Gamma_{x,x'}))D_{B,\mu}f(x') - D_{B,\mu}f(x)| \tag{5.5}$$

is the Hölder norm of a field on the unit lattice, taken with the background B of the propagator for scalar fields and with $B = 0$ for vector fields, and extended variable by variable to functions of several points; δ_0 depends on d only and C on α_0 and $\bar{n}$.[7]

The proof proceeds as follows. The lines joining vertices whose cubes do not meet, and the differences of propagators of nested domains, are moved into the external fields, which the bounds (3.8)–(3.9) permit; the localization cubes of a connected component of a class then lie in a cube of side $\ell_{\bar{n}}$ depending on $\bar{n}$ only. This cube is enclosed in two concentric cubes $\square_1 \subset \square$ of large blocks, of sides $r(L^k\varepsilon)$ and $3r(L^k\varepsilon)$. The propagator $G_k(\Omega, B)$ is written $G_k(\square, B) + \delta G_k(\Omega, \square, B)$, the difference being treated as an external field; the field B is replaced on $\square$ by the constant $B_0 = B(c)$, c the centre of $\square$, with $|B - B_0| \leqq O(r(L^k\varepsilon)p(L^k\varepsilon))$, and the

[5] Propositions 2.1–2.3 of [Ba 1], the inequalities (2.24)–(2.26) there, and Proposition 2.2 of [Ba 2], (2.58). The constant is written R_0 in those papers, where it is also the lower bound imposed on R in (2.7) of [Ba 2].

[6] Sect. 3 of [Ba 1], (3.12)–(3.13) and (3.53)–(3.54); Sect. 2 of [Ba 1]; (2.7) of [Ba 2].

[7] Proposition 1 of [Ba 3], the inequality (1.33) there; the norm is (1.32) there. The classes are defined in Sect. 2 of [Ba 3], and the hypotheses on the domains and on the external vector field are stated in Sect. 1 there.

difference $B' = B - B_0$ is expanded; B_0 is removed by a gauge transformation, and $G_k(\square, 0)$ is replaced by the propagator $G_k(0)$ of the torus, the difference being again an external field.[8] The cube $\square$ must lie in Ω, since the Neumann propagator $G_k(\square, B)$ is built from the bonds of $\square$, on which B is defined only if $\square \subset \Omega$. The vertices of a class lie at depth exceeding $5w_k$ in Ω, five corridors of the step $k-1$, so that the cube of side $3r(L^k\varepsilon)$ fits at every small $L^k\varepsilon$ only if $\frac{3}{2}L \leqq 5$, which fails at $L = 5$.

The cubes are replaced by concentric cubes of sides

$$r^\flat\,, \qquad 3r^\flat\,, \tag{5.6}$$

where $r^\flat$ is a constant, a multiple of M_b with

$$r^\flat \geqq \max\{3\ell^\sharp,\ R_1 + 2,\ \ell^\sharp + 2M_b\}\,, \qquad \ell^\sharp = 3\ell_{\bar{n}} + 4N\,. \tag{5.7}$$

Here N is the number of insertions on a line in the expansion in B' to the order $\bar{n}$, which produces at most N new vertices on a line, each lying within two units of the preceding one, so that $\ell^\sharp$ depends on $\bar{n}$ only (Appendix D). The cube of side $r^\flat$ is a cube of large blocks containing the cube of side $\ell^\sharp$ about the localization cube; it exists since $r^\flat \geqq \ell^\sharp + 2M_b$, the large blocks meeting a cube of side $\ell^\sharp$ spanning at most $\ell^\sharp + 2M_b$ in each direction, and its centre lies within $\frac{1}{2}(\ell^\sharp + M_b)$ of every old vertex, the centres of the cubes of large blocks of side $r^\flat$ forming a lattice of spacing M_b. The cube of side $3r^\flat$ is concentric with it.

With these cubes the proof goes through, with a constant depending on α_0, $\bar{n}$, d, a, M_b and R_1 of the same form as before, for every class whose old vertices, the vertices present before the expansion in B', lie at depth exceeding

$$D^\flat = \tfrac{3}{2}r^\flat + 2M_b + \ell^\sharp \tag{5.8}$$

in the domain of the propagator; this is Lemma D.3. The bound on $|B - B_0|$ becomes $O(r^\flat p(L^k\varepsilon))$, smaller than before, and the norms (5.5) of $\delta G_k(\Omega, \square, B)$ on the inner cube are bounded by $C\exp[-\delta_0(r^\flat - 2)]$ by the propagator bounds at depth R_1 for the pair $\square \subset \Omega$, the points and the sources lying at depth at least $r^\flat - 2 \geqq R_1$ in $\square$.

The basic estimate is stated under the hypothesis that the external vector field is supported at depth exceeding $2r(L^k\varepsilon)$ in the domain of the propagator, and it is used twice.[9] Where it is stated, the vertices of a class lie within one unit of $B^k(\Lambda_7^{(k-1)\prime})$ or of the support of the external field, the former at depth exceeding $5w_k - 1 = r(L^{k-1}\varepsilon) - M_b - 1 \geqq R - M_b - 1$ in Ω. In its second use, on the domain $B^k(\Lambda_0^{(k)})$ after the translation of the vector field at the step $k+1$, the classes are the terms of the expansion of the normalization factors of the earlier steps in the field $\theta_{k+1}A'^{(k)} + \tilde{B}^{(k)}$, so that every connected component of a class carries a leg of that field, and its old vertices lie within $\ell_{\bar{n}}$ of a unit cube of the support of that field (Appendix D), at depth exceeding $2r(L^k\varepsilon) - 3 - M_\theta - 2\eta - \ell_{\bar{n}} \geqq 2R - 3 - M_\theta - 2\eta - \ell_{\bar{n}}$ in $B^k(\Lambda_0^{(k)})$, M_θ being the width of the slice on which the function θ_{k+1} varies (Appendix B). The λ- and δm^2-vertices carry no leg of that field and are covered by the bound on their component. So the two lower bounds

$$R \geqq D^\flat + M_b + 1\,, \qquad R \geqq M_\theta + \ell_{\bar{n}} + 2 \tag{5.9}$$

suffice for both uses, and $D^\flat$ depends on $\bar{n}$, d, R_1 and M_b only. The basic estimate therefore holds with its hypothesis on the external field replaced by the condition that the old vertices of a class lie at depth exceeding $D^\flat$ in the domain of the propagator, which is all that its proof uses (Lemma D.3).

[8] Sect. 3 of [Ba 3], pp. 432–433.

[9] The hypothesis is stated in Sect. 1 of [Ba 3], p. 412; the second use is at (2.89) of [Ba 2], on the domain $B^k(\Lambda_0^{(k)})$ of (2.80) there, the classes being the terms of the expansion of the factors (2.52) of [Ba 2].

Second, the cut-off in the background field. The background field of the step k is $A^{(k)}(x) = a_k \sum_y \zeta^{(k)}(x,y)(G_k Q_k^*)(x,y)A(y)$, where A is the average field of the previous step and the cut-off $\zeta^{(k)}(\cdot, y)$ equals one within $\frac{1}{2}r(L^k\varepsilon)$ of y, vanishes at distance $r(L^k\varepsilon) - 2M_b$ from it and has first differences bounded by one.[10] The bounds on the background field, $A^{(k)}(x) = A(y) + O(p(L^k\varepsilon))$ for $x \in B^k(y)$ and $|\partial^\eta A^{(k)}| \leqq Cp(L^k\varepsilon)$ on $B^k(\Lambda_2^{(k-1)\prime})$, use the restrictions (3.2) on A at the points of the support of $\zeta^{(k)}(\cdot, y)$, which must therefore lie inside the region $\Lambda_{-1}^{(k-1)\prime}$ on which they hold; and in the split of the background field at the step $k+1$ the support of $\zeta^{(k+1)}$ about the support of θ_{k+1} must lie inside the region of the previous scale, which at $L = 5$ it does only if the radius of the support is at most $\frac{2}{5}r(L^{k+1}\varepsilon)$.[11] Here M_θ, the width of the slice on which θ_{k+1} varies, is measured in the units of $T_1^{(k)}$, which only strengthens the conditions below. The cut-off is replaced by one with

$$\begin{aligned} \zeta^{(k)}(x) = 1 \quad \text{for} \quad |x-y| \leqq \tfrac{1}{5}r(L^k\varepsilon)\,, \\ \operatorname{supp}\zeta^{(k)} \subset \{x:\ |x-y| < \tfrac{2}{5}r(L^k\varepsilon) - 2M_b\}\,. \end{aligned} \tag{5.10}$$

We choose $R \geqq 10(M_b + 1)$, a further lower bound on R (Appendix B); since $r(L^k\varepsilon) \geqq R$, the two radii differ by $\frac{1}{5}r(L^k\varepsilon) - 2M_b \geqq 2$, and at $L = 5$ the support of $\zeta^{(k+1)}$ about a point within M_θ of $B^k(\Lambda_2^{(k)})$ reaches at most $\frac{2}{5}r(L^{k+1}\varepsilon) - 2M_b + M_\theta + 1$ beyond $\Lambda_2^{(k)}$, all lengths in the units of $T_1^{(k+1)}$, where M_θ is less than M_b by the definition of θ_{k+1}.[12] This is at most $\frac{2}{5}r(L^k\varepsilon) - \frac{2}{5}M_b = 2w_{k+1}$, the width of two corridors of the step k in the units of the next step, r decreasing with its argument, while the support of $\zeta^{(k)}$ about a point of $\Lambda_2^{(k-1)\prime}$ lies within the three corridors between $\Lambda_2^{(k-1)\prime}$ and $\Lambda_{-1}^{(k-1)\prime}$, since $\frac{2}{5}r(L^k\varepsilon) - 2M_b \leqq \frac{3}{5}(r(L^{k-1}\varepsilon) - M_b) = 3w_k$.

The error terms in the bounds on the background field, which carry the decay from the plateau of $\zeta^{(k)}$, then decay as $e^{-\delta r(L^k\varepsilon)/5}$, still faster than every power of $L^k\varepsilon$; the proof is otherwise unchanged.

Third, the boxes in the proof of the corresponding bounds on the scalar field. For $y \in \Lambda_7^{(k-1)\prime}$ and $x \in B^k(y)$ the field $\phi^{(k)} = a_k G_k(\Omega, A^{(k)}) Q_k^*(A^{(k)})\Lambda_6\phi$, $\Lambda_6\phi$ the restriction of the scalar field ϕ of the previous step to $\Lambda_6^{(k-1)\prime}$, satisfies $\phi^{(k)}(x) = U(A^{(k)}(\Gamma_{x,y}))\phi(y) + O(p(L^k\varepsilon))$, and its covariant derivative on the bonds of $B^k(\Lambda_7^{(k-1)\prime})$ is $O(p(L^k\varepsilon))$; this is Lemma D.1.[13] The proof uses two auxiliary boxes, the large blocks of $\Lambda_7^{(k-1)\prime}$ within $2r(L^k\varepsilon)$ and $4r(L^k\varepsilon)$ of the large block of y, and applies the propagator bounds at depth R_1 to them as to full boxes at depth in a rectangular parallelepiped, which they are not near the boundary of $\Lambda_7^{(k-1)\prime}$; an untruncated box of radius $4r(L^k\varepsilon)$ inside the region Ω would require $4L \leqq 5$, which fails for every block size $L \geqq 2$. We replace them by a single box of large blocks of radius $\min\{4r(L^k\varepsilon),\ 5w_k - 2M_b\}$ about the large block of y, a rectangular parallelepiped inside the region, with the source restricted to the concentric cube of half the radius about y; at $L = 5$ the radius is at least $r(L^k\varepsilon) - 3M_b$, since $5w_k = r(L^{k-1}\varepsilon) - M_b \geqq r(L^k\varepsilon) - M_b$.

On this box the proof is repeated in Appendix D: it uses the decay of the propagators and of their differences on nested domains and the expansion about the constant value of the field at the block, and it gives the two bounds with constants independent of ε, of the scale and of the position of the block relative to the boundary of the region. The conditions on R and ε_0 under which the box and the partition below exist are listed in Appendix B.

Fourth, the partition. The combinatorial estimate which bounds the number of sequences of regions uses a regular partition of $T_1^{(k)}$ into a lattice of cubes of large blocks with side in

[10] (2.44) of [Ba 2].

[11] Lemma 2.3 and the split (2.89) of [Ba 2]; the restrictions are (2.55) there.

[12] Sect. 2 of [Ba 2], after (2.44).

[13] Lemma 2.4 of [Ba 2], (2.65)–(2.66) there.

$(r, 2r)$, $r = r(L^k\varepsilon)$.[14] It is replaced by a partition of each side of $T_1^{(k)}$ into integer multiples of M_b lying in

$$(r,\ 2r + 3M_b]\,. \tag{5.11}$$

Such a partition exists since the side $2L_\mu L^{K-k}$ of $T_1^{(k)}$, a multiple of M_b because $L_\mu M^{-1} \in N$ (Sect. 2.3; the weaker condition $2L_\mu M^{-1} \in N$ would suffice here), exceeds $r(L^k\varepsilon) + M_b$ for $k \leqq K - J$ and J sufficiently large; it suffices to split the side into multiples of M_b lying in $[n_0M_b, 2n_0M_b]$, $n_0 = [r/M_b] + 1$.

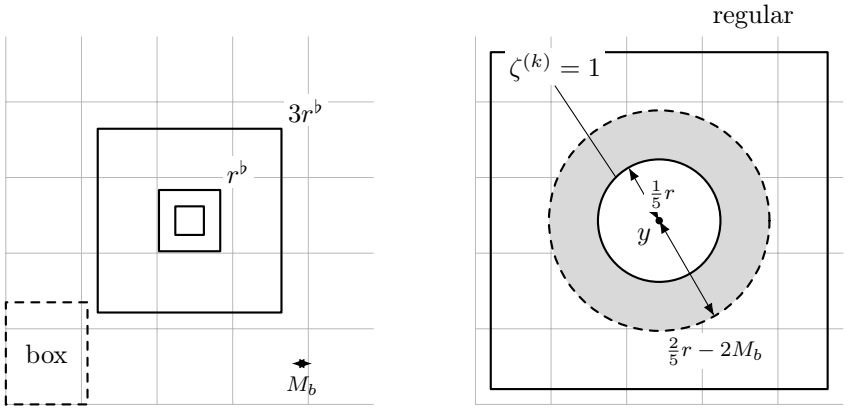


Fig. 6. The regions of Sect. 5.1 at $L = 5$ on the lattice of large blocks of side M_b: (a) the cubes of sides $r^\flat$, $3r^\flat$ about the localization cube of a class, and one box of the partition (dashed); (b) the cut-off $\zeta^{(k)}$ about y, equal to one in the white disc, with its support (shaded) inside the region where the vector field is regular; $r = r(L^k\varepsilon)$

The statements are unchanged: the bounds (3.4)–(3.5),[15] and the basic estimate (5.4), whose classes contain lines of scales below $L^k\varepsilon$ only, for the renormalised classes with at least one external leg or curvature vertex and for the vacuum classes of non-positive degree. Sources do not occur in [Ba 3]; the classes with a source vertex and an external leg are treated in Sect. 3.5.

The field-free class of order e^2 linear in h is not bounded uniformly in ε. By explicit computation of the one-loop graphs of Sect. 3.3, the scalar self-energy Σ, the one particle irreducible two-point function, which enters the inverse propagator additively, satisfies

$$\Sigma(p) - \Sigma(0) = -\tfrac{1}{4}e^2|p| + O(e^2(m + \mu_0)) \quad \text{for} \quad m, \mu_0 \ll |p| \ll \varepsilon^{-1}\,. \tag{5.12}$$

Hence for $N = 2$ the value of the corresponding term of the perturbation expansion on εZ^d is, with a constant depending on m and μ_0,

$$(4\pi^2)^{-1}e^2 \ln \varepsilon^{-1} \int h + O(1)\,. \tag{5.13}$$

Here $\int h$ denotes $\sum_x \varepsilon^d h(x)$. The class is treated in Sect. 2.1, through the constant a_ε of (2.8), and in Appendix A.

The polynomial E_1 of (3.53) is defined with the full propagators C^η. Its connected vacuum graphs of positive degree, those of order $\alpha + 2\beta \geqq 8$ in the notation of (3.50), contain lines of all scales, and for some of them the inequality (3.98) fails (Sect. 3.5). For the graph of order

[14] (3.43)–(3.45) of [Ba 2].

[15] The lower bound (1.14) of [Ba 1] and the upper bound (1.3) of [Ba 2].

λ^5 with vertices a, b, c, d, e and lines $ab, ab, ac, bc, ad, bd, ce, ce, de, de$, which has no divergent subgraph, the terms with at least one line of index $\geqq k$ sum, in absolute value, to at least

$$c\lambda^5 m^{-2} (L^k \varepsilon)^3 \tag{5.14}$$

per site of $T_1^{(k)}$ for $k \leqq K - J$: the full graph has the positive propagator C^η on every line and no divergent subgraph, converges as $\varepsilon \to 0$ to a positive multiple of $\lambda^5 m^{-2}$ per unit volume, by dominated convergence with the bounds of Appendix D, and is positive at every ε, its terms being positive, so that it exceeds a positive constant depending on m for every $\varepsilon \leqq 1$, while the term with all lines of index below k is at most $C\lambda^5 (L^k \varepsilon)^5$ per site by the kernel bounds of Theorem 3.3 and the power counting of Sect. 3.4, every subgraph having positive degree and the five vertices (3.34) each carrying the running coupling $\lambda L^k \varepsilon$; the lower bound holds for $k \leqq K - J$ with J depending on m as in Sect. 5.3. So the basic estimate (5.4), whose right side carries the power $(L^k \varepsilon)^{o/2}$ of the running couplings (5.21), o the order of the class, does not extend in general to the terms of these graphs with lines of index $\geqq k$. These terms are constants independent of the fields and are removed in the normalized action $S_N^{(k),1}$ of Sect. 3.2. (3.98) holds for the vacuum classes of non-positive degree, those of order $\alpha + 2\beta \leqq 6$.

Three further points need modification.

The first is the count. With the partition into boxes of large blocks described above, each box being a union of whole large blocks with side in (5.11), the bound on the number of boxes meeting the large field region, 3^d times the number of large blocks on which a field is large, and the bound on the sum over the configurations of that region hold with constants changed by factors depending on d only, the ratio of the sides of a box being at most $2 + 3M_b / r(L^k \varepsilon) \leqq 3$ for $R \geqq 3M_b$, and the combinatorial estimate follows as before.[16]

The second is the estimate

$$\delta m^2 = O(\varepsilon^{-1}) \quad \text{for each graph of (3.52)}\,, \tag{5.15}$$

the bound on the counterterm of the kth step, and the bounds on the counterterm vertices used in the large field estimates, all of which follow from the explicit form (3.51) of the counterterm by power counting at each scale.[17] For the two-loop graphs with the one-loop counterterm inserted, the tadpoles with a λ-tadpole or a seagull tadpole on their loop and their insertions cancel exactly by (3.51), the tadpole with the vertex (3.38), $m = 2$, vanishing at zero momentum, since for a constant field ϕ at zero background $D\phi = 0$, and its insertion being bounded uniformly in ε by the summation by parts of Appendix A applied to the inserted line.

The same power counting does not bound the constant E_{fin}, the difference between E_1 and the sum E_K of the same graphs, of orders at most $\bar{n}$, with the block spin propagators of the last step at zero background in place of the full propagators C_s, C_v on every line, the coefficients δm^2 of the counterterm vertices, which are not propagators, being unchanged.[18] With the coefficients recomputed from the block spin propagators the difference $E_1 - E_K$ would not be bounded uniformly in ε (Appendix D). The differences between the full and the block spin propagators are bounded kernels, uniformly in ε (by (D.7) and Theorem 3.3), but the other lines of a graph of E_{fin} carry propagators whose values at coinciding points are of order ε^{-1}, so that the bound $|E_{\text{fin}}| \leqq C|T|$ is not obtained graph by graph.

Writing each full propagator as the block spin propagator plus the difference and expanding, E_{fin} becomes a sum over connected skeleton graphs whose nodes are renormalized classes at the last step, with the block spin propagators on their lines and with external legs, and whose edges carry the difference kernels; it is the sum over all graphs of a class, and not a single piece of a graph, which the basic estimate bounds. Each node is bounded by the basic estimate

[16] (3.45), the first inequality of (3.46), and (3.42) of [Ba 2].
[17] (1.28) of [Ba 3]; (2.11)–(2.13) of [Ba 2].
[18] (1.25)–(1.26) of [Ba 3], which we read in this way.

at the last step and the edges are summed by the row sums of the block covariance in the representation (D.7) of the difference kernels, which gives $|E_{\rm fin}| \leqq C|T|$ uniformly in ε, as (2.24) requires (Lemma D.2); at order λ, for instance, the figure-eight and the tadpole with $\delta m^2(x;0,1)$ inserted leave, up to sign, $\lambda N(N+2)(C_s - G_{K-J}(0))(x,x)^2$ per site in $E_1 - E_{K-J}$. $E_{\rm fin}$ is a constant independent of the fields and is carried through the induction as in [Ba 3].

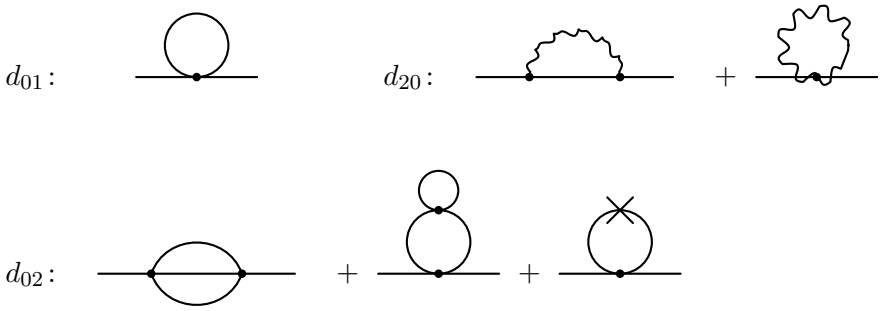
The third is the proof of the basic estimate, which is sketched in [Ba 3]. The sketch goes through with the changes above and with the expansion order $\bar{n} = 13$ of Appendix A; the only places where the changes enter are the cube step, treated in Lemma D.3, and the partition.

5.2. The Mass Counterterm. The counterterm $\delta m^2(x)$ of (3.50)–(3.51) on the torus T does not depend on x, the model (2.4) on T being translation invariant. As the torus grows it converges, at fixed ε, to the counterterm defined by the same equations with the propagators of εZ^d. Every coefficient of the counterterm, (5.16) below, is a finite combination of sums over one or two momenta of T of continuous periodic functions, all masses being positive, and these sums are Riemann sums of the corresponding integrals over the Brillouin zone and its square. The latter counterterm is the one used on the cubes of [II].

We write $\delta m^2_\varepsilon(M)$ for it, exhibiting the dependence on the scalar mass M, so that $\delta m^2_\varepsilon = \delta m^2_\varepsilon(m)$. By (3.50)–(3.51) and the remark following (3.50),

$$\delta m^2_\varepsilon(M) = \lambda d_{01} + e^2 d_{20} + \lambda^2 d_{02} + e^2\lambda d_{21} + e^4 d_{40}\,, \tag{5.16}$$

where $d_{\alpha\beta} = d_{\alpha\beta}(M;\varepsilon)$ is the sum of the values at zero momentum of the one particle irreducible two-point graphs of order $e^\alpha\lambda^\beta$, with the counterterms of lower order inserted. The graphs of orders λ, e^2 and λ^2 are the following, in the notation of Sect. 3.3, the two short lines being the external scalar legs:



(5.17)

Here the cross denotes the insertion of $-\delta m^2(x;0,1)$ of (3.51). The tadpole with a tadpole on its loop and the tadpole with this insertion cancel exactly at every ε, as in Sect. 5.1, so that d_{02} is the sunset graph alone; the graphs of orders $e^2\lambda$ and e^4, two-loop graphs with one, respectively two, vector field lines and their insertions, are not drawn. The terms of order λ and λ^2 contain no vector field line and do not depend on μ_0; the terms of order e^2, $e^2\lambda$ and e^4 do.

Theorem 5.1. *Let $d = 3$ and $N = 2$, with the action (2.4), the quartic term $\lambda|\phi|^4$, $|\phi|^2 = \phi_1^2 + \phi_2^2$, and the vector field propagator $\delta_{\mu\nu}C_v$. (i) For $0 < M_1 < M_2$ the difference $\delta m^2_\varepsilon(M_2) - \delta m^2_\varepsilon(M_1)$ converges as $\varepsilon \to 0$. The limit is a polynomial in e^2, λ of the form (5.16), whose coefficients are convergent subtracted continuum Feynman integrals, and it is differentiable in M_2^2. (ii) To one-loop order,*

$$\begin{aligned}&\lim_{\varepsilon\to 0} \tfrac{1}{2}\big[(m^2 + \delta m^2_\varepsilon(m)) - (M_1^2 + \delta m^2_\varepsilon(M_1))\big] \\ &\quad = \tfrac{1}{2}(m^2 - M_1^2) + \frac{2\lambda(m - M_1)}{\pi} - e^2\,\frac{(m-M_1)(m\mu_0 + mM_1 + M_1\mu_0)}{8\pi(\mu_0 + M_1)(m+\mu_0)}\,.\end{aligned} \tag{5.18}$$

The subtractions are those of the renormalization theorem of [R 2], which we now describe; the theorems of [R 1, R 2] are stated there for $d = 4$, and their form in $d = 3$, with the degrees

of (3.49), is Lemma C.1 of Appendix C. The difference of the counterterms at two masses is, order by order, a sum of lattice Feynman integrals in which one propagator has been replaced by

$$(M_1^2 - M_2^2)\, C_{M_2} C_{M_1}\,, \qquad C_M = (-\Delta^\varepsilon + M^2)^{-1}\,, \tag{5.19}$$

so that the overall degree of divergence is negative. The counterterm insertions are combined with the self-energy subgraphs they subtract, the lines of each such subgraph being taken consecutively, which is possible since these subgraphs are nested or disjoint, without common lines, the proper self-energy subgraphs of a two-loop graph being one-loop, and the two lines of a scalar bubble being taken consecutively as well; a self-energy subgraph containing the difference line has ultra-violet degree -1 and is subtracted at zero momentum by the difference of the inserted counterterms, a subtraction which the renormalization theorem admits though the degree does not require it, and one not containing it is subtracted at zero momentum at its own mass.

The one-loop terms have no subdivergences, and the power counting theorem, Lemma C.1(i), gives their convergence, the integrands being of the class considered there: the propagators have positive masses, and the vertices are polynomials in the momenta with periodic coefficients.

At two loops the scalar self-energy subgraphs are subtracted at zero momentum by the counterterms of lower order inserted in (5.16), which is the subtraction to degree one, since the linear term vanishes by the symmetry $p \to -p$ of the lattice two-point function. The vector field self-energy subgraphs, of degree one, satisfy

$$\Pi_{\mu\nu}(0) = 0\,, \qquad \Pi_{\mu\nu}(-p) = \Pi_{\mu\nu}(p) \tag{5.20}$$

on εZ^d, the first since a constant field is a gauge transform of zero on εZ^d, so that the one-loop vacuum energy per site does not depend on a constant field.

The $\phi\phi A$ vertex subgraphs of degree zero are a seagull or a λ-vertex joined to a derivative vertex by two lines, and the triangle with its three derivatives on internal lines. Their values at zero momenta vanish under the reflection $x_\mu \to -x_\mu$, $A_\mu \to -A_\mu$, so the subtractions are zero; the terms of the triangle with a derivative on an external leg are convergent and need no subtraction.

The subtraction terms at these subgraphs, their values at zero external momenta integrated over the loop momentum of the subgraph, are therefore the factors $\Pi_{\mu\nu}(0) = 0$, $\partial\Pi_{\mu\nu}(0) = 0$ and the vanishing vertex values, summed over the graphs of (5.16) with a common reduced graph; the subtraction term of a forest factorizes over its elements, the loop integral of $t_\gamma F_\gamma$ vanishing identically in the external momentum of γ, hence also after the Taylor operators of the elements containing γ. Scalar loops with one photon leg vanish, their integrand at zero photon momentum being odd under $k \to -k$; loops with three photon legs do not occur in graphs of order at most e^4.

The line momenta of each graph of (5.16), and of each of its subgraphs, form a natural set in the sense of Appendix C: the graphs have at most two loops, and the lines carry $\pm k_1$, $\pm k_2$ and, on the lines common to both cycles, all one of $\pm(k_1 + k_2)$ or all one of $\pm(k_1 - k_2)$, for the chord momenta k_1, k_2 of a spanning tree, the shared lines forming a path traversed consistently. So every pair of independent line momenta has coefficient matrix of determinant ± 1, and the internal momentum of every one-loop subgraph is, up to sign, the momentum of its chord, hence one of these.

The subtraction degrees are the ultra-violet degrees, the negatives of the degrees (3.49): 1 for the self-energy subgraphs and the one-loop two-point graphs, 0 for the vertex subgraphs and for the two-loop two-point graphs. For a graph with the difference line (5.19) we take the degree 0 for a scalar self-energy subgraph containing it, which is the subtraction effected by the difference of the inserted counterterms, the subtraction degree -1, that is no subtraction, for the whole graph, for a vector self-energy subgraph and for a vertex subgraph containing it, and the degrees above for the subgraphs not containing it.

Since the degree of a graph is the sum of the degrees of a subgraph and of the reduced graph, and the difference line lowers the degree of every subgraph containing it by 2, the two conditions (C.2) on the subtraction degrees hold, with equality for the subgraphs not containing the difference line and for the forests whose maximal element containing the difference line is a scalar self-energy or a vertex subgraph. The hypotheses of Lemma C.1(ii) are therefore satisfied for the graphs of (5.16).

The difference $\delta m_\varepsilon^2(M_2') - \delta m_\varepsilon^2(M_2)$, divided by $M_2'^2 - M_2^2$, is by (5.19) a sum of integrals with one line $C_{M_2} C_{M_2'}$, which converge as $\varepsilon \to 0$ by the same argument to the corresponding continuum integrals; these are absolutely convergent and continuous in M_2' by dominated convergence. Letting $M_2' \to M_2$, the limit is differentiable in M_2^2, its derivative being the continuum integrals with one line $C_{M_2}^2$, since $\partial C_M / \partial M^2 = -C_M^2$. This gives (i), and with it the statement that the divergent parts of δm_ε^2, of the orders given by the degrees (3.49), are the same for all M. Explicit computation gives (5.18), which is (ii). This completes the proof of Theorem 5.1.

Part (i) is used in Sect. 6 of [II], where the models with two scalar masses are compared, and (5.18) gives the one-loop form of the condition on the coupling constants stated there.

5.3. The Range of the Coupling Constants. No restriction is placed on e, λ, m, μ_0 beyond $e, \lambda, m, \mu_0 > 0$.

The smallness conditions imposed in [Ba 1–4] and used in Sect. 3 are, apart from the choices of Sect. 5.1, conditions on the running coupling constants

$$e(L^k \varepsilon)^{2-d/2}\,, \qquad \lambda (L^k \varepsilon)^{4-d} \tag{5.21}$$

at the scales where the perturbation expansion of the effective action is used; the assumption that the scalar mass be bounded, made in the lower bound on the quadratic form of the action, is an upper bound on ε_0 for the fixed mass (Appendix B).[19] They hold once

$$L^k \varepsilon \leqq \varepsilon_0\,, \qquad \varepsilon_0 = \varepsilon_0(e, \lambda, m, \mu_0)\,, \tag{5.22}$$

and the integer J of Theorem 3.1 is chosen accordingly, after the coupling constants. The conditions, and the order in which the constants are fixed, are listed in Appendix B.

The constants C of Theorems 2.1 and 3.1–3.4 depend on the coupling constants, through ε_0 and through the explicit factors $(\mu_0 L^k \varepsilon)^{-1}$, $\lambda^{-1/4}$ of (3.2) and (3.113), and the counterterm (3.51) is always computed at the photon mass μ_0 of the model. The only condition on the coupling constants in [II] concerns the generating functional with a scalar source and is stated there.

5.4. Relation to the Earlier Paper. This paper follows [K 1]. Its results are stated for $d = 3$; the two-dimensional case of [K 1] is not treated. The statements corrected are Theorems 2.1, 3.4 and 3.5, Proposition 3.6, the Wick ordering (2.8), the localization (3.54)–(3.55), the bounds (3.4), (3.63), (3.117) and (4.22), the expansion (3.66), and the scope of (3.60) and of (3.97)–(3.98). The proofs of Theorem 2.1 and of (3.139), (4.38) are replaced.

The following are added. Proposition 3.11 compares, scale by scale, the determinants of the covariances of the scalar fluctuation fields for the models on T_{ε_K} and $T_{\varepsilon_{K+1}}$; it is used in the proof of Theorem 3.4 in Sect. 3.6. Section 5 collects the results on the renormalization transformation with block size $L = 5$, on the mass counterterm and on the range of the coupling constants which are used here and in [II]: Sect. 5.1 gives the modifications of the arguments of [Ba 1–3] and of Theorem 3.1 required by the block size $L = 5$; Sect. 5.2 gives the dependence of the mass counterterm (3.50)–(3.51) on the scalar mass, Theorem 5.1; Sect. 5.3 gives the range of the coupling constants.

[19] Proposition 3.1 of [Ba 2].

Appendix B lists the conditions of [Ba 1–4] and of Theorem 3.1 which involve the block size or the coupling constants. Appendix C gives the form in three dimensions of the power counting and renormalization theorems of [R 1, R 2]. Appendix D proves the bounds on the scalar field of Sect. 5.1 on the box introduced there (Lemma D.1), the bound on the constant E_{fin} (Lemma D.2) and the cube step of the basic estimate with the cubes of Sect. 5.1 (Lemma D.3), together with the bounds on the propagator of a box at a constant field which these proofs use.

Fig. 1 and the graphs drawn in Sects. 3.3 and 3.5 are those of [K 1]; Figs. 2–6 and the graphs of (5.16) are added.

Besides the papers cited in [K 1] we cite the power counting theorems of [R 1, R 2] (Sect. 5.2), which postdate it. The displays are numbered consecutively within each section, so that their numbers differ from those of [K 1] where displays have been added. The footnotes give the places in [Ba 1–4] and [R 1, R 2] from which the imported statements are taken. The succeeding paper [II] follows [K 2] in the same way.

Appendix A

We wish to modify the proof of the lower bound presented in [Ba 1] by using the following bounds on the fields at each step:

$$\begin{aligned} |A_\mu^{(k)}(x)|\,, \qquad |\Delta^\eta A_\mu^{(k)}(x)| &\leqq C_1 p(L^k\varepsilon)(\mu_0(L^k\varepsilon))^{-1}\,, \\ |\phi^{(k)}(x)|\,, \qquad |\Delta^\eta_{A^{(k)}}\phi^{(k)}(x)| &\leqq C_1 p(L^k\varepsilon)[\lambda(L^k\varepsilon)^{4-d}]^{-1/4}\,, \end{aligned} \tag{A.1}$$

where C_1 is a fixed constant $O(1)$. Examining the paper [Ba 1], we see that the expansion (3.59) of [Ba 1] must be taken to order $\bar{n} = 13$, and that (3.52) of [Ba 1] is replaced by

$$|\psi(y)| \leqq Cp(L^k\varepsilon)[\lambda(L^k\varepsilon)^{4-d}]^{-1/4}\,, \tag{A.2}$$

but the relations (3.54) of [Ba 1] are still sufficient. With these modifications the lower bound holds as before.

Next we want to justify replacing the bound on $D_{\bar{A}^{(k)}}\phi_{(k)}(b)$ in [Ba 2] by (3.2). Recall that

$$\bar{A}_\mu^{(k)}(x) = \sum_{j=0}^{L^k-1} \eta A^{(k)}(\langle x + j\eta e_\mu, x + (j+1)\eta e_\mu\rangle)\,. \tag{A.3}$$

Furthermore, denoting by 1 the constant function, we have

$$\begin{aligned} a_k G_k Q_k^* \cdot 1 &= a_k G_k \cdot 1 \\ &= G_k(-\Delta^\eta + \mu_0^2(L^k\varepsilon)^2 + a_k Q_k^* Q_k)\cdot 1 - \mu_0^2(L^k\varepsilon)^2 G_k \cdot 1\,, \end{aligned} \tag{A.4}$$

and therefore

$$a_k G_k Q_k^* \cdot 1 = (1 + \mu_0^2(L^k\varepsilon)^2 a_k^{-1})^{-1} Q_k^* \cdot 1\,. \tag{A.5}$$

So for $x \in B^k(y)$, it follows from (A.5) and the bounds (3.2) that

$$\begin{aligned} A_\mu^{(k)}(x) &= a_k G_k Q_k^* \cdot A_{k,\mu}(y) + a_k G_k Q_k^*(A_{k,\mu}(x) - A_{k,\mu}(y)) \\ &= A_{k,\mu}(y) + C(L^k\varepsilon)^\alpha + Cp(L^k\varepsilon)\,, \end{aligned} \tag{A.6}$$

where $\alpha > 0$. Therefore from (A.3) we see that $\bar{A}_\mu^{(k)}(x) = A_{k,\mu}(y) + Cp(L^k\varepsilon)$ for $x \in B^k(y)$, and so the expansion (3.65) and the bounds (3.2) imply

$$D_{\bar{A}^{(k)}}\phi_k(b) = D_{A_k}\phi_k(b) + C(L^k\varepsilon)^\alpha\,, \qquad \alpha > 0\,. \tag{A.7}$$

Finally we wish to show that the bounds (3.2) imply (A.1), allowing us to replace (A.1) by (3.2) in the lower bound of Theorem 3.1. For instance, we get

$$(-\Delta^\eta_{A^{(k)}})\phi^{(k)} = a_k Q_k^*(A^{(k)})\phi_k - m^2(L^k\varepsilon)^2\phi^{(k)} - a_k Q_k^*(A^{(k)})Q_k(A^{(k)})\phi^{(k)}\,. \tag{A.8}$$

So from (3.2) and Theorem 3.3 we get

$$|\Delta^{\eta}_{A^{(k)}}\phi^{(k)}| \leqq Cp(L^k\varepsilon)[\lambda(L^k\varepsilon)^{4-d}]^{-1/4} + C(L^k\varepsilon)^{\alpha}\,. \tag{A.9}$$

Therefore for C_1 large enough, we get the desired result. The other bounds in (A.1) follow similarly. The inclusion of sources g, h is straightforward, apart from a field-independent constant in the term in h, which we discuss below. From (3.36), (3.37) we see that their field-dependent parts are small perturbations in the effective action, and so may be bounded when making a perturbative expansion. Furthermore, in the small field region these parts are proportional to $(L^k\varepsilon)^{\alpha}$, $\alpha > 0$, and so they do not affect the leading order positivity of the action, which is provided by (3.34).

The term in h also contains a field-independent constant, which we now discuss. By (3.48), (3.49) the field-independent graphs of order e^2 linear in h built from the vertices (3.37), (3.38), (3.39) have degree $6-2d$. Those containing (3.42), (3.43) have degree at least $8-d$, the factor L^{-kd} of Q_k being counted as η^d, so that their index sums are bounded, and the vertex (3.41), which carries the external field ϕ_k, enters no field-independent graph; with the factor $(L^k\varepsilon)^{6-d}$ carried by the vertices and $\sum_x \eta^d|h(x)| \leqq (L^k\varepsilon)^{-d}\|h\|_1$, they are bounded by $C\|h\|_1$ uniformly in k and ε for $d=3$.

Three graphs of order e^2 built from (3.37)–(3.39) occur, besides the insertion of δm^2. The graph with the vertex (3.38), $m=2$, in the loop is not affected by δm^2, its value at zero momentum vanishing, the sum over the bonds of a lattice derivative being zero. The vertex carries the factor η and a lattice derivative on one scalar leg, and summation by parts along the bonds, at the cost of a second factor η, puts a lattice derivative on the other leg, up to a term carrying the difference of the coefficient $G^v_k(b,b)$ of the photon loop along the bond together with the factor η.

The first term is bounded by $C\eta^2 \sup_b G^v_k(b,b) \sum_b \eta^d |\partial G_k(0)(b,z)|^2 \leqq C$ for $d=3$. In the second the diagonal of the free propagator C_v is constant on the torus, and the remaining part $F_{k,v} = C_v - G^v_k$ varies on the scale of the blocks, so that by (3.8) its difference along a bond is bounded by $C\eta$, as in Sect. 3.3; the second term is therefore bounded by $C\eta \sum_x \eta^d G_k(0)(x,z)^2 \leqq C\eta$ for $d=3$. So the graph is bounded uniformly in k and ε, and it is contained in the constant a_ε with the other graphs of order e^2.

The graph with the vertex (3.39) is cancelled by the term of order e^2 in δm^2, up to the replacement in the loop of the propagator C_v of the counterterm by G^v_k, the difference being the part $F_{k,v} = C_v - G^v_k$ of C_v carrying the scales not yet integrated. The same term also removes from the exchange graph, by (3.51), the value of the self-energy at zero momentum, up to the same replacement of C_v, C_s by G^v_k, $G_k(0)$, and the remainder still has degree $6-2d$.

So for $d=3$ this constant contains a term of order e^2 and of order zero in λ, linear in h, which in perturbation theory grows like $e^2 k \int h$ after k steps, where $\int h$ denotes $\sum_x \varepsilon^d h(x)$, that is like $\ln \varepsilon^{-1}$ at the last step, as noted after (2.8). Its divergence is cancelled, at order e^2 and at order zero in λ in perturbation theory on εZ^d, by the term a_ε in (2.8), a_ε being defined there as the term of order e^2 and of order zero in λ in $\langle|\phi(0)|^2\rangle$ on εZ^d with the term of order e^2 of δm^2 inserted.

On T_ε, of smallest side ℓ, the same term of the perturbation expansion, with δm^2 computed by (3.51) on T_ε, differs from its value on εZ^d by at most

$$Ce^{-c\ell} \sum_x \varepsilon^d |h(x)| \tag{A.10}$$

uniformly in ε. To see this, recall that the propagators of T_ε are the periodizations $\sum_n C(u+\ell_n)$ of those of εZ^d, ℓ_n running over the periods of T_ε; the convolution on T_ε of two periodized kernels is the periodization of their convolution, and the sum over T_ε of a periodized kernel is the sum of the kernel over εZ^d.

Of the three graphs of order e^2, the seagull graph is cancelled on T_ε, as on εZ^d, exactly by the term of δm^2 of order e^2 proportional to $dC_v(0)$, the counterterm being computed with the propagators of the lattice at hand. The graph with the vertex (3.38), $m = 2$, carries the coefficient $C_v(b,b) = C_v(0)$, independent of b on either lattice, so that the summation by parts of Sect. 3.3 produces no difference term and the graph is a constant times $C_v(0)\sum_b \varepsilon^{d+2}|\partial C_s(x,\cdot)(b)|^2$ summed against $\sum_x \varepsilon^d h(x)$. On T_ε it differs from its value on εZ^d by the term in $C_v^T(0) - C_v(0) = \sum_{n\neq 0} C_v(\ell_n)$, bounded by $Ce^{-c\ell}$ times $\sum_b \varepsilon^{d+2}|\partial C_s|^2 \leqq C\varepsilon$ for $d = 3$, and by the cross terms of the periodized product $\partial C_s^T\, \partial C_s^T$, in which the near factor carries one derivative and is summable uniformly in ε while the far factor is bounded by $Ce^{-c\ell/2}$, the two factors ε compensating $C_v^T(0) \leqq C\varepsilon^{-1}$. There remains the exchange graph with its counterterm.

The exchange term on T_ε differs from the term on εZ^d only by the values of the kernel of the latter at the periods $\ell_n \neq 0$, and by the cross terms of the periodized products of propagators, in the self-energy inside the graph and in the counterterm (3.51) computed on T_ε. Summing by parts over the positions of the current vertices, we put the derivatives of the vertices (3.38), $m = 1$, on the lines of the loop, so that the graph is $\sum_{u,u'} C_s(x-u)\Sigma(u-u')C_s(u'-x)$ with Σ the amputated self-energy; the function $C_s * C_s$ being even, only the even part of Σ contributes, and by (3.51) the subtracted self-energy has zero sum.

So the first contribution involves the second differences $\partial C_s * \partial C_s$ of the function $C_s * C_s$ at the periods, and since $|\Sigma(u)| \leqq C|u|^{-4}e^{-c|u|}$ for $u \neq 0$ uniformly in ε, by the bounds on C_s, C_v established in Appendix D, the sum $\sum_u \varepsilon^d |u|^2|\Sigma(u)|$ over $|u| \leqq \ell/4$ is bounded by $C\ell$ uniformly in ε for $d = 3$, the factor ℓ being absorbed in $e^{-c\ell}$, while for $|u| > \ell/4$ the decay of Σ and $|u| + |u+\ell_n| \geqq |\ell_n|$ bound the sum by $C\sum_{n\neq 0} e^{-c'|\ell_n|}$.

In a cross term one factor is evaluated at a distance at least $\ell/2$ and the other is near, $|w + \ell_n| \geqq \ell/2$ for w in the fundamental domain and $n \neq 0$. Inside the graph the near factor carries at most one lattice derivative, except in the terms in which both derivatives of the current vertices act on the near line of the loop, the scalar line or the photon line; there we sum one of them by parts at its vertex, over the position of the vertex on T_ε and before the propagators are written as sums over images, onto the outer propagator and onto the far factor, after which every near factor is summable uniformly in ε and each outer propagator carries at most one derivative.

In the counterterm the external legs are constant, so that in the exchange term both derivatives act on the scalar line of the loop, and a cross term is the sum over εZ^d of $\partial\partial C_s(v)$ against $\Phi(v) = \sum_{\ell_m \neq 0} C_v(v + \ell_m)$, or, what is the same sum with the summation variable shifted by a period, of $C_v(v)$ against the same sum of $\partial\partial C_s$; so it suffices to bound the first.

We sum one derivative by parts onto Φ. For $|v| \leqq \ell/2$ every argument $v + \ell_m$ has modulus at least $\ell/2$, so that $|\partial\Phi(v)| \leqq Ce^{-c\ell}$, while $\sum_v \varepsilon^d|\partial C_s(v)| \leqq C$ uniformly in ε for $d = 3$; for $|v| > \ell/2$ the factor $\partial C_s(v)$ is bounded by $Ce^{-c|v|}$, and its sum against $|\partial C_v(v+\ell_m)|$ is bounded by $Ce^{-c'|\ell_m|}$, since $|\partial C_v(w)| \leqq C|w|^{-2}e^{-\delta|w|}$ for some $\delta > 0$, uniformly in ε, by the bounds on C_v established in Appendix D, so that $e^{-c|v|/2}e^{-\delta|v+\ell_m|/2} \leqq e^{-c'|\ell_m|}$ with $c' = \frac{1}{2}\min\{c,\delta\}$ by the triangle inequality, while $\sum_v \varepsilon^d|v+\ell_m|^{-2}e^{-\delta|v+\ell_m|/2} \leqq C$ for $d = 3$; the sum over the periods is then geometric. The rates c, δ being positive, every term of the difference is bounded by $Ce^{-c'\ell}$ uniformly in ε.

Theorem 2.1(i) is stated for $h = 0$. The generating functional with a scalar source in a finite cube is treated in Sect. 6 of [II], with the constant a_ε of (2.8).

The constant does not enter the bounds on the fields in the first terms of (3.4) and (3.5), where it is a term of $S^{(k),1}$. The second term of (3.5) is a bound on the integral over the large field region with the sources.

For the source g it follows from the bound at $g = 0$ by the Schwarz inequality. The integral of $\exp F(g)$ over that region is at most the square root of its measure times the square root of $Z^\varepsilon(T_\varepsilon, 2g, 0)$. The measure is bounded by the large field term of (3.5) at $g = 0$, which in

[Ba 2] is a bound on that integral itself. Furthermore $Z^{\varepsilon}(T_{\varepsilon}, 2g, 0) \leqq \exp[2\|g\|^2] Z^{\varepsilon}(T_{\varepsilon}, 0, 0) \leqq \exp[2\|g\|^2 + C|T|]$, $\|g\|$ the norm of (2.24), by the second inequality of (2.24), which Sect. 3.2 proves for every real g and every K without use of Theorem 3.1, and by the first. The large field term of (3.5) carries the factor $\exp[C|T|]$, so that the square roots of the two factors give that term with its exponent $cp(L^k \varepsilon_K)^2$ halved and the additional factor $\exp[\|g\|^2]$, which the constant C, depending on g, absorbs. Theorem 3.1 is used at $h = 0$.

Among the terms of $P^{(k),1}$ which are whole graphs of degree zero, two have no counterterm for their degree as whole graphs. The term of order τ^2, quadratic in g and free of fields, is a whole graph of degree zero whose photon line carries a curl at each end; summing by parts at one end puts the curl on g, and the term is then bounded as in Sect. 3.5 and retained in $S_N^{(k),1}$. The term of order e^2 linear in h is the one whose index sum is not bounded, and Theorem 3.5 does not apply to it.

The h-linear constants of order e^0 and of order λ are cancelled by the Wick ordering constant and by the term of order λ in δm^2 respectively, up to the part $F_{k,s} = C_s - G_k(0)$ of C_s carrying the scales not yet integrated. The remainder of order λ is bounded uniformly in k. The remainder of order e^0,

$$N\theta(L^k\varepsilon)^2 \sum_x \eta^d h(x) F_{k,s}(x,x)\,, \tag{A.11}$$

is of order $(L^k\varepsilon)^{-1} \int h$ for $d = 3$, since $F_{k,s}(x,x) = a_k[C_s Q_k^* Q_k G_k(0)](x,x)$ is bounded uniformly in k, by (3.8) applied to the characteristic function of a block and the integrability of C_s over a block; it is the Wick constant of the scales not yet integrated, and is compensated, at order zero in the coupling constants, by the term $-\theta(L^k\varepsilon)^2 \sum_x \eta^d h(x) |\phi^{(k)}(x)|^2$ of $S^{(k),1}$ when the remaining fluctuation integrals are performed. Both remainders are field-independent terms of $S^{(k),1}(T^{(k)}, 0, 0, 0, h)$, and do not enter the bounds on the fields.

The field-independent vacuum graphs of order $\alpha + 2\beta \leqq 6$, α, β being the powers of e and λ, are cancelled in the same way by E_1, up to their parts with indices $j \geqq k$, and the convergent vacuum graphs of E_1 of positive degree are field-independent constants. All of these cancel in the ratio $Z^{\varepsilon_K}(T, g, 0) Z^{\varepsilon_K}(T, 0, 0)^{-1}$ of Theorem 2.1, being terms of the constant $S^{(k),1}(T^{(k)}, 0, 0, 0, 0)$, which does not depend on g and is removed in the normalized action $S_N^{(k),1}$ of Sect. 3.2. So Theorem 3.4 and the proof of Theorem 2.1(i) in Sect. 3.2 are stated for $h = 0$, and the field-dependent terms, the g-dependent terms and the h-dependent terms of $P^{(k),1}$ other than the constants discussed above are the renormalised graphs treated by Theorem 3.5 and Sect. 3.5.

Appendix B

We collect here the conditions on the constants of the proofs of [Ba 1–4] and of Theorem 3.1 which involve the block size L or the coupling constants, in the form in which Sects. 5.1 and 5.3 use them, and the order in which the constants are fixed. The notation is that of Sect. 5.1: M_b is the side of a large block in the units of the kth lattice, $r(L^k\varepsilon)$ is the corridor width (5.1), with $\varrho = 2$, w_k is the width (5.2) of a corridor of the previous step in the units of the kth lattice, and $p(L^k\varepsilon)$ is the function (3.1), with the exponent $p = 6$.

B.1. The places where L enters. Every place where the value of L enters the proofs of the bounds (3.4)–(3.5), of the basic estimate (5.4) and of Theorem 3.1 is of one of seven kinds.[20]

First, constants depending on L. The bounds obtained by transporting the restrictions (3.2)

[20] Sects. 2–3 of [Ba 1], Sects. 2–3 of [Ba 2] and Sects. 1–3 of [Ba 3].

along the contours inside a block, $2Ldp$, $3Ldp$, $4Ldp$, are $10dp$, $15dp$, $20dp$ at $L = 5$;[21] the ratios

$$\frac{p(L^{k-1}\varepsilon)}{p(L^k\varepsilon)} \leqq (1+\ln L)^6 \ (= 315.7)\,, \qquad \frac{r(L^{k-1}\varepsilon)}{r(L^k\varepsilon)} \leqq (1+\ln L)^2 \ (= 6.81)\,, \qquad \frac{a_\infty}{a} = 1 - L^{-2} \ (= 0.96) \tag{B.1}$$

at $L = 5$, for $L^k\varepsilon \leqq 1$, enter the restrictions on the fields of the previous step, the proof of Lemma D.1 and the coefficients (2.14) respectively;[22] and the constants of Theorem 3.3 depend on L and are finite for every $L \geqq 2$.

Second, rates of decay. Where a decay rate δ is used across β corridors of the previous scale, as in the bounds on the terms coupling distant regions in the final density, with $\beta = 4$, it becomes $\delta\beta/L$ in $r(L^{k-1}\varepsilon)$, up to the constant $e^{\delta\beta M_b/L}$, that is $4\delta/5$ at $L = 5$.[23]

Third, scale sums and scale counts. The sums $\sum_{j\geqq 1} L^{-j}$, $\sum_{j\geqq 1} L^{-2j}$ and $\sum_{m\geqq 0} L^{-m}(1 + m\ln L)^2$ of the combinatorial estimate are $\frac{1}{4}$, $\frac{1}{24}$ and $\frac{L}{L-1} + \frac{2L\ln L}{(L-1)^2} + \frac{L(L+1)\ln^2 L}{(L-1)^3}$ $(= 3.47)$ at $L = 5$, and the number of scales is at most $\ln\varepsilon^{-1}/\ln L$.[24]

Fourth, conditions on the constants b_0, b_1, R and on the large block size, which are fixed after L. The conditions under which the restrictions on the fields of one step imply those of the next are $p - 1 \geqq p_1$ and

$$b_0\, p \ln L \geqq c_1 b_1 (1+\ln L)^{p_1}\,, \tag{B.2}$$

where p_1 is the exponent and b_1 the constant of the large field bound on the vector field, and $c_1 = 4d$ at the first step;[25] at $L = 5$ and $p_1 = 1$ the factor $(1+\ln L)^{p_1}/\ln L$ is $(1+\ln 5)/\ln 5$ $(= 1.62)$, against $(1+\ln 2)/\ln 2$ $(= 2.44)$ at $L = 2$, and the right side is finite for every $L \geqq 2$, so that this is a condition on b_0 and not on L. The condition that the large block size be sufficiently large and the conditions $\varrho > 1$ and $R > R_1$ on (5.1) have right sides which may depend on L and on the large block size but are finite for every $L \geqq 2$; they are conditions on constants fixed after L, not restrictions on L.[26]

Fifth, conditions on ε_0. The first is $\frac{1}{2}b_0 p(p-1)\ln L \geqq c_2\varepsilon_0^\alpha$, from the same implication between the restrictions of successive steps.[27] The second is $\varepsilon_0 < e^{1-\varrho} = e^{-1}$, which gives the statement that a distance measured on $T_{L^{-j}}$ exceeds $L^{k-j}r(L^k\varepsilon) > r(L^j\varepsilon)$ for $j < k$, the function $\eta \mapsto \eta r(\eta)$ being increasing for $\eta < e^{-1}$.[28] The third is the condition, in the combinatorial estimate, that $p(L^k\varepsilon)^2$ exceed a multiple of $r(L^k\varepsilon)^{d+1}$; since $p(\eta)^2 r(\eta)^{-d-1} = b_0^2 R^{-d-1}(1+\ln\eta^{-1})^{2p-(d+1)\varrho}$ this is an upper bound on ε_0 when $2p > (d+1)\varrho$, as here.[29] The fourth are the conditions "ε_0 sufficiently small" of the final steps of the lower and of the upper bound.[30] Each is an upper bound on ε_0 whose right side is positive once $2p > (d+1)\varrho$.

Sixth, a divisibility statement, the regular partition of the combinatorial estimate, which fails on some tori and is replaced in Sect. 5.1.

Seventh, the containments between regions of different scales. An object of size $\alpha r(L^k\varepsilon)$ built at the step k must fit inside β corridors built at the step $k-1$, each of width exceeding w_k in the units of the kth lattice (Sect. 5.1), while the corridors of the kth step have width exceeding $r(L^k\varepsilon) - 1$. Since

$$\frac{r(L^{k-1}\varepsilon)}{r(L^k\varepsilon)} = \Big(1 + \frac{\ln L}{1 + \ln(L^k\varepsilon)^{-1}}\Big)^\varrho \to 1 \qquad \text{as } L^k\varepsilon \to 0\,, \tag{B.3}$$

[21] (2.16)–(2.17) of [Ba 2].
[22] The restrictions are (2.55) of [Ba 2].
[23] (3.8), (3.10), (3.12) of [Ba 2].
[24] (3.51) of [Ba 2].
[25] (3.12)–(3.13) and (3.53)–(3.54) of [Ba 1].
[26] Sect. 2 of [Ba 1]; (2.7) of [Ba 2], where ϱ is written r and R_1 is written R_0.
[27] (3.54) of [Ba 1].
[28] (2.104) of [Ba 2].
[29] (3.54) of [Ba 2].
[30] (3.66)–(3.69) of [Ba 1] and (2.116)–(2.118) of [Ba 2].

such a containment holds at every scale, for R large, if and only if $\alpha L \leqq \beta$ (in the case $\alpha L = \beta$ because $r(L^{k-1}\varepsilon) - r(L^k\varepsilon) \geqq 2R \ln L$ for $\varrho = 2$, so that $R \geqq (M_b + c/\alpha)/(2 \ln L)$ suffices when the object carries an additive constant c), and when $\alpha L > \beta$ it fails for every $L^k\varepsilon$ below a value depending on α, β, ϱ and L only, for every R.

The containments are four. In the cube step of the basic estimate the cube of side $3r(L^k\varepsilon)$ must lie inside the region on which the propagator is defined, and the vertices of a class carrying no leg of the external field lie five corridors deep in it: $\alpha = \frac{3}{2}$, $\beta = 5$. In the bounds on the background field the support of $\zeta^{(k)}$ about a point of $\Lambda_2^{(k-1)\prime}$ must lie inside $\Lambda_{-1}^{(k-1)\prime}$: $\alpha = 1$, $\beta = 3$. In the bounds on the scalar field the boxes about the block are the large blocks of $\Lambda_7^{(k-1)\prime}$ within $2r(L^k\varepsilon)$ and $4r(L^k\varepsilon)$ of it, and the propagator bounds at depth R_1 are used on them as on full boxes; a full box of radius $4r(L^k\varepsilon)$ would have to lie inside the same region at depth five corridors: $\alpha = 4$, $\beta = 5$, which fails at every $L \geqq 2$. In the split of the background field at the next step the supports of $\zeta^{(k+1)}$ about the support of θ_{k+1} must lie inside $\Lambda_0^{(k)\prime}$; with a support of radius $\theta r(L^{k+1}\varepsilon)$ this requires $L\theta \leqq 2$, which the choice $\theta = 1$ made there violates at every $L \geqq 3$ and which $\theta = \frac{2}{5}$ satisfies at $L = 5$, the inner radius being reduced with it (Sect. 5.1).[31] These are the four containments replaced in Sect. 5.1; none of them holds at $L = 5$ in the form the proofs require.

The hypothesis of the basic estimate on the support of the external vector field,

$$\operatorname{dist}(\operatorname{supp} \tilde{A}, \partial\Omega) > 2r(L^k\varepsilon)\,, \qquad \Omega = B^k(\Lambda_2^{(k-1)\prime})\,, \tag{B.4}$$

holds where the estimate is stated, with $\Omega = B^k(\Lambda_2^{(k-1)\prime})$, and involves no comparison of two scales. In its second use the domain of the propagator is $B^k(\Lambda_0^{(k)})$, and the field $\theta_{k+1}A'^{(k)} + \tilde{B}^{(k)}$ is supported in $\operatorname{supp}\theta_{k+1}$, within M_θ of $B^k(\Lambda_2^{(k)})$, M_θ being the width of the slice on which θ_{k+1} varies, as in B.2 below; $\Lambda_2^{(k)}$ lies two corridors inside $\Lambda_0^{(k)}$, each of width exceeding $r(L^k\varepsilon) - 1$, so that the support lies at depth exceeding $2r(L^k\varepsilon) - 2 - M_\theta - 2\eta$ in that domain, the fine blocks costing 2η, and a vertex within a unit cube of it at depth exceeding $2r(L^k\varepsilon) - 3 - M_\theta - 2\eta$. This is less than $2r(L^k\varepsilon)$.

The hypothesis enters the proof of the basic estimate through the propagator bounds at depth R_1 at the vertices, at the points of their localization cubes and their η-neighbourhoods, at depth $R_1 + 2$, and in the cube step, where the cubes about a vertex must lie in the domain (Appendix D). With the cubes of Sect. 5.1 the latter is the depth $D^\flat$ of (5.8), which exceeds the former, $D^\flat \geqq r^\flat \geqq R_1 + 2$. In the second use every connected component of a class carries a leg of the field $\theta_{k+1}A'^{(k)} + \tilde{B}^{(k)}$ and its old vertices lie within $\ell_{\bar{n}}$ of a unit cube of its support (Appendix D), at the depth above less $\ell_{\bar{n}}$, and $2R - 3 - M_\theta - 2\eta - \ell_{\bar{n}} \geqq D^\flat$ once $R \geqq D^\flat + M_b + 1$ and $R \geqq M_\theta + \ell_{\bar{n}} + 2$, the two lower bounds (5.9) on R, which are of the fourth kind. No value of L enters.

B.2. The parameter conditions at $L = 5$. The constants are fixed in the order d; a; the expansion order $\bar{n}$; the Hölder exponent; L; the large block size; $R > R_1$; the exponents ϱ and p with $2p \geqq (d+1)\varrho$, here $\varrho = 2$ and $p = 6$ with $2p > (d+1)\varrho$; b_0, b_1, p_1; and last ε_0. The replacements of Sect. 5.1 add conditions of the same kinds, each a lower bound on R or an upper bound on ε_0 whose right side depends only on the data fixed before it.

The cube step (the first replacement of Sect. 5.1) uses two concentric cubes of constant sides $r^\flat$, $3r^\flat$, where $r^\flat$ is a multiple of M_b satisfying (5.7), $\ell^\sharp$ there being the side of the cube containing a renormalized class after the expansion in B' to the order $\bar{n}$, which depends on $\bar{n}$ only, and R_1 the depth of the propagator bounds; it requires every old vertex of the class to lie at depth exceeding $D^\flat = \frac{3}{2}r^\flat + 2M_b + \ell^\sharp$ in the domain of the propagator (Lemma D.3). At $L = 5$ a vertex anchored at $\Lambda_7^{(k-1)\prime}$ has depth exceeding $5w_k - 1 = r(L^{k-1}\varepsilon) - M_b - 1 \geqq R - M_b - 1$ in

[31] In the order stated: Sect. 3 of [Ba 3]; Lemma 2.3 of [Ba 2]; the proof of Lemma 2.4 of [Ba 2]; the split (2.89) of [Ba 2].

$B^k(\Lambda_2^{(k-1)\prime})$, and in the second use of the basic estimate every connected component of a class carries a leg of the external field and its old vertices lie at depth exceeding $2R-3-M_\theta-2\eta-\ell_{\bar{n}}$ in the domain there, as in B.1, so that the two lower bounds (5.9) suffice; $D^\flat$ depends on $\bar{n}$, d, R_1 and M_b only, and the depths are measured in the supremum norm.

The cut-off (the second replacement) requires $R \geqq 10(M_b+1)$, where M_θ, the width of the slice on which the function θ_k varies, is less than M_b by its definition.[32] The cube step and the box also require $e(L^k\varepsilon)^{2-d/2}p(L^k\varepsilon) \leqq 1$ and the smallness of the error (D.4), upper bounds on ε_0. The box of Lemma D.1 (the third) requires $r(\varepsilon_0) \geqq R_1 + (3+\sqrt{3})M_b$, a lower bound on R since $r(\varepsilon_0) \geqq R$, the ratio bound $r(L^{k-1}\varepsilon) \leqq 2Lr(L^k\varepsilon)$ of (B.1), and an upper bound on ε_0 making the decay across one corridor smaller than the power of $L^k\varepsilon$ required there; the mass term in the identity (D.6) enters the constants only (Appendix D).

The partition (the fourth) requires that the sides of $T_1^{(k)}$ exceed $r(L^k\varepsilon)$, with an upper bound on ε_0 from the number of sequences of regions. Finally we take the sides $2L_\mu$ of T to exceed $2(8r(\varepsilon_0)+M_b)\varepsilon_0$ and $2(3r^\flat+2M_b)\varepsilon_0$, the factor 2 covering the rounding of the boxes to large blocks, so that no cube or box of the proofs wraps around the torus; for the tori (2.22) these are lower bounds on J. None of these conditions restricts L, and together with those of B.1 they can be imposed in the order stated.

B.3. The smallness conditions on the coupling constants. Every smallness condition in [Ba 1–4] and in Theorem 3.1 is of one of five kinds.

First, the running charge: "$e(L^k\varepsilon)$ sufficiently small" in the propagator bounds at depth R_1 and in their corollaries, where $e(L^k\varepsilon)^{2-d/2}$ is the charge of the kth effective theory, and "e sufficiently small" in the decay and regularity results for the operators of Theorem 3.3, where e is the charge of the unit lattice model of [Ba 4], which is $e(L^k\varepsilon)^{2-d/2}$ when its bounds are applied at the kth step.[33] The thresholds depend on d, a, L, the Hölder exponent and the large block size. Since the charge $e(L^k\varepsilon)^{2-d/2}$ increases with k for $d<4$, each is implied by the condition $e\varepsilon_0^{2-d/2} \leqq e_0$ at the stopping scale, where the charge is largest.

Second, the scale: "ε_0 is some fixed sufficiently small number", "$L^{k+1}\varepsilon \leqq 1$" and "$L^k\varepsilon \leqq \varepsilon_0$", "$L^k\varepsilon$ sufficiently small".[34] These are upper bounds on ε_0 whose right sides depend on the coupling constants and on the constants fixed before.

Third, the units. In the lower bound the model is rescaled by $\lambda^{1/(4-d)}$, which in $d=3$ sends the spacing to $\lambda\varepsilon$, the masses to m/λ and μ_0/λ, the charge to $e\lambda^{-1/2}$ and the quartic coupling to one, and leaves the running charge unchanged; in the upper bound the restrictions on the fields and the quartic term carry the running coupling $\lambda(L^k\varepsilon)^{4-d}$ explicitly.[35] In either form the conditions "ε small" are conditions on $\lambda\varepsilon_0$, with constants which may depend on e, m, μ_0, so that the dependence on λ is a dependence of constants and not a condition on the coupling constants.

Fourth, the mass: the assumption $m^2 \leqq O(1)$ made in the lower bound on the quadratic form of the action, whose constant γ_0 depends on d and a only.[36] In that inequality, and in the inequality on the unit lattice of the kth step from which it is derived, the mass enters through the terms $m^2(L^k\varepsilon)^2|\phi_k|^2$ on the right side, while the form on the left is at most a_k. A constant γ_0 independent of the mass therefore requires $m^2(L^k\varepsilon)^2 \leqq a_k/\gamma_0$ at the steps where the inequality is used, and since $L^k\varepsilon \leqq \varepsilon_0$ there this is the upper bound $\varepsilon_0^2 m^2 \leqq a(1-L^{-2})/\gamma_0$ on ε_0, the mass being fixed before ε_0: a dependence of the constants on m, not a condition on the coupling constants.

[32] Sect. 2 of [Ba 2], after (2.44).

[33] (2.23) and the theorem of Sect. 2 of [Ba 1]; Proposition 2.2 of [Ba 2]; (1.8), (2.25)–(2.26), (2.33) and Corollary 2.3 of [Ba 4].

[34] Sect. 3 of [Ba 3]; (3.54) of [Ba 1]; Sects. 2–3 of [Ba 2].

[35] Sect. 2 of [Ba 1]; (2.55) and (2.56) of [Ba 2].

[36] Proposition 3.1 of [Ba 2], the inequalities (3.26) and (3.29) there; the proof of (3.29) is referred there to [Ba 4].

Fifth, the number of steps: "for some fixed integer J, for all $K \geqq J$ and all $k \leqq K - J$" of Theorem 3.1. The stopping scale is $\varepsilon_0 = L^{-J}$, the largest power of L^{-1} admitted by the first four kinds.

No other smallness condition on the coupling constants occurs in these papers: the remaining conditions there are $\lambda > 0$, $\mu_0 > 0$, $m > 0$, conditions on the constants fixed before ε_0, such as $\bar{n} = 13$, the conditions on the exponent p of (3.1), $p > 2$ and $2p \geqq (d+1)\varrho$, $p - 1 \geqq p_1$ of B.1, and the Hölder exponent, the lower bounds on R and b_0 of B.1 and B.2, and the choice of L and of the large block size.[37]

The matrix q is assumed in the lower bound to satisfy $\|q\| \leqq 1$, and the case $N = 2$, $q^2 = -1$ singled out there as the most important one is ours, an antisymmetric 2×2 matrix with $\|q\| = 1$ having $q^2 = -1$; the estimates involving q, the expansion of $U(A)$ to order $\bar{n}$ and the bound on its remainder, use only the unitarity of $U(A)$ and $\|q\| \leqq 1$, so the normalization of q is not a condition on the coupling constants. The counterterm δm^2 depends on q^2 and is a matrix unless q^2 is a scalar; this concerns its form, not its size.[38]

Accordingly the constants are fixed in the order of B.2, with ε_0 as the smallest of the upper bounds of the first four kinds, of the fifth kind of B.1 and of B.2, each of which has a positive right side at every $e > 0$ and $\lambda, m, \mu_0 > 0$; and finally J by the fifth kind. This is what Sect. 5.3 uses.

Appendix C

The theorems of [R 1] and [R 2] are stated for four dimensions. We use them in $d = 3$ in the following form, with the notions of those papers and with d-vectors in place of four-vectors. For an integrand F with loop momenta k and a Zimmermann subspace H of dimension h, given by a basis $t_1, \ldots, t_h, v_1, \ldots, v_{m-h}$ of a natural set of integer linear forms in the loop momenta (the line momenta of F in (i) below, the set named in (ii)), the degree of H is

$$\omega(H) = dh + \mathrm{degr}_t F\,, \tag{C.1}$$

degr_t the ultra-violet degree of [R 1] with respect to t, and the degree of a one particle irreducible subgraph γ with $m(\gamma)$ loops is $\omega(\gamma) = \sum_L \omega(\Delta_L) + \sum_B \omega(V_B) + d\,m(\gamma)$, with the degrees of the lines and vertices of [R 2]. The two conditions on the subtraction degrees $\delta(\gamma)$ are[39]

$$\delta(\gamma) \geqq \omega(\gamma)\,, \qquad \delta(\gamma) \geqq \omega(\bar{\gamma}(U)) + \sum_i \delta(\gamma_i)\,, \tag{C.2}$$

for every one particle irreducible γ and every forest U, $\gamma_1, \ldots, \gamma_r$ being the maximal elements of U properly contained in γ and $\bar{\gamma}(U) = \gamma / \{\gamma_1, \ldots, \gamma_r\}$ the reduced graph.

Lemma C.1. (i) *Let $F = V/C$ be a lattice Feynman integrand of the class of* [R 1] *with positive masses and a numerator periodic in the loop momenta, let its line momenta form a natural set, and let $\omega(H) < 0$ for every Zimmermann subspace H. Then the integral of F over the Brillouin zone converges as $\varepsilon \to 0$ to the absolutely convergent continuum integral, in $d = 3$ as in $d = 4$.*

(ii) *Let Γ be one particle irreducible with positive masses, its internal momenta assigned by the rule of* [R 2]*, and let the subtraction degrees satisfy the two conditions* (C.2) *with the degrees computed in $d = 3$.*

Then $R_\Gamma F$ has negative degree on every Zimmermann subspace formed from the line momenta of Γ and the internal momenta of its one particle irreducible subgraphs, and if the line momenta of Γ and the internal momenta of its subgraphs with $\delta(\gamma) \geqq 0$, the flows of [R 2] *with the external momenta of the subgraph set to zero, form a natural set, the integral of $R_\Gamma F$ converges as $\varepsilon \to 0$.*

[37] The condition $p > 2$ is imposed in Sect. 2 of [Ba 2].
[38] Sect. 1 of [Ba 1].
[39] Sect. 4 of [R 2].

Proof. (i) It is remarked in [R 1] that the considerations there extend to other dimensions, and the proof of the power counting theorem uses the dimension only as the number of components of a momentum: through the measure d^dk, through the decomposition of the domain of integration around the points of $(2\pi/\varepsilon)Z^d$ and the translations by vectors of $(2\pi/\varepsilon)Z^{dm}$, through the counts of components, through the term dh of (C.1) and through the weight of the outer integration in the induction which proves the auxiliary theorem.[40] In each of these places the dimension is a parameter, and the proof holds verbatim for $d = 3$.

(ii) In the proof of the renormalization theorem of [R 2] the dimension enters in the degrees, through the term $d\,m(\gamma)$, in the quantities $M_U(\gamma)$, in the inequality $M_U(\Gamma) \geqq 4h$ and in the target $4h + \mathrm{degr}_t R_\Gamma F < 0$.[41] Here $M_U(\gamma) = 4N_U(\gamma)$, where $N_U(\gamma) = \sum m(\bar\tau(U))$, the sum over the elements τ of U contained in γ, γ included, whose reduced graph $\bar\tau(U)$ is variable on H; the numbers of loops add over a forest, $m(\gamma) = \sum_\tau m(\bar\tau(U))$ with the sum over all the elements τ of U contained in γ, γ included, so that $N_U(\gamma) \leqq m(\gamma)$ counts the loops of γ belonging to variable reduced graphs, and $M_U(\gamma)$ is the number of their real components, $dN_U(\gamma)$ in d dimensions.

The lemma which gives the inequality states $M_U(\gamma) \geqq 4\,\mathrm{rank}_t T^*_\gamma$, the rank of the coefficient matrix of the variables t in the chord momenta of γ. Its proof is an induction over the forest U in which the dimension enters only as the factor four. For a minimal element γ of U, $N_U(\gamma)$ is the number $m(\gamma)$ of chords of γ when $\bar\gamma(U) = \gamma$ is variable, and zero when it is constant, in which case no chord momentum of γ depends on t. At the step, $N_U(\gamma) = m(\bar\gamma(U)) + \sum_i N_U(\gamma_i)$ when $\bar\gamma(U)$ is variable, since the chord sets $T^*_{\gamma_i}$ are disjoint subsets of T^*_γ,[42] so that $T^*_\gamma \setminus \bigcup_i T^*_{\gamma_i}$ has $m(\gamma) - \sum_i m(\gamma_i) = m(\bar\gamma(U))$ elements, and the rank is subadditive, $\mathrm{rank}_t T^*_\gamma \leqq m(\bar\gamma(U)) + \sum_i \mathrm{rank}_t T^*_{\gamma_i}$. When $\bar\gamma(U)$ is constant, $N_U(\gamma) = \sum_i N_U(\gamma_i)$; a chord c of γ which is a chord of no γ_i carries the momentum k^γ_c, which is a combination of the momenta k^γ_L, $L \in \bar\gamma(U)$, constant on H, and, if c lies in γ_i, of its internal momentum in γ_i, a combination of the chord momenta of γ_i; hence $\mathrm{rank}_t T^*_\gamma \leqq \sum_i \mathrm{rank}_t T^*_{\gamma_i}$. Dividing by four gives $N_U(\gamma) \geqq \mathrm{rank}_t T^*_\gamma$, the rank being that of a real matrix, which does not depend on the number of components of the t_j. So $M_U(\gamma) = dN_U(\gamma) \geqq d\,\mathrm{rank}_t T^*_\gamma$ in d dimensions, and at $\gamma = \Gamma$ the rank is h, the map $(t, v) \mapsto k$ being invertible.

The degree lemma[43] then goes through with d in place of 4, with one supplement given below. Its induction uses the degree rules, which do not refer to the dimension, and the two conditions (C.2). The degrees are additive over a forest, the vertices of $\bar\gamma(U)$ obtained by contracting the γ_i carrying degree zero,

$$\omega(\gamma) = \omega(\bar\gamma(U)) + \sum_i \omega(\gamma_i)\,, \tag{C.3}$$

which holds in every dimension since the numbers of loops add, so that for instance the degrees $\delta(\gamma) = \omega(\gamma)$ satisfy (C.2).

The supplement concerns the elements γ of a forest, and Γ itself, whose subtraction degree is negative: there no subtraction is made, and the rule[44] $\mathrm{degr}_t(1 - t^\delta)Y \leqq \mathrm{degr}_{tq} Y - \delta - 1$ is used with $\delta \leqq -1$, where $1 - t^\delta = 1$; what is needed there is $\mathrm{degr}_t Y \leqq \mathrm{degr}_{tq} Y$, from which the rule follows since $-\delta - 1 \geqq 0$. The rule requires, for every δ, that each factor of the denominator of Y depending on q depend on t as well; this is shown below where the rule is used.

For an element γ of a forest U and a line L of γ we write $l_L = k^\gamma_L + q^\gamma_L$, k^γ_L the internal momentum of L in γ and q^γ_L the flow of the external momenta of γ through L. Classifying the elements of U from Γ inwards, we call $\bar\gamma(U)$ constant on H if no k^γ_L, $L \in \bar\gamma(U)$, depends on t; otherwise let σ be the smallest element of U properly containing γ whose reduced graph is

[40] Theorem 1 of [R 1]; the counts of components are in Sect. 9 there.
[41] Sect. 5 and Appendix C of [R 2].
[42] Appendix B of [R 2].
[43] Lemma 5.2 of [R 2].
[44] Lemma 3.1 of [R 2].

constant, and call $\bar\gamma(U)$ variable if every l_L, $L \in \bar\gamma(U)$, depends on t when $q^\sigma = 0$, or, when there is no such σ, when the external momenta of Γ are held fixed on H.

If $\bar\gamma(U)$ is constant, the external momenta of the maximal elements γ_i of U below γ are, at $q^\gamma = 0$, integer combinations of the k_L^γ of their legs in $\bar\gamma(U)$ and do not depend on t; so for a line L of such a γ_i the dependence on t of l_L at $q^\gamma = 0$ is that of $k_L^{\gamma_i}$. This is the sense in which the forest formula[45] writes $R_\Gamma F$ on H as a sum, over the forests U complete on H, of terms in which an element γ_i of U enters through $1 - t^{\delta(\gamma_i)}$ when $\bar\gamma_i(U)$ is variable and through $-t^{\delta(\gamma_i)}$ otherwise, the momenta of the lines of an element of the second kind being Taylor expanded about $q^{\gamma_i} = 0$.

In the computation of $\mathrm{degr}_{tq^\gamma} Y_\gamma$ the external momenta q^γ are independent variables, and a line of γ depends on t through its internal momentum k_L^γ alone; inside a constant reduced graph the external momenta of its maximal subelements are constant on H, so that the two notions agree for the lines of the variable reduced graphs at the two places where the rule is used.

The inequality $\mathrm{degr}_t Y \leqq \mathrm{degr}_{tq} Y$ holds for $Y = P/C$ of the class of [R 1] whenever every factor of C which depends on the external momenta q of γ depends on t as well, since then $\mathrm{degr}_t C = \mathrm{degr}_{tq} C$ while $\mathrm{degr}_t P \leqq \mathrm{degr}_{tq} P$, the scaling in t alone being obtained from the scaling in (t, q) by a Taylor expansion in q, which can only lower the leading power.

At the two places where the rule is used the factors of the denominator of Y_γ are of two kinds. First, the propagators of the lines of the variable reduced graphs $\bar\tau(U)$, $\tau \subseteq \gamma$, with no constant element of U between τ and γ; their momenta are $k_L^\gamma + q_L^\gamma$, and k_L^γ depends on t by the definition above, since the frame of the definition for these lines is that of γ itself, when no element of U containing γ is constant, or that of the constant element next above γ, in which the external momenta of γ do not depend on t.

Second, the powers of $e(k_L^\tau) + \mu^2$ produced by the Taylor operators, which carry, for every τ whose Taylor operator has acted, variable or constant, the internal momenta k_L^τ of the lines of τ and do not depend on q^γ, since the internal momentum of a line of τ is an integer combination of the chord momenta of τ, the chords of τ being chords of γ, on which the line momentum is the internal momentum of γ, the flow of the external momenta of γ vanishing off its tree; the Taylor coefficients are numerators. The hypothesis of the rule is therefore satisfied there.

When $\bar\Gamma(U)$ is constant on H the strict inequality comes from the elements below Γ: if $\bar\gamma(U)$ is constant and $M_U(\gamma) > 0$ then $\mathrm{degr}_t Y_\gamma \leqq -M_U(\gamma) - 1$, since the external momenta of the maximal elements γ_i of U below γ are then constant on H at $q^\gamma = 0$, so that each variable γ_i contributes $-M_U(\gamma_i) - 1$ by the rule and each constant γ_i with $M_U(\gamma_i) > 0$ contributes the same by induction, while at least one γ_i has $M_U(\gamma_i) > 0$ because $M_U(\gamma) = \sum_i M_U(\gamma_i)$; at $\gamma = \Gamma$ this gives $-M_U(\Gamma) - 1 \leqq -dh - 1$.

The lemma gives $\mathrm{degr}_t R_\Gamma F \leqq -dh - 1$ on every Zimmermann subspace. The convergence then follows from (i) applied to $R_\Gamma F$, which is periodic in the loop momenta and of the class of (i) by the construction of the subtracted integrand,[46] its denominators carrying the line momenta of Γ and the internal momenta of its subgraphs with $\delta(\gamma) \geqq 0$, which is the natural set of the hypothesis; it contains the loop momenta, so that its Zimmermann subspaces are among those of the first part of the proof, on which the degrees are negative. This completes the proof of Lemma C.1.

Appendix D

We give here the proofs of three steps of Sect. 5.1: the bounds on the scalar field on the box of the third replacement (Lemma D.1), the bound on the constant E_{fin} (Lemma D.2), and the cube step of the basic estimate (5.4) with the cubes of the first replacement (Lemma D.3). The bounds on the propagator of a box at a constant field which these proofs use are collected in

[45] Lemma 5.1 of [R 2], in Sect. 5 there.
[46] Sects. 4 and 5 of [R 2].

D.4.

D.1. The bounds on the scalar field on the box of Sect. 5.1. We use the notation of Sect. 5.1 on the unit lattice $T_1^{(k)}$ of the kth step: $\Lambda_j = \Lambda_j^{(k-1)\prime}$ are the regions of the previous step, consecutive ones separated by a corridor of width exceeding w_k, $\Omega = B^k(\Lambda_2)$ is the domain of the propagator $G_k(\Omega, A^{(k)})$, and ϕ is the scalar field of the previous step, which satisfies the restrictions (3.2) with $k-1$ in place of k on Λ_{-1}, the region on which they are imposed: $|\phi| \leqq \Upsilon = p(L^{k-1}\varepsilon)[\lambda(L^{k-1}\varepsilon)^{4-d}]^{-1/4}$ on the sites and $|D_A\phi| \leqq p(L^{k-1}\varepsilon)$ on the bonds of Λ_{-1}, the transporter of a bond being that of the field $A^{(k)}$ along its contour; only the restriction $\Lambda_6\phi$ of ϕ to Λ_6 enters the source. The background $A^{(k)}$ satisfies $|\partial^\eta A^{(k)}| \leqq Cp(L^k\varepsilon)$ on $B^k(\Lambda_2)$, the bound of Sect. 5.1 with the cut-off of the second replacement, and $A^{(k)}$, for $L^k\varepsilon$ small, and every constant field are regular in the sense of Definition 3.2.

Two facts about the propagators are used below. For the propagator at the background $A^{(k)}$ we use the propagator bounds at depth R_1 of Sect. 5.1, which hold on a domain $\Omega = B^k(\Omega^{(k)})$, $\Omega^{(k)}$ any union of large blocks, for a regular field, at points at depth at least R_1 in Ω, with no condition on the shape of Ω. For the propagator at a constant field on a rectangular parallelepiped of large blocks with Neumann conditions we use the bound (3.8) up to its faces, proved in D.4. Depths are measured in the supremum norm; the Euclidean depth is at least as large.

Lemma D.1. *Under the restrictions* (3.2) *of the step* $k-1$*, for* $y \in \Lambda_7$*,* $x \in B^k(y)$ *and bonds* $b \subset B^k(\Lambda_7)$*,*

$$\phi^{(k)}(x) = U(A^{(k)}(\Gamma_{x,y}))\phi(y) + O(p(L^k\varepsilon))\,, \qquad D^\eta_{A^{(k)}}\phi^{(k)}(b) = O(p(L^k\varepsilon))\,, \tag{D.1}$$

where $\phi^{(k)} = a_k G_k(\Omega, A^{(k)})Q_k^*(A^{(k)})\Lambda_6\phi$*, with constants independent of* ε*, of* k *and of the position of* y *in* Λ_7*.*[47]

Proof. Let $s = \min\{4r(L^k\varepsilon),\, 5w_k - 2M_b\}$, let $\square$ be the cube of large blocks of radius s about the large block containing y, in the supremum norm, in which the corridors are measured, so that $\square \subset \Omega$, Λ_7 lying five corridors inside Λ_2; and let $\square_1\phi$ be the restriction of $\Lambda_6\phi$ to the cube of radius $s/2$ about y. The cube $\square$ is a rectangular parallelepiped and a union of large blocks, and the point x and the bonds of $B^k(y)$ lie at Euclidean depth at least $s - \sqrt{d}\,M_b$ in $\square$ and at least $5w_k - \sqrt{d}\,M_b$ in Ω. We assume that these depths are at least R_1, so that the propagator bounds at depth R_1 hold at x on both domains; at $L = 5$ this follows from $r(\varepsilon_0) \geqq R_1 + (3+\sqrt{3})M_b$, since $5w_k = r(L^{k-1}\varepsilon) - M_b \geqq r(\varepsilon_0) - M_b$ for $k \leqq K - J$ and r decreases with its argument. We also assume $r(L^{k-1}\varepsilon) \leqq 2L\, r(L^k\varepsilon)$, which (B.1) gives at $L = 5$; then $s/2 \geqq w_k$, since $w_k \leqq r(L^{k-1}\varepsilon)/L \leqq 2r(L^k\varepsilon)$ and $3w_k \geqq \frac{3}{5}(R_1 + (2+\sqrt{3})M_b) \geqq 2M_b$.

First, the source is cut. The field $\Lambda_6\phi - \square_1\phi$ vanishes within $s/2$ of y, so that by the bound (3.8) at depth R_1 applied to $G_k(\Omega, A^{(k)})Q_k^*(A^{(k)})$ at the point x, which lies at depth at least R_1 in Ω, and by the bound on ϕ

$$|a_k(G_k(\Omega, A^{(k)})Q_k^*(A^{(k)})(\Lambda_6\phi - \square_1\phi))(x)| \leqq Ca_k\Upsilon(1+s)^{d-1}e^{-\delta_0(s/2-\sqrt{d}\,M_b)}\,. \tag{D.2}$$

Second, the domain is changed. We write $G_k(\Omega, A^{(k)}) = G_k(\square, A^{(k)}) - \delta G_k(\square, \Omega, A^{(k)})$, where the last operator is the difference of the two propagators; by the bound with the factor (3.11) on the difference of the propagators of the nested domains $\square \subset \Omega$, its kernel at x applied to $Q_k^*\square_1\phi$ is bounded by $Ca_k\Upsilon(1+s)^{d-1}e^{-\delta_0(3s/2-2\sqrt{d}\,M_b)}$, x lying at depth $s - \sqrt{d}\,M_b$ and the source at depth $s/2 - \sqrt{d}\,M_b$ in $\square$. The same holds for the covariant derivative. Hence $\phi^{(k)}(x) = a_k(G_k(\square, A^{(k)})Q_k^*(A^{(k)})\square_1\phi)(x)$ up to the error (D.2).

Third, the background is replaced by its constant value $A_0 = A^{(k)}(y)$. On the convex cube $\square$ we have $|A^{(k)}(z) - A_0| \leqq Cp(L^k\varepsilon)\, d\,(s + 2M_b)$, along a rectilinear path, and $|\partial^\eta A^{(k)}| \leqq Cp(L^k\varepsilon)$

[47] Lemma 2.4 of [Ba 2], the equalities (2.65)–(2.66) there; the five steps of the proof below correspond to (2.67), (2.68), (2.69)–(2.74), (2.75)–(2.76) and (2.77) there.

by the bound on the background. Write $G = G_k(\Box, A^{(k)})$, $G_0 = G_k(\Box, A_0)$, $\psi = \Box_1\phi$ and $V = G^{-1} - G_0^{-1}$. The identity $G = G_0 - GVG_0$ gives

$$\begin{aligned} a_k(GQ_k^*(A^{(k)})\psi)(x) = a_k(G_0Q_k^*(A_0)\psi)(x) + a_k(G_0[Q_k^*(A^{(k)}) - Q_k^*(A_0)]\psi)(x) \\ - a_k(GVG_0Q_k^*(A^{(k)})\psi)(x)\,. \end{aligned} \tag{D.3}$$

The constant field A_0 is regular, and by (3.8) on the parallelepiped $\Box$ up to its faces, as in D.4, the kernels of G_0 and of $D_{A_0}G_0$ are summable over $\Box$ uniformly in ε and k, so that $h = G_0Q_k^*(A^{(k)})\psi$ satisfies $\sup_\Box |h| + \sup_\Box |D_{A_0}h| \leqq C\Upsilon$. Writing $D_{A^{(k)}} = D_{A_0} + E$, where by (2.1) $(Ef)(b) = \epsilon(b)f(b_+)$ with $|\epsilon(b)| \leqq e(L^k\varepsilon)^{2-d/2}|A^{(k)}(b) - A_0|$ and $|\epsilon(b) - \epsilon(b')| \leqq e(L^k\varepsilon)^{2-d/2}\eta|\partial^\eta A^{(k)}|$ on neighbouring bonds, we have $V = D_{A_0}^*E + E^*D_{A_0} + E^*E + a_k[P_k(A^{(k)}) - P_k(A_0)]$, $P_k = Q_k^*Q_k$, and the difference of the averaging operators is bounded through $|U(A^{(k)}(\Gamma)) - U(A_0(\Gamma))| \leqq e(L^k\varepsilon)^{2-d/2}|\Gamma| \sup_\Box |A^{(k)} - A_0|$ on the contours Γ of the blocks. So $\sup_\Box |Vh| \leqq Ce(L^k\varepsilon)^{2-d/2}p(L^k\varepsilon)(1+s)^2\Upsilon$ for $L^k\varepsilon$ so small that $e(L^k\varepsilon)^{2-d/2}p(L^k\varepsilon) \leqq 1$, and the same bound holds for $\sup_\Box |[Q_k^*(A^{(k)}) - Q_k^*(A_0)]\psi|$.

The outer factor G in (D.3) is evaluated at x only, where (3.8) at the point x, at depth at least $s - \sqrt{d}\,M_b \geqq R_1$ in $\Box$, gives $|a_k(GVh)(x)| \leqq Ca_k \sup_\Box |Vh|$; the second term of (D.3) is bounded in the same way, by (3.8) for G_0. Hence

$$\begin{aligned} a_k(GQ_k^*(A^{(k)})\psi)(x) &= a_k(G_0Q_k^*(A_0)\psi)(x) + E_3\,, \\ |E_3| &\leqq Ce(L^k\varepsilon)^{2-d/2}p(L^k\varepsilon)(1+s)^2\Upsilon \leqq C(L^k\varepsilon)^{\kappa_0}\,, \end{aligned} \tag{D.4}$$

for every $\kappa_0 < \frac{1}{4}$ when $d = 3$, since $e(L^k\varepsilon)^{1/2}\Upsilon = e\lambda^{-1/4}L^{1/4}(L^k\varepsilon)^{1/4}p(L^{k-1}\varepsilon)$ and $s \leqq 4r(L^k\varepsilon)$. The expansion is taken in the order which keeps $G_k(\Box, A^{(k)})$ at the point x, where the propagator bounds at depth R_1 apply. The same identity, with $D^\eta_{A^{(k)}}$ at a bond $b \subset B^k(y)$ acting on the outer propagator, and with $D_{A^{(k)}} = D_{A_0} + E$ in the first two terms, the terms in E being bounded like E_3, gives the derivative version with the same bound.

Fourth, the constant field is removed by a gauge transformation: $A_0 = \partial^\eta\lambda_0$ with $\lambda_0(z) = A_0(\Gamma_{z,y})$ on $\Box$, which does not wrap around T, its diameter being at most $8r(L^k\varepsilon)+3M_b$, which the sides of T chosen in B.2 exceed, so that $G_k(\Box, A_0) = U(\lambda_0)G_k(\Box, 0)U(\lambda_0)^*$, $Q_k(A_0) = U(\lambda_0)Q_k(0)U(\lambda_0)^*$ and, with $\phi' = U(\lambda_0)^{-1}\phi$,

$$a_k(G_k(\Box, A_0)Q_k^*(A_0)\Box_1\phi)(x) = U(A_0(\Gamma_{x,y}))\, a_k(G_k(\Box, 0)Q_k^*\Box_1\phi')(x)\,. \tag{D.5}$$

Fifth, the constants are reproduced. From $G_k(\Box, 0)^{-1} = -\Delta^{\eta,N} + m^2(L^k\varepsilon)^2 + a_kQ_k^*Q_k$ with Neumann conditions on $\partial\Box$ we have $a_kG_k(\Box, 0)Q_k^*1 = 1 - m^2(L^k\varepsilon)^2G_k(\Box, 0)Q_k^*1$, so that

$$\begin{aligned} a_k(G_k(\Box, 0)Q_k^*\Box_1\phi')(x) - \phi'(y) = a_k(G_k(\Box, 0)Q_k^*[\Box_1(\phi' - \phi'(y))])(x) \\ - \phi'(y)\, a_k(G_k(\Box, 0)Q_k^*1_{\Box\setminus\Box_1})(x) \\ - \phi'(y)\, m^2(L^k\varepsilon)^2(G_k(\Box, 0)Q_k^*1)(x)\,. \end{aligned} \tag{D.6}$$

The first term is $O(p(L^k\varepsilon))$ up to an exponentially small error: the cube of radius $s/2$ about y lies within three corridors of Λ_7, since $s/2 \leqq \frac{5}{2}w_k - M_b$, hence in Λ_{-1}, where $|D_A\phi| \leqq p(L^{k-1}\varepsilon)$ with the transporter $U(A^{(k)}(\Gamma_b))$ on the bond b, so that along a rectilinear lattice path from y to z in this cube, bond by bond, $|\phi'(b_+) - \phi'(b_-)| = |\phi(b_+) - U(A_0(\Gamma_b))\phi(b_-)| \leqq p(L^{k-1}\varepsilon) + e(L^k\varepsilon)^{2-d/2} \sup_\Box |A^{(k)} - A_0|\,\Upsilon = O(p(L^k\varepsilon))$, hence $|\phi'(z) - \phi'(y)| \leqq O(p(L^k\varepsilon))|z - y|$, and the kernel of $a_kG_k(\Box, 0)Q_k^*$ decays exponentially, while $\Box_1 \setminus \Lambda_6$ lies at distance at least $w_k - \sqrt{d}\,M_b$ from x, so that its contribution, $-\phi'(y)\, a_k(G_k(\Box, 0)Q_k^*1_{\Box_1\setminus\Lambda_6})(x)$, is bounded by $Ca_k\Upsilon(2s + 2M_b)^d e^{-\delta_0(w_k - \sqrt{d}\,M_b)}$ and is absorbed in the exponential errors below.

The second term is exponentially small, $\Box\setminus\Box_1$ lying at distance at least $w_k - \sqrt{d}\,M_b$ from x, since $s/2 \geqq w_k$. The third is the mass term, which is $O(p(L^k\varepsilon))$ with a constant depending on

m and λ, since $m^2(L^k\varepsilon)^2\Upsilon = m^2\lambda^{-1/4}L^{1/4}(L^k\varepsilon)^{7/4}p(L^{k-1}\varepsilon)$ for $d=3$ and $(G_k(\Box,0)Q_k^*1)(x)$ is bounded. With (D.2)–(D.5) and $U(A_0(\Gamma_{x,y}))\phi(y) = U(A^{(k)}(\Gamma_{x,y}))\phi(y) + O((L^k\varepsilon)^{\kappa_0})$ this gives the first statement of (D.1).

Finally, a bond $b \subset B^k(y)$ lies at depth at least $s - \sqrt{d}\,M_b$ in $\Box$, the derivative bound in (3.8) applies at b, and the steps above with $D^{\eta}_{A^{(k)}}$ at b acting on the outer propagator give the second statement of (D.1). Every exponential error above is bounded by $Ca_k\Upsilon(1 + r(L^k\varepsilon))^d e^{-\delta_0(w_k - \sqrt{d}\,M_b)}$, since $s/2 \geqq w_k$, and this is smaller than any power of $L^k\varepsilon$ once $L^k\varepsilon$ is small, w_k growing like $r(L^{k-1}\varepsilon)$ and ϱ being larger than one; this is the restriction on ε_0 from the decay across one corridor stated in B.2. The constants depend on d, L, M_b, on the coupling constants and on the constants of the propagator bounds, and not on ε, on k or on the position of y in Λ_7; the value of L enters only through w_k and the conditions stated. This completes the proof of Lemma D.1.

D.2. The bound on $E_{\rm fin}$. Let $\bar{k} = K - J$ be the last step, so that $L^{\bar{k}}\varepsilon = \varepsilon_0$, and let the lattice be $T_1^{(\bar{k})}$, on which the lines of E_1 are the full propagators C_s, C_v of (3.47) and the lines of $E_{\bar{k}}$ are $G_{\bar{k}}(0)$ and the vector propagator $G^v_{\bar{k}}$, the same operator with μ_0 in place of m and d components. We write $S = \max\{(m\varepsilon_0)^{-2}, (\mu_0\varepsilon_0)^{-2}\} + a_{\bar{k}}^{-1}$. From $G_{\bar{k}}(0)^{-1} = C_s^{-1} + a_{\bar{k}}Q^*_{\bar{k}}Q_{\bar{k}}$ we have $C_s - G_{\bar{k}}(0) = a_{\bar{k}}G_{\bar{k}}(0)Q^*_{\bar{k}}Q_{\bar{k}}C_s = a_{\bar{k}}C_sQ^*_{\bar{k}}Q_{\bar{k}}G_{\bar{k}}(0)$, and therefore

$$C_s - G_{\bar{k}}(0) = H\Gamma H^*\,, \qquad H = a_{\bar{k}}G_{\bar{k}}(0)Q^*_{\bar{k}}\,, \qquad \Gamma = Q_{\bar{k}}C_sQ^*_{\bar{k}} + a_{\bar{k}}^{-1}\,, \tag{D.7}$$

and the same for the vector propagators with μ_0 in place of m. The matrix Γ on the blocks of $T_1^{(\bar{k})}$ is positive and has nonnegative entries, the kernel of C_s being nonnegative, as the terms of its expansion in the hopping term are, and $\Gamma 1 = [(m\varepsilon_0)^{-2} + a_{\bar{k}}^{-1}]1$, since $C_s1 = (m\varepsilon_0)^{-2}1$ on the torus; so its entries and its row sums are at most S. The field $H(x,B) = a_{\bar{k}}(G_{\bar{k}}(0)1_B)(x)$ of a block B and its first differences are bounded by $Ce^{-\delta_0 \mathrm{dist}(x,B)}$, and its localized Hölder norms, the norms (5.5) on a unit cube $\Box$, by $Ce^{-\delta_0\mathrm{dist}(\Box,B)}$, by the propagator bounds (3.8)–(3.9) on the torus, where the depth condition is void.

Lemma D.2. *Let $E_{\bar{k}}$ be the sum of the vacuum graphs of E_1 with the propagators $G_{\bar{k}}(0)$, $G^v_{\bar{k}}$ in place of C_s, C_v, without the block vertices* (3.41)–(3.43)*, and with the coefficients δm^2 of the counterterm vertices unchanged, and let $E_{\rm fin} = E_1 - E_{\bar{k}}$. Then*

$$|E_{\rm fin}| \leqq C(1+S)^N |T_1^{(\bar{k})}| \leqq C(\varepsilon_0)|T|\,, \tag{D.8}$$

with N depending on $\bar{n}$ only; the constant depends on ε_0, on the coupling constants and on the constants of the basic estimate, and not on ε, K or T.[48]

Proof. Let ξ be the Gaussian field on the blocks with covariance Γ, let $\zeta = H\xi$, and let ξ' be the Gaussian field with the covariances $G_{\bar{k}}(0)$, $G^v_{\bar{k}}$, independent of ξ; then $\xi' + \zeta$ has the covariances C_s, C_v by (D.7). Let V be the interaction built from the vertices (3.34), (3.35), (3.38), (3.39) at the step $\bar{k}$, the vertices of E_1 in (3.53), with the external fields set to zero and the counterterm vertices carrying the coefficients δm^2 of (3.51) computed with C_s, C_v. The constants E_1 and $E_{\bar{k}}$ are the sums of the connected vacuum graphs of $\ln E\, e^{-V}$ with the respective lines, so that by the independence of ξ and ξ'

$$E_{\rm fin} = E_1 - E_{\bar{k}} = \ln E_\zeta \exp W(\zeta)\,, \qquad W(\zeta) = \ln E_{\xi'} e^{-V(\xi'+\zeta)} - \ln E_{\xi'} e^{-V(\xi')}\,, \tag{D.9}$$

as power series in e, λ truncated at the order $\bar{n}$ in the sense of (3.53). We write the coefficients as $\delta m^2_{\rm fin} + \delta m^2_{\bar{k}}$, the second computed with the lines $G_{\bar{k}}$, the counterterm vertices of lower order inside the graphs of $\delta m^2_{\bar{k}}$, of $E_{\bar{k}}$ and of the nodes below carrying the full coefficients δm^2. The block vertices do not occur in E_1; the vacuum graphs which they generate at the step $\bar{k}$ belong to the constant $S^{(k),1}(T^{(k)},0,0,0,0)$ removed in the normalized action of Sect. 3.2, not to $E_{\rm fin}$.

[48] $E_{\bar{k}}$ and $E_{\rm fin}$ are the constants E_K and $E_{\rm fin}$ of (1.26) of [Ba 3], read as in Sect. 5.1.

The full coefficients are essential. If the vacuum counterterm were computed with $\delta m^2_{\bar{k}}$ alone, the difference would contain the vacuum graphs with lines $G_{\bar{k}}$ and a vertex $\delta m^2_{\mathrm{fin}}$, among them the graph of order λe^2 in which the loop closed by the vertex $\delta m^2_{\mathrm{fin},\lambda}$ carries the subtracted exchange self-energy, a graph of degree zero without counterterm, whose value per site grows like $\delta m^2_{\mathrm{fin},\lambda} e^2 \ln(\varepsilon_0/\varepsilon)$ by (5.12). The bound (D.8) is a bound on $E_1 - E_{\bar{k}}$ with the same coefficients δm^2 in both.

The expansion of W in powers of ζ is the sum of the connected graphs with lines $G_{\bar{k}}(0)$, $G^v_{\bar{k}}$ and $n \geqq 1$ external legs carrying ζ, and at fixed order and type this sum is a renormalized class at the step $\bar{k}$, which contains every partner of each of its graphs. The linked cluster theorem for the field ζ then writes E_{fin}, at each order, as a finite sum over connected skeleton graphs: multigraphs whose nodes are these classes, every leg of a node being an end of exactly one edge, and whose edges carry the covariance $H\Gamma H^*$; by (D.7) each leg carries a field $H(\cdot, B)$ and each edge a factor $\Gamma(B, B')$, summed over the blocks B, B'.

A node, with the fields $H(\cdot, B_l)$ on its legs and its vertices localized in unit cubes $\{\square(v)\}$, one of them fixed, is bounded by the basic estimate (5.4) at the step $\bar{k}$, which applies on the torus, a union of large blocks whose sides exceed the cubes of its proof (Sect. 5.1, Appendix B), the conditions on the domains being satisfied there,[49] by $C \exp[-\delta\, d(\{\square(v)\})] \prod_l \|h_l H(\cdot, B_l)\|_{1,\alpha_0}$ with the norms (5.5), hence by $C \exp[-\delta\, d(\{\square(v)\})] \prod_l e^{-\delta_0 \mathrm{dist}(\square(v_l), B_l)}$, v_l being the vertex at which the leg l sits.

The counterterm vertices of a node carry the full coefficients $\delta m^2 = \delta m^2_{\mathrm{fin}} + \delta m^2_{\bar{k}}$; the part $\delta m^2_{\bar{k}}$, computed with the lines $G_{\bar{k}}$, is the counterterm of the class, matched by its two-point subgraphs as in Sect. 3.5 and covered by the basic estimate, while the finite part is a two-point vertex with the coefficient $\delta m^2_{\mathrm{fin}}$, bounded uniformly in ε by the estimate below.

Its degree in the sense of (3.48)–(3.49) is that of a two-point vertex, $d + 2(1 - \frac{d}{2}) = 2$, so that inserting it on a line raises the degree of every subgraph containing it by 2, and since the divergent subgraphs of (3.50) with external legs have degree at least -1 in $d = 3$, apart from the current tadpole of degree -2, which vanishes by $\mathrm{tr}\, q = 0$ with or without the insertion, $\delta m^2_{\mathrm{fin}}$ being a multiple of the identity for $q^2 = -1$, every subgraph containing it has positive degree, as holds for every graph containing a counterterm vertex with a finite coefficient;[50] the proof of the basic estimate covers the classes containing it, the coefficient entering the constant.

At order two, $\delta m^2_{\mathrm{fin}}$ is the difference of the three graphs of (3.51), the two tadpoles and the exchange bubble, with the lines C_s, C_v and with the lines $G_{\bar{k}}(0)$, $G^v_{\bar{k}}$; these graphs have no divergent proper subgraph, so each may be bounded separately, and the difference is a sum of terms with one or two lines replaced by $H\Gamma H^*$, each at most $C(S + S^2)$ by the bounds on H and its first differences and by the entries and row sums of Γ, one lattice derivative of the scalar line of the exchange bubble having been summed by parts onto the factor $H\Gamma H^*$, whose first differences are bounded by CS.

At order four, $\delta m^2_{\mathrm{fin}}$ is the difference of the zero-momentum values of the one particle irreducible two-point functions of order four with the two kinds of lines. The reducible two-point graphs of order four are the chains of two two-point graphs of order two joined by one scalar line; their zero-momentum value is $\Sigma^{(2)}(0)^2 C(0)$ with the respective lines, which vanishes for the lines C_s, C_v by the definition of δm^2 at order two, and is at most $(\delta m^2_{\mathrm{fin},(2)})^2 a^{-1}_{\bar{k}}$ for the lines $G_{\bar{k}}$, since $G_{\bar{k}}(0)1 = (m^2\varepsilon_0^2 + a_{\bar{k}})^{-1}1$. The remaining difference is that of the connected two-point functions at zero momentum, that is with the constant external field one on the two amputated legs. Writing each line C_s, C_v as $G + H\Gamma H^*$ by (D.7), it is the sum over the connected two-point skeleton graphs of order four with at least one edge, whose nodes are renormalized classes at the step $\bar{k}$ with the lines $G_{\bar{k}}(0)$, $G^v_{\bar{k}}$, the field one on the amputated legs and the fields $H(\cdot, B)$ on the legs which carry edges. The two-point subgraphs of order two

[49] They are stated before (1.25) of [Ba 3].

[50] Corollary 2.3 of [Ba 3].

inside a node carry the full coefficient $\delta m^2_{(2)}$, of which $\delta m^2_{k,(2)}$ is the counterterm of the class and $\delta m^2_{\mathrm{fin},(2)}$ the bounded vertex above; no graph of a node is taken separately, the tadpole of order λ with lines G and its counterterm leaving $-\lambda F(u,u)$ only, $F = C_s - G_{\bar{k}}(0)$. Each node is bounded by the basic estimate with $\|h \cdot 1\|_{1,\alpha_0} \leqq 1$, the tree decay sums the positions of all vertices but one, and the gluing below gives a factor S for each edge; the number of skeleton graphs being finite, the difference is bounded by $C(1+S)^{N_4}$ with a fixed integer N_4.

Finally the nodes are glued. Summing the bound of each node over the positions of its vertices and over the blocks at the ends of its legs, with one vertex of one node fixed, the tree decays of the nodes and of the fields H and the row sums and the entries of Γ give a factor $CS^{|E|}$ for a skeleton graph with $|E|$ edges, the sum over the fixed vertex gives $|T_1^{(\bar{k})}|$, and the number of skeleton graphs at each order is finite. This gives (D.8) with N depending on $\bar{n}$ only, since $|T_1^{(\bar{k})}| = \varepsilon_0^{-d}|T|$. No decay of the difference kernels is used: the infrared is controlled by the row sums of Γ alone, which is where ε_0 enters. This completes the proof of Lemma D.2.

D.3. The cube step of the basic estimate. We use the notation of Sect. 5.1 on the unit lattice of the kth step: $\Omega = B^k(\Lambda_2^{(k-1)\prime})$ is the domain of the propagator $G_k(\Omega, B)$, B the regular vector field of the estimate, and the vertices of a renormalized class lie within a unit cube of $B^k(\Lambda_7^{(k-1)\prime})$ or of the support of the external field.

In the cube step the localization cubes of a connected component of the class, which lie in a cube of side $\ell_{\bar{n}}$ depending on $\bar{n}$ only, the lines of a component being localized in cubes at distance zero from one another after the lines between distant localizations have been moved into the external fields, are enclosed in concentric cubes $\square_1 \subset \square$ of large blocks; $G_k(\Omega, B)$ is written $G_k(\square, B) + \delta G_k(\Omega, \square, B)$ with the difference treated as an external field, B is replaced on $\square$ by a constant field B_0 and the difference $B' = B - B_0$ is expanded, B_0 is removed by a gauge transformation, and $G_k(\square, 0)$ is replaced by $G_k(0)$, the difference being again an external field (Sect. 5.1). The step presupposes $\square \subset \Omega$, the Neumann operator $G_k(\square, B)$ being built from the bonds of $\square$, on which B is defined only if $\square \subset \Omega$.

Lemma D.3. *Let $\square_1 \subset \square$ be concentric cubes of large blocks of the constant sides $r^\flat$, $3r^\flat$ of (5.7). Then the proof of the basic estimate (5.4) goes through, with a constant depending on α_0, $\bar{n}$, d, a, M_b and R_1 of the same form as before, for every renormalized class whose old vertices, the vertices present before the expansion in B', lie at depth exceeding $D^\flat$ of (5.8) in Ω. In particular the estimate holds with its hypothesis on the support of the external vector field replaced by this condition, in both of its uses, once R satisfies (5.9).*

Proof. The side of the cubes enters the proof in the following places.

First, the containment. For an old vertex x_v of the component,

$$\operatorname{dist}(x_v, \Omega^c) > \tfrac{3}{2}r^\flat + 2M_b + \ell^\sharp \quad \text{implies} \quad \square \subset \Omega \quad \text{and} \quad \operatorname{dist}(\square_1, \square^c) \geqq r^\flat\,, \tag{D.10}$$

the cubes $\square_1$, $\square$ being concentric unions of large blocks of sides $r^\flat$, $3r^\flat$; this is the triangle inequality, the center of $\square$ lying within $\frac{1}{2}(\ell^\sharp + M_b)$ of x_v, the centers of the cubes of large blocks of side $r^\flat$ forming a lattice of spacing M_b. This is the only place where the side $r(L^k\varepsilon)$ of the original cubes is compared with the depth of a vertex, and $r^\flat$ serves by hypothesis.

Second, the classes with an old vertex lie in $\square_1$. The expansion in B' uses the expansions of the transporters and of the propagator in the vector field to the order $\bar{n}$,[51] and only lines between localizations whose cubes meet remain internal lines of a class (Sect. 5.1); so along an expanded line a new vertex advances by at most two units, the expansion of a line to the order $\bar{n}$ produces at most N new vertices, N depending on $\bar{n}$ only, the new vertices of a component lie within $2N$ of its cube, and the localization cubes of all graphs of a connected component lie in a cube of side $3\ell_{\bar{n}}$ concentric with the cube of one member.

[51] (3.14), (3.44) and (3.45) of [Ba 1].

The members of a class arise from one graph by replacing divergent subgraphs by counterterm vertices with the same localization or by adding the partners of a renormalized subgraph, the localizations of the graphs of a class being in agreement by the construction of the classes,[52] operations which keep every localization cube within one unit of a cube of the given graph, and $3\ell_{\bar n} \geqq \ell_{\bar n} + 2$. Hence the class lies in a cube of side $\ell^\sharp = 3\ell_{\bar n} + 4N$, and $3\ell^\sharp \leqq r^\flat$.

Third, the two differences of propagators. The operator $\delta G_k(\Omega, \square, B)$ is evaluated at points within two units of $\square_1$, the unit cubes of the localization and their η-neighbourhoods, hence at depth at least $r^\flat - 2 \geqq R_1$ in $\square$, with sources in the same region; by the propagator bounds at depth R_1 for the pair $\square \subset \Omega$, whose bounds on the kernel, its first covariant differences and their Hölder quotients carry the factor $\exp[-\delta_0(\mathrm{dist}(x, \square^c) + \mathrm{dist}(x', \square^c))]$ at points at depth R_1 in $\square$, its norms (5.5) are bounded by $Ce^{-\delta_0(r^\flat - 2)}$.

The operator $\delta G_k(\square, 0) = G_k(\square, 0) - G_k(0)$, $G_k(0)$ the propagator of $T^{(k)}$ at the field zero, is bounded in the same norms by the same bounds for the pair $\square \subset T^{(k)}$, both arguments lying at depth at least $r^\flat - 2 \geqq R_1$ in $\square$, the torus being B^k of a union of large blocks. Both are external fields of dimension zero in the power counting,[53] and a bound on their norms is all that is used of them; the small factors $e^{-\delta r(L^k\varepsilon)}$ which the proof produces at changes of domain come from the lines removed before the cube step and depend on Ω only.

Fourth, the constant field. For $x \in \square$ and c the center of $\square$ we have $|B(x) - B(c)| \leqq d(\frac{3}{2}r^\flat + M_b) \sup_\square |\partial^\eta B| = O(r^\flat p(L^k\varepsilon))$, and we take $B_0 = B(c)$; this replaces the bound $O(r(L^k\varepsilon)p(L^k\varepsilon))$ of the original cube. Every use of this bound is an upper bound increasing in the size of B': the vertices carry $e(L^k\varepsilon)^{2-d/2}|B'|$, the remainder of the expansion of the propagator carries a power of $e(L^k\varepsilon)^{2-d/2} \sup |B'|$,[54] and the comparison of the norms (5.5) at B_0 and at B costs a factor $1 + O(r^\flat p(L^k\varepsilon))$ for each leg, absorbed as before; so the smaller cube gives the stronger bound.

Fifth, the gauge transformation. We have $B_0 = \partial^\eta \lambda_0$ with $\lambda_0(x) = B_0 \cdot (x - c)$ on $\square$; the covariant Laplacian with Neumann conditions on the bonds of $\square$ and the averaging operators Q_k, whose contours lie inside blocks contained in $\square$, transform covariantly, $G_k(\square, B_0) = U(\lambda_0)G_k(\square, 0)U(\lambda_0)^*$, and every class is invariant when the external scalar fields are transformed at the same time, as in the proof of Lemma D.1; this is exact as long as $\square$ does not wrap around the torus (Appendix B).

Sixth, the rest of the proof. After these operations every old vertex lies in $\square_1 \subset \square \subset \Omega$ and every new vertex in $\square$, so that the characteristic function of Ω equals one wherever a vertex or the end of a line occurs; every object depending on Ω or on the cubes, the two differences of propagators, the remainder of the expansion of the propagator and the gauge factors, has been moved into the external fields, which enter the rest of the proof through their norms at the points of $\square_1$ only.[55]

The localization function of the vector field legs and the localized counterterm, the objects which depend on $B^k(\Lambda_7^{(k-1)\prime})$, remain in the vertices, fixed before the step, and enter only the identities between a graph and its counterterm with the same localization.[56] In the rest of the proof every propagator is $G_k(0)$ or one of its slices, every sum runs over the lattice, the localization cubes or the torus, the Taylor formula is applied to the legs,[57] and the λ-vertices are matched against counterterms with the same localization; neither the side of the cubes nor the distance to the boundary of Ω occurs there.

Seventh, the chains without old vertices. A class without old vertices is a chain $v_1, \ldots, v_m$ of new vertices of one product $(G_0V)^m G_0$, $G_0 = G_k(\square, B_0)$, with the cut lines at its ends as

[52] Sect. 2 of [Ba 3].
[53] (2.1) of [Ba 3].
[54] (1.16) of [Ba 3].
[55] The norms in (3.3) of [Ba 3].
[56] The function g_k and the counterterm $\delta m^2(e', g_k, \lambda', \Omega_1; x)$ of Sect. 1 of [Ba 3].
[57] (3.10) of [Ba 3].

external scalar fields; the vertices are those which carry legs of the external field B'.[58] These chains are bounded in [Ba 3] by the estimate for the graphs all of whose subgraphs have positive degree, whose summation by parts is performed on the torus;[59] on the box with Neumann conditions it produces a term at the faces, where the localization of B' does not vanish, and we proceed differently. With the norm $\|\psi\|_E^2 = \langle \psi, G_0^{-1}\psi\rangle$ we have $\|D_{B_0}\psi\| \leqq \|\psi\|_E$ and $\|\psi\| \leqq \gamma_1^{-1/2}\|\psi\|_E$, where

$$G_0^{-1} \geqq \gamma_1 = \min\{(2C_P)^{-1},\ \tfrac{3}{8}a\}\,, \tag{D.11}$$

C_P being the constant of the Poincaré inequality $\|\psi - Q_k\psi\|^2 \leqq C_P\|\partial^\eta\psi\|^2$ on a unit block, the inverse of the first non-zero Neumann eigenvalue of the discrete cube of side one, which is bounded uniformly in the mesh and depends on d only; indeed $\|\psi\|^2 \leqq 2\|\psi - Q_k\psi\|^2 + 2\|Q_k\psi\|^2$, the bonds inside the blocks of $\square$ are bonds of $\square$, $a_k \geqq (1 - L^{-2})a \geqq \frac{3}{4}a$ by (2.14), and the constant field is removed by the gauge transformation.

Each vertex is then a bounded form in this norm. The vertex with one covariant derivative is a sum over the bonds of $\square$ of $(D_{B_0}\phi)(b)\cdot M(b)\chi(b_-)$ with $|M(b)| \leqq C(e(L^k\varepsilon)^{2-d/2}\sup|B'|)^{n'}$, bounded by $C\|\phi\|_E\gamma_1^{-1/2}\|\chi\|_E$; the two vertices with the remainder factor $R_{\bar n+1}$ likewise, the remainder being at most one; the vertex without derivative uses $\gamma_1^{-1/2}$ on both legs; and a block pair uses $|B'(\Gamma)| \leqq 2d\sup|B'|$ on its contours and $\sum_{x\in B^k(y)}\eta^d|\phi(x)| \leqq \|\phi\|$.[60]

Hence the value of the chain,

$$\langle G_0^{1/2}K_1^*\Phi_1,\ \hat K_2\cdots\hat K_{m-1}G_0^{1/2}K_m\Phi_m\rangle\,, \qquad \hat K_v = G_0^{1/2}K_vG_0^{1/2}\,, \tag{D.12}$$

is bounded by the product of the form norms of its vertices and of the norms $\|h\Phi\|_E$ of the fields at its ends, h being equal to one on the 2η-neighbourhood of the end cube and to zero at distance at least one from it, with $|\partial^\eta h| \leqq c_h$. Each form norm is at most $C(e(L^k\varepsilon)^{2-d/2}\sup|B'|)^{n'_v}$, and $\|h\Phi\|_E^2 = \|D_{B_0}(h\Phi)\|^2 + m^2(L^k\varepsilon)^2\|h\Phi\|^2 + a_k\|Q_k(B_0)h\Phi\|^2 \leqq C(1 + c_h)^2\|h\Phi\|_{1,0}^2$ by the Leibniz rule on a cube of side at most four, $\|\cdot\|_{1,0}$ denoting the supremum of a field and of its first differences. So the supremum of an end field and of its first differences on the end cube is all that is used of it.

The cut lines of a chain are the propagators G_0 between the η-neighbourhoods of unit cubes which do not meet, hence between points at distance at least $1 - 2\eta \geqq \frac{3}{5}$; by (D.13) they are bounded, with their first differences in each argument, by $Ce^{-\delta|x-x'|}$ at every pair of points of $\square$, the faces included. A cut line K contracted between two functionals A, B, each bounded by the supremum and the first differences of its field, satisfies $|A(x \mapsto B(K(x,\cdot)))| \leqq \|A\|\,\|B\|\max_{a,b\in\{0,1\}}\sup|D_x^aD_{x'}^bK(x,x')|$, since $x \mapsto B(K(x,\cdot))$ is linear in K and its differences are $B(D_xK(x,\cdot))$ bond by bond; so no hypothesis on the distance of the two supports beyond $\frac{3}{5}$ is needed.

A cut line between the end of a chain and a class with old vertices is an external scalar field of that class, read through its norm after localization to the η-neighbourhood of the cube of the class by a function h'; by the same linearity its contribution is bounded by $\|A\|\,\|B\|\max_a\sup_x\|h'D_x^aK(x,\cdot)\|_{1,\alpha_0}$ with the norm (5.5) in the last factor, which (D.14) bounds by $C(1 + c_h)^2e^{-\delta\,\mathrm{dist}(x,\Delta')}$, Δ' the cube of the class, at every x of the end region, the faces of $\square$ included. The forms of the vertices are sums over the bonds of $\square$, so that vertices at the faces need no separate treatment.

The tree factor of (5.4) is obtained as before: it comes from the decay of the cut lines and of the external fields, the bound on a component carrying no decay;[61] the lines between the cubes of the old vertices and the end cubes of the chains are cut, and they carry the decay of (D.13) and (D.14).

[58] (1.8)–(1.11) and (1.13)–(1.15) of [Ba 3].
[59] Proposition 2.1 of [Ba 3] and the formula (2.8) there.
[60] The vertices (1.8), (1.9) and (1.11), (1.10), and (1.13)–(1.14) of [Ba 3], in this order.
[61] (3.1)–(3.2) and (2.5) of [Ba 3].

So the side $r(L^k\varepsilon)$ of the original cubes is used only in the containment, where $r^\flat$ serves by hypothesis, and the constant acquires the factors of the third and fourth places, both of the kind occurring before; this proves the first statement. The remaining steps of the proof use the propagator bounds at depth R_1 at the vertices and at points within two units of them, which require the depth $R_1 + 2 \leqq r^\flat \leqq D^\flat$. For the last statement, a vertex within a unit cube of $B^k(\Lambda_7^{(k-1)\prime})$ lies at depth exceeding $5w_k - 1$ in Ω, $\Lambda_7^{(k-1)\prime}$ lying five corridors inside $\Lambda_2^{(k-1)\prime}$; in the second use of the estimate, where the domain is $B^k(\Lambda_0^{(k)})$, the classes are the terms of the expansion of the normalization factors in the field $\theta_{k+1}A'^{(k)} + \tilde{B}^{(k)}$, so that every connected component of a class carries a leg of that field, and its old vertices lie within $\ell_{\bar{n}}$ of a unit cube of the support of that field, at depth exceeding $2r(L^k\varepsilon) - 3 - M_\theta - 2\eta - \ell_{\bar{n}}$ (Appendix B). For $R \geqq D^\flat + M_b + 1$ and $R \geqq M_\theta + \ell_{\bar{n}} + 2$ both depths exceed $D^\flat$, since $5w_k = r(L^{k-1}\varepsilon) - M_b \geqq R - M_b$ and $2r(L^k\varepsilon) - 3 - M_\theta - 2\eta - \ell_{\bar{n}} \geqq (R - 1 - 2\eta) + (R - M_\theta - \ell_{\bar{n}} - 2) \geqq D^\flat + M_b - 2\eta \geqq D^\flat$, as $M_b \geqq 2\eta$. This completes the proof of Lemma D.3.

D.4. The propagator of a box at a constant field. Let $\square$ be a rectangular parallelepiped of large blocks of the unit lattice $T_1^{(k)}$ which does not wrap around the torus, B_0 a constant field, and $G_0 = G_k(\square, B_0)$ the propagator with Neumann conditions on the faces of $\square$, $G_0^{-1} = D_{B_0}^* D_{B_0} + m^2(L^k\varepsilon)^2 + a_k Q_k(B_0)^* Q_k(B_0)$ on the functions on $\square_\eta$. We write D_x^a, $a = 0, 1$, for the identity or a covariant forward difference on a bond of $\square$ at x. Then, for $d = 3$, with constants depending on L, a and the masses only, for $x, x' \in \square$ and $a, b \in \{0, 1\}$,

$$|D_x^a D_{x'}^b G_0(x, x')| \leqq C(\eta + |x - x'|)^{-(d-2+a+b)} e^{-\delta|x-x'|} \tag{D.13}$$

at every pair of points of $\square$, the bonds of its faces included; in particular the bound (3.8) holds for G_0 on $\square$ up to its faces, and $|D_x^a D_{x'}^b G_0(x, x')| \leqq C e^{-\delta|x-x'|}$ for $|x - x'| \geqq \frac{3}{5}$ when $d = 3$.

Moreover, if Δ, Δ' are two unit cubes of $\square$ at distance at least one, h' a function on $\square$ with $0 \leqq h' \leqq 1$, equal to one on the η-neighbourhood of Δ' and to zero at distance at least $\frac{1}{2}$ from Δ', with first and second differences bounded by c_h, and x lies within η of Δ, then for $a = 0, 1$

$$\|h'(\cdot)\, D_x^a G_0(x, \cdot)\|_{1,\alpha_0} \leqq C(1 + c_h)^2 e^{-\delta \operatorname{dist}(x, \Delta')}\,, \tag{D.14}$$

the norm being (5.5), the supremum of the field, of its first differences and of the Hölder quotients of exponent α_0 of the first differences on unit balls, taken over all of $\square$.

The operators on ηZ^d used below are the limits of those on the tori as the sides grow, the masses being positive, and the bounds of Theorem 3.3 and Proposition 3.7, whose constants depend by [Ba 4] on d, a, L and the large block size only and not on the torus, hold for them as well.

First, the constant field is removed: with $\lambda_0(x) = B_0 \cdot (x - c)$ on $\square$ we have $G_k(\square, B_0) = U(\lambda_0) G_k(\square, 0) U(\lambda_0)^*$ and $D_{B_0} U(\lambda_0)\phi = U(\lambda_0)\partial^\eta \phi$ bond by bond, so that every quantity in (D.13)–(D.14) equals the one at $B_0 = 0$.

Second, the Neumann propagator is written by images. Let $\square = \prod_\mu [\alpha_\mu, \beta_\mu)$, let $T_\square$ be the torus of periods $2(\beta_\mu - \alpha_\mu)$, each a multiple of $2M_b$, and let $W_\square$ be the group generated by the reflections in the planes $x_\mu = \alpha_\mu - \eta/2$, which map sites to sites, bonds to bonds and large blocks to large blocks of $T_\square$ and commute with the averages Q_k. Every bond of $T_\square$ crossing one of these planes joins a site to its image, where an even function has zero difference, and the remaining bonds of $T_\square$ inside $\square$ are the bonds of $\square$; so the Neumann operator on $\square$ is the restriction of the periodic operator on $T_\square$ to the even functions, and

$$G_k(\square, 0)(x, x') = \sum_{g \in W_\square} G_k(T_\square, 0)(x, gx')\,, \qquad x, x' \in \square\,, \tag{D.15}$$

every image satisfying $\operatorname{dist}_{T_\square}(x, gx') \geqq |x - x'|$.

Third, the periodic propagator is the periodization $\sum_n G_k(\eta Z^d, 0)(x, x' + 2\ell n)$ of the propagator on ηZ^d, ℓ the vector of the sides of $\square$, with $|x - x' - 2\ell n| \geqq \operatorname{dist}_{T_\square}(x, x')$ for every n and

at most $(2+j)^d$ terms with $|x-x'-2\ell n| < j+1$; so (D.13) follows from the same bound on ηZ^d with twice the rate.

Fourth, on ηZ^d at zero field $G_k(\eta Z^d, 0) = \sum_{j<k} G^\eta_{(j)}$ by (2.18), and each slice with $j \geqq 1$ factorizes, $G^\eta_{(j)} = a_j G_j Q_j^* C^{(j)} Q_j G_j$, so that with the legs $\ell_j(\cdot, u) = a_j (G_j Q_j^* 1_u)(\cdot)$ of the blocks u of the scale $L^j\eta$

$$D^a_x D^b_{x'} G^\eta_{(j)}(x,x') = a_j^{-1} \sum_{u,v} (D^a \ell_j)(x,u)\, C^{(j)}(u,v)\, (D^b \ell_j)(x',v)\,, \tag{D.16}$$

each variable sitting in its own leg. In the units of the scale $L^j\eta$ the legs and their first differences are bounded by $Ce^{-\delta_0 \mathrm{dist}(x,u)}$ at every site and bond, by (3.8) after the rescaling (2.21), and $|C^{(j)}(u,v)| \leqq Ce^{-\delta_0|u-v|}$ by (3.7); two geometric sums give $|D^a_x D^b_{x'} G^\eta_{(j)}(x,x')| \leqq Ce^{-\delta|x-x'|}$ in those units, that is $C(L^j\eta)^{-(d-2+a+b)} \exp[-\delta(L^j\eta)^{-1}|x-x'|]$ in the units of the step. The slice $j=0$, the propagator of the first step, is not of the form (D.16); its kernel and first differences are bounded by (3.81) with $j=0$, and a further difference costs at most the factor $2\eta^{-1}$, so that it obeys the bound of the other slices with $L^j\eta = \eta$ and a rate and constant depending on L.

Summing over j, the slices with $L^j\eta \geqq \max\{|x-x'|, \eta\}$ contribute at most $C\max\{|x-x'|,\eta\}^{-(d-2+a+b)}$, the sum being geometric, and the slices with $L^j\eta < |x-x'|$ at most $C|x-x'|^{-(d-2+a+b)}$ times $\sum_t t^{d-2+a+b} e^{-\delta t/2}$ over the powers t of L, each term carrying besides the factor $e^{-\delta|x-x'|/2}$; this is (D.13) on ηZ^d.

The same argument applies to the slices $j<K$ of the full propagators C_s, C_v of (2.18) on εZ^d; the remaining part $C_s - G_K(0)$ is a propagator of the unit lattice with the mass m, and it and its first and second differences are bounded by $Ce^{-\delta|u|}$ with δ depending on m, uniformly in ε, and likewise for C_v with μ_0. Together these give $|\partial^a\partial^b C_s(u)|$, $|\partial^a\partial^b C_v(u)| \leqq C|u|^{-(d-2+a+b)} e^{-\delta|u|}$ for $u \neq 0$ and $a,b \in \{0,1\}$ on εZ^d, uniformly in ε, with $\delta > 0$ depending on the masses; these are the bounds which Appendix A uses.

Finally, (D.14). The reflections and the translations above are isometries of the lattice mapping bonds to bonds, so that each clause of the norm of $x' \mapsto D^a_x G_k(\square, 0)(x,x')$ is bounded by the sum over the images of the same clause for $x' \mapsto D^a_x G_k(\eta Z^d, 0)(x,x')$ at points at least as far from x.

For the latter, by (D.16), in the units of the scale $L^j\eta$ the slice j of this function is the leg of the block data $a_j^{-1}\beta_j$, $\beta_j(v) = \sum_u (D^a\ell_j)(x,u) C^{(j)}(u,v)$, with $|\beta_j(v)| \leqq Ce^{-\delta \mathrm{dist}(x,v)}$; the supremum, the first differences and the Hölder quotients of the leg of bounded block data decay from the support of the data, by (3.8)–(3.9) at the scale $L^j\eta$ after the rescaling (2.21), the data being split into the part within $\rho'/2$ of x, whose support is at distance at least $\rho'/2$ from the points considered, and the rest, of size at most $Ce^{-\delta\rho'/2}$; so at points at distance at least ρ' from x every clause of the slice j is bounded by $C\exp[-\delta(L^j\eta)^{-1}\rho']$ in those units, and the sum over j is bounded as before by $C(\eta+\rho')^{-(d-1+a+\alpha_0)} e^{-\delta\rho'} \leqq Ce^{-\delta\rho'}$ for $\rho' \geqq \frac{1}{20}$.

The function $h' D^a_x G_0(x,\cdot)$ and its first differences vanish outside the $(\frac{1}{2}+\eta)$-neighbourhood of Δ', whose points lie at distance at least $\frac{1}{2} - 2\eta \geqq \frac{1}{10}$ from x for $k \geqq 1$, where $\eta \leqq \frac{1}{5}$, and at distance $\rho' \geqq \mathrm{dist}(x,\Delta') - 1$, which gives the exponential factor of (D.14) with the constant enlarged by e^δ; the Leibniz rule with the bounds on h' gives (D.14), a Hölder pair with one point within $\frac{1}{20}$ of x being bounded by a constant times a first difference, since the first difference vanishes at that point.

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